

# Optimization

# Quadratic Forms and Definite Matrices

The natural starting point for the study of optimization problems is the simplest such problem: the optimization of a *quadratic form*. There are a number of good reasons for studying *quadratic* optimization problems first. Quadratic forms are the simplest functions after linear ones. Like linear functions, they have matrix representations, so that studying the properties of a quadratic form reduces to studying properties of a symmetric matrix. Quadratic forms provide an excellent introduction to the vocabulary and techniques of optimization problems. Furthermore, the second order conditions that distinguish maxima from minima in economic optimization problems are stated in terms of quadratic forms. Finally, a number of economic optimization problems have a quadratic objective function, for example, risk minimization problems in finance, where riskiness is measured by the (quadratic) variance of the returns from investments.

**Example 16.1** Among the functions of one variable, the simplest functions with a unique global extremum are the pure quadratics:  $y = x^2$  and  $y = -x^2$ . The former has a global minimum at  $x = 0$ ; the latter has a global maximum at  $x = 0$ , as illustrated in Figure 16.1.

## 16.1 QUADRATIC FORMS

Recall the definition of a quadratic form on  $\mathbf{R}^n$  from Section 13.3.

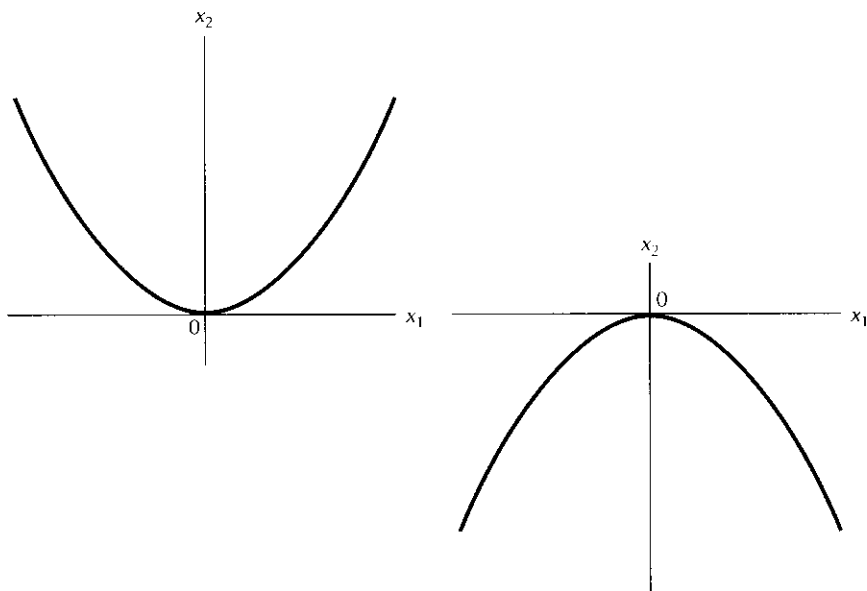
**Definition** A **quadratic form** on  $\mathbf{R}^n$  is a real-valued function of the form

$$Q(x_1, \dots, x_n) = \sum_{i \leq j} a_{ij} x_i x_j, \quad (1)$$

in which each term is a monomial of degree two.

The presentation in Section 13.3 showed that each quadratic form  $Q$  can be represented by a *symmetric* matrix  $A$  so that

$$Q(\mathbf{x}) = \mathbf{x}^T \cdot A \cdot \mathbf{x}. \quad (2)$$



**Figure**  
**16.1**

*The functions of  $f(x) = x^2$  and  $f(x) = -x^2$ .*

For example, the general two-dimensional quadratic form

$$a_{11}x_1^2 + a_{12}x_1x_2 + a_{22}x_2^2 \quad (3)$$

can be written as

$$(x_1 \ x_2) \begin{pmatrix} a_{11} & \frac{1}{2}a_{12} \\ \frac{1}{2}a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

And the general three dimensional quadratic form

$$a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 + a_{12}x_1x_2 + a_{13}x_1x_3 + a_{23}x_2x_3 \quad (4)$$

can be written as:

$$(x_1 \ x_2 \ x_3) \begin{pmatrix} a_{11} & \frac{1}{2}a_{12} & \frac{1}{2}a_{13} \\ \frac{1}{2}a_{12} & a_{22} & \frac{1}{2}a_{23} \\ \frac{1}{2}a_{13} & \frac{1}{2}a_{23} & a_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

## 16.2 DEFINITENESS OF QUADRATIC FORMS

A quadratic form always takes on the value zero at the point  $\mathbf{x} = \mathbf{0}$ . Its distinguishing characteristic is the set of values it takes when  $\mathbf{x} \neq \mathbf{0}$ . In this chapter, we

focus on the question of whether  $\mathbf{x} = \mathbf{0}$  is a max, a min or neither of the quadratic forms under consideration.

The general quadratic form of one variable is  $y = ax^2$ . If  $a > 0$ , then  $ax^2$  is always  $\geq 0$  and equals 0 only when  $x = 0$ . Such a form is called **positive definite**;  $x = 0$  is its global *minimizer*. If  $a < 0$ , then  $ax^2$  is always  $\leq 0$  and equals 0 only when  $x = 0$ . Such a quadratic form is called **negative definite**;  $x = 0$  is its global *maximizer*. Figure 16.1 illustrates these two situations.

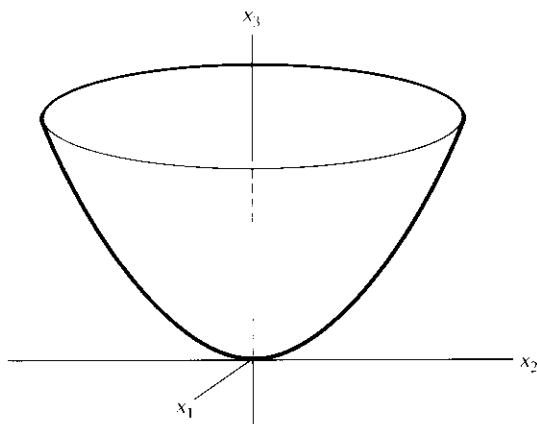
In two dimensions, the quadratic form  $Q_1(x_1, x_2) = x_1^2 + x_2^2$  is always greater than zero at  $(x_1, x_2) \neq (0, 0)$ . So, we call  $Q_1$  **positive definite**. Quadratic forms like  $Q_2(x_1, x_2) = -x_1^2 - x_2^2$ , which are strictly negative except at the origin, are called **negative definite**. Quadratic forms like  $Q_3(x_1, x_2) = x_1^2 - x_2^2$ , which take on both positive and negative values ( $Q_3(1, 0) = +1$  and  $Q_3(0, 1) = -1$ ) are called **indefinite**.

There are two intermediate cases: a quadratic form which is always  $\geq 0$  but may equal zero at some nonzero  $\mathbf{x}$ 's is called **positive semidefinite**. This property is illustrated by the quadratic form

$$Q_4(x_1, x_2) = (x_1 + x_2)^2 = x_1^2 + 2x_1x_2 + x_2^2,$$

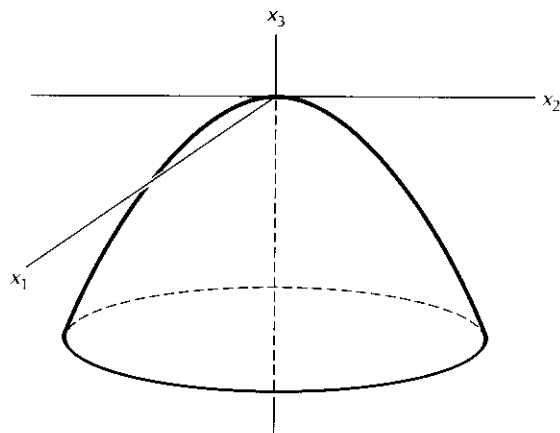
which is never negative but which equals zero at nonzero points such as  $(x_1, x_2) = (1, -1)$  or  $(-2, 2)$ . A quadratic form like  $Q_5(x_1, x_2) = -(x_1 + x_2)^2$ , which is never positive but can be zero at points other than the origin, is called **negative semidefinite**.

Figures 16.2 through 16.6 present the graphs of the above five quadratic forms  $Q_1, \dots, Q_5$ . Every quadratic form on  $\mathbf{R}^2$  has a graph similar to one of these five. For example, every positive definite quadratic on  $\mathbf{R}^2$  has a bowl shaped graph as in Figure 16.2, and every indefinite quadratic on  $\mathbf{R}^2$  has a saddle-shaped graph as in Figure 16.4.



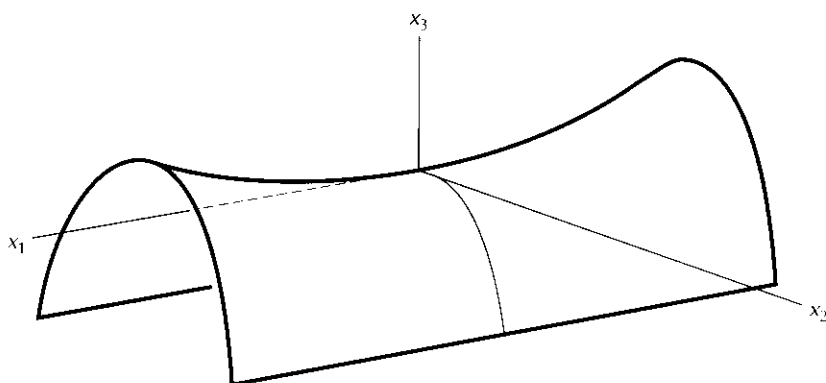
Graph of the positive definite form  $Q_1(x_1, x_2) = x_1^2 + x_2^2$ .

**Figure**  
**16.2**



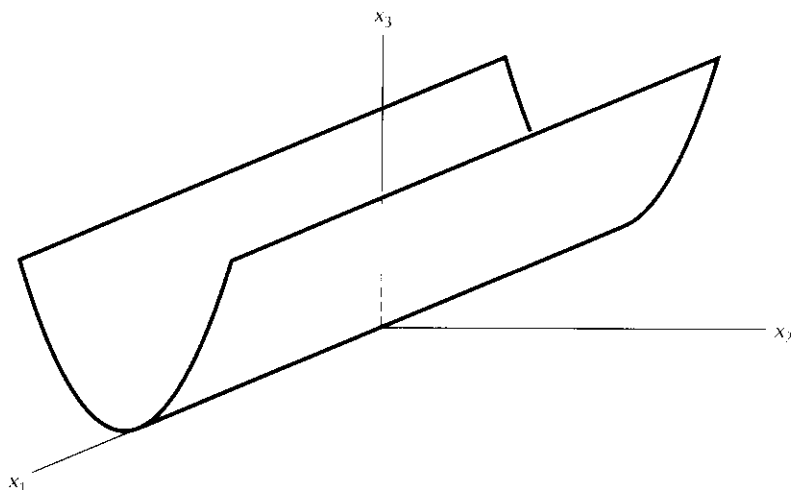
**Figure  
16.3**

*Graph of the negative definite form  $Q_2(x_1, x_2) = -x_1^2 - x_2^2$ .*



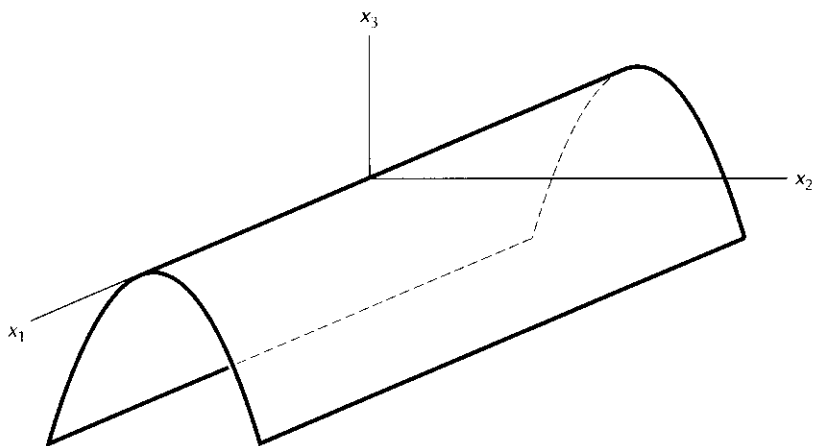
**Figure  
16.4**

*The graph of the indefinite form  $Q_3(x_1, x_2) = x_1^2 - x_2^2$ .*



**Figure  
16.5**

*The graph of the positive semidefinite form  $Q_4(x_1, x_2) = (x_1 + x_2)^2$ .*



**Figure  
16.6**

*The graph of the negative semidefinite form  $Q_5(x_1, x_2) = -(x_1 + x_2)^2$ .*

## Definite Symmetric Matrices

A symmetric matrix is called positive definite, positive semidefinite, negative definite, etc., according to the definiteness of the corresponding quadratic form  $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ . Since we will usually be applying this terminology to symmetric matrices directly, we focus on such matrices for our formal definitions of definiteness.

**Definition** Let  $A$  be an  $n \times n$  symmetric matrix, then  $A$  is:

- (a) **positive definite** if  $\mathbf{x}^T A \mathbf{x} > 0$  for all  $\mathbf{x} \neq \mathbf{0}$  in  $\mathbf{R}^n$ ,
- (b) **positive semidefinite** if  $\mathbf{x}^T A \mathbf{x} \geq 0$  for all  $\mathbf{x} \neq \mathbf{0}$  in  $\mathbf{R}^n$ ,
- (c) **negative definite** if  $\mathbf{x}^T A \mathbf{x} < 0$  for all  $\mathbf{x} \neq \mathbf{0}$  in  $\mathbf{R}^n$ ,
- (d) **negative semidefinite** if  $\mathbf{x}^T A \mathbf{x} \leq 0$  for all  $\mathbf{x} \neq \mathbf{0}$  in  $\mathbf{R}^n$ , and
- (e) **indefinite** if  $\mathbf{x}^T A \mathbf{x} > 0$  for some  $\mathbf{x}$  in  $\mathbf{R}^n$  and  $< 0$  for some other  $\mathbf{x}$  in  $\mathbf{R}^n$ .

**Remark** A matrix that is positive (negative) definite is automatically positive (negative) semidefinite. Otherwise, every symmetric matrix falls into one of the above five categories.

## Application: Second Order Conditions and Convexity

The definiteness of a symmetric matrix plays an important role in economic theory and in applied mathematics in general. For example, for a function  $y = f(x)$  of one variable, the sign of the second derivative  $f''(x_0)$  at a critical point  $x_0$  of  $f$  gives a necessary condition and a sufficient condition for determining whether  $x_0$  is a maximum of  $f$ , a minimum of  $f$ , or neither. The generalization of this second derivative test to higher dimensions involves checking whether the

second derivative matrix (or Hessian) of  $f$  is positive definite, negative definite, or indefinite at a critical point of  $f$ .

In a similar vein, a function  $y = f(x)$  of one variable is concave if its second derivative  $f''(x)$  is  $\leq 0$  on some interval. The generalization of this result to higher dimensions states that a function is concave on some region if its second derivative matrix is *negative semidefinite* for all  $\mathbf{x}$  in the region.

### Application: Conic Sections

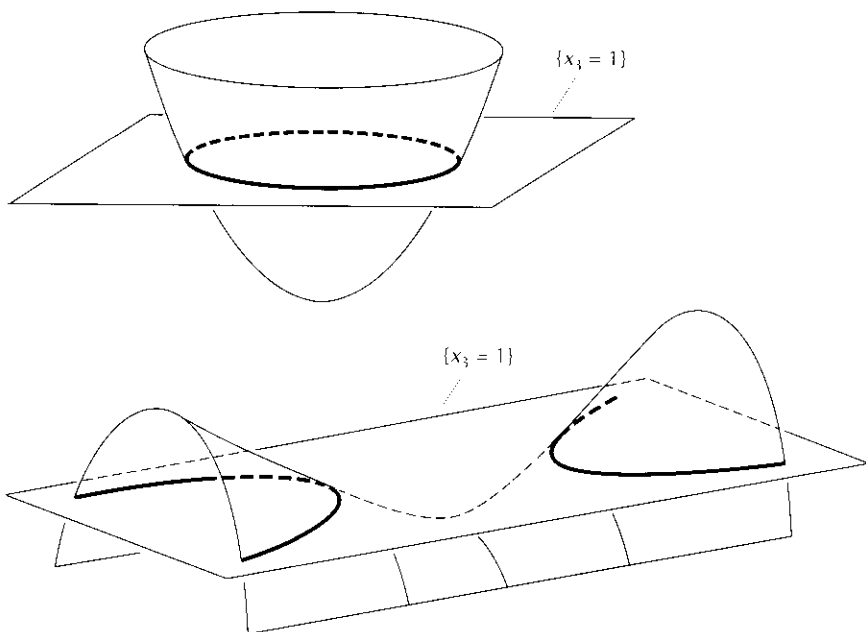
In plane geometry, the conic section described by the level curve

$$Q(x_1, x_2) \equiv a_{11}x_1^2 + a_{12}x_1x_2 + a_{22}x_2^2 = 1 \quad (5)$$

is completely determined by the definiteness of  $Q$  or of its associated matrix

$$A = \begin{pmatrix} a_{11} & \frac{1}{2}a_{12} \\ \frac{1}{2}a_{12} & a_{22} \end{pmatrix}.$$

Figure 16.7 illustrates the connection. The horizontal plane  $\{x_3 = 1\}$  cuts the graph in Figure 16.2 in an ellipse or circle. Therefore, if  $A$  is positive definite, the set (5) is an ellipse or circle. Since  $\{x_3 = 1\}$  cuts the graph in Figure 16.4 in a hyperbola, as also illustrated in Figure 16.7, equation (5) describes a hyperbola if  $A$  is



**Figure**  
**16.7**

*Levels sets of graphs of quadratic forms.*

indefinite. Since  $\{x_3 = 1\}$  cuts the graph of Figure 16.5 in a pair of parallel lines, equation (5) defines two lines if  $A$  is positive semidefinite but not positive definite. Finally, since the plane  $\{x_3 = 1\}$  lies strictly above the graphs in Figures 16.3 and 16.6, the set (5) is empty if  $A$  is negative definite or even negative semidefinite.

## Principal Minors of a Matrix

In this section we will describe a simple test for the definiteness of a quadratic form or of a symmetric matrix. To describe this algorithm, we need some more vocabulary.

**Definition** Let  $A$  be an  $n \times n$  matrix. A  $k \times k$  submatrix of  $A$  formed by deleting  $n - k$  columns, say columns  $i_1, i_2, \dots, i_{n-k}$  and the same  $n - k$  rows, rows  $i_1, i_2, \dots, i_{n-k}$ , from  $A$  is called a  $k$ th order **principal submatrix** of  $A$ . The determinant of a  $k \times k$  principal submatrix is called a  $k$ th order **principal minor** of  $A$ .

*Example 16.2* For a general  $3 \times 3$  matrix

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

there is one third order principal minor:  $\det(A)$ . There are three second order principal minors:

- (1)  $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$ , formed by deleting column 3 and row 3 from  $A$ ;
- (2)  $\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$ , formed by deleting column 2 and row 2 from  $A$ ;
- (3)  $\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$ , formed by deleting column 1 and row 1 from  $A$ .

There are three first order principal minors:

- (1)  $|a_{11}|$ , formed by deleting the last 2 rows and columns,
- (2)  $|a_{22}|$ , formed by deleting the first and third rows and the first and third columns, and
- (3)  $|a_{33}|$ , formed by deleting the first 2 rows and columns.

It is important to understand why no other submatrix of  $A$  is a principal submatrix. For practice, list all the principal minors of a general  $4 \times 4$  matrix.

Among the  $k$ th order principal minors of a given matrix, there is one special one that we want to highlight.



**Definition** Let  $A$  be an  $n \times n$  matrix. The  $k$ th order principal submatrix of  $A$  obtained by deleting the *last*  $n - k$  rows and the *last*  $n - k$  columns from  $A$  is called the  $k$ th order **leading principal submatrix** of  $A$ . Its determinant is called the  $k$ th order **leading principal minor** of  $A$ . We will denote the  $k$ th order leading principal submatrix by  $A_k$  and the corresponding leading principal minor by  $|A_k|$ .

An  $n \times n$  matrix has  $n$  leading principal submatrices — the top-leftmost  $1 \times 1$  submatrix, the top-leftmost  $2 \times 2$  submatrix, etc. For the general  $3 \times 3$  matrix of Example 16.2, the three leading principal minors are

$$|a_{11}|, \quad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \quad \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$$

The following theorem provides a straightforward algorithm which uses the leading principal minors to determine the definiteness of a given matrix. We present its proof in the Appendix of this chapter. We will present other criteria for the definiteness of a symmetric matrix in Section 23.7.

**Theorem 16.1** Let  $A$  be an  $n \times n$  symmetric matrix. Then,

- (a)  $A$  is positive definite if and only if all its  $n$  leading principal minors are (strictly) positive.
- (b)  $A$  is negative definite if and only if its  $n$  leading principal minors alternate in sign as follows:

$$|A_1| < 0, \quad |A_2| > 0, \quad |A_3| < 0, \quad \text{etc.}$$

The  $k$ th order leading principal minor should have the same sign as  $(-1)^k$ .

- (c) If some  $k$ th order leading principal minor of  $A$  (or some pair of them) is nonzero but does not fit either of the above two sign patterns, then  $A$  is indefinite. This case occurs when  $A$  has a *negative*  $k$ th order leading principal minor for an *even* integer  $k$  or when  $A$  has a *negative*  $k$ th order leading principal minor and a *positive*  $\ell$ th order leading principal minor for two distinct *odd* integers  $k$  and  $\ell$ .

One way that the leading principal minor test of Theorem 16.1 can fail for a given symmetric matrix  $A$  is that some leading principal minor of  $A$  is *zero* while the nonzero ones fit the sign pattern in either *a)* or *b)* of Theorem 16.1. When this occurs, the matrix  $A$  is *not* definite and it may or it may not be semidefinite. In this case, to check for semidefiniteness, one no longer has the luxury of checking only the  $n$  *leading* principal minors of  $A$ , but must check the sign of *every* principal minor of  $A$ , using the test described by the following theorem.

**Theorem 16.2** Let  $A$  be an  $n \times n$  symmetric matrix. Then,  $A$  is positive semidefinite if and only if every principal minor of  $A$  is  $\geq 0$ ;  $A$  is negative semidefinite if and only if every principal minor of odd order is  $\leq 0$  and every principal minor of even order is  $\geq 0$ .

*Example 16.3* Suppose  $A$  is a  $4 \times 4$  symmetric matrix and, as usual, write  $|A_i|$  for its  $i$ th order *leading* principal minor.

- (a) If  $|A_1| > 0$ ,  $|A_2| > 0$ ,  $|A_3| > 0$ ,  $|A_4| > 0$ , then  $A$  is positive definite (and conversely).
- (b) If  $|A_1| < 0$ ,  $|A_2| > 0$ ,  $|A_3| < 0$ ,  $|A_4| > 0$ , then  $A$  is negative definite (and conversely).
- (c) If  $|A_1| > 0$ ,  $|A_2| > 0$ ,  $|A_3| = 0$ ,  $|A_4| < 0$ , then  $A$  is indefinite because of  $A_4$ .
- (d) If  $|A_1| < 0$ ,  $|A_2| < 0$ ,  $|A_3| < 0$ ,  $|A_4| < 0$ , then  $A$  is indefinite because of  $A_2$  (and  $A_4$ ).
- (e) If  $|A_1| = 0$ ,  $|A_2| < 0$ ,  $|A_3| > 0$ ,  $|A_4| = 0$ , then  $A$  is indefinite because of  $A_2$ .
- (f) If  $|A_1| > 0$ ,  $|A_2| = 0$ ,  $|A_3| > 0$ ,  $|A_4| > 0$ , then  $A$  is not definite. It is not negative semidefinite, but it may be positive semidefinite. To check for positive semidefiniteness, one must check all 15 principal minors of  $A$ , not just the four leading principal ones. If none of the principal minors are negative, then  $A$  is positive semidefinite. If at least one of them is negative,  $A$  is indefinite.
- (g) If  $|A_1| = 0$ ,  $|A_2| > 0$ ,  $|A_3| = 0$ ,  $|A_4| > 0$ , then  $A$  is not definite, but it may be positive semidefinite or negative semidefinite. To decide, one must again check all 15 of its principal minors.

To motivate these two theorems and to understand their algorithms better, we will examine them in some detail for the simplest symmetric matrices — diagonal matrices and then  $2 \times 2$  matrices.

### The Definiteness of Diagonal Matrices

The simplest  $n \times n$  symmetric matrices are the diagonal matrices. They also correspond to the simplest quadratic forms since

$$\begin{aligned}
 (x_1 \quad x_2 \quad \cdots \quad x_n) & \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \\
 &= a_1 x_1^2 + a_2 x_2^2 + \cdots + a_n x_n^2,
 \end{aligned} \tag{6}$$

a sum of squares. Clearly, this quadratic form will be positive definite if and only if all the  $a_i$ 's are positive and negative definite if and only if all the  $a_i$ 's are negative. It will be positive semidefinite if and only if all the  $a_i$ 's are  $\geq 0$  and negative semidefinite if and only if all the  $a_i$ 's are  $\leq 0$ . If there are two  $a_i$ 's of opposite signs, this form will be indefinite.

Since all the principal submatrices are diagonal matrices, their determinants — the principal minors — are just products of the  $a_i$ 's. If all the  $a_i$ 's are positive, then all their products are positive and so all the leading principal minors are positive. If all the  $a_i$ 's are negative (so the form is negative definite), then products of odd numbers of the  $a_i$ 's will be negative and products of even numbers of the  $a_i$ 's will be positive. This corresponds to the alternating signs condition in *b*) of Theorem 16.1 and indicates why we should expect such an alternating sign condition instead of an all negative condition in the test for negative definiteness.

If  $a_1$  is zero in (6), the form cannot be definite since it will be zero when evaluated at  $(1, 0, \dots, 0)$ . Notice that in this diagonal case, all the *leading* principal minors of (6) will also be zero, independent of the signs of the other  $a_i$ 's. In order to check that all the  $a_i$ 's have the proper sign when some of them are zero, one must check much more than just the leading principal minors.

### The Definiteness of $2 \times 2$ Matrices

One can verify Theorems 16.1 and 16.2 directly for  $2 \times 2$  symmetric matrices by completing the square in the corresponding quadratic form. Consider the general quadratic form on  $\mathbf{R}^2$ :

$$\begin{aligned} Q(x_1, x_2) &= \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ &= ax_1^2 + 2bx_1x_2 + cx_2^2. \end{aligned} \tag{7}$$

For ease of notation, we are using  $a$ ,  $b$ , and  $c$  in this example in place of  $a_{11}$ ,  $a_{12}$ , and  $a_{22}$ , respectively. If  $a = 0$ , then  $Q$  cannot be positive or negative definite because  $Q(1, 0) = 0$ . So, assume now that  $a \neq 0$  and complete the square in (7) by adding and subtracting  $b^2x_2^2/a$  in expression (7):

$$\begin{aligned} Q(x_1, x_2) &= ax_1^2 + 2bx_1x_2 + cx_2^2 + \frac{b^2}{a}x_2^2 - \frac{b^2}{a}x_2^2 \\ &= a \left( x_1^2 + \frac{2b}{a}x_1x_2 + \frac{b^2}{a^2}x_2^2 \right) - \frac{b^2}{a}x_2^2 + cx_2^2 \\ &= a \left( x_1 + \frac{b}{a}x_2 \right)^2 + \frac{(ac - b^2)}{a}x_2^2. \end{aligned} \tag{8}$$

If both coefficients,  $a$  and  $(ac - b^2)/a$  in (8) are positive, then  $Q$  will never be negative. It will equal zero only when

$$x_1 + \frac{b}{a}x_2 = 0 \quad \text{and} \quad x_2 = 0,$$

that is, when  $x_1 = 0$  and  $x_2 = 0$ . In other words, if

$$|a| > 0 \quad \text{and} \quad \det A = \begin{vmatrix} a & b \\ b & c \end{vmatrix} > 0,$$

then  $Q$  is positive definite. Conversely, in order for  $Q$  to be positive definite, we need both  $a$  and  $\det A = ac - b^2$  to be positive.

Similarly,  $Q$  will be negative definite if and only if both coefficients in expression (8),  $a$  and  $(ac - b^2)/a$ , are negative. This situation occurs if and only if  $a < 0$  and  $ac - b^2 > 0$ , that is, when the leading principal minors alternate in sign.

If  $(ac - b^2)$ , the second order principal minor, is negative, then the two coefficients in (8) have opposite signs. In particular,

$$Q\left(\frac{b}{a}, -1\right) = \frac{ac - b^2}{a} \quad \text{and} \quad Q(1, 0) = a$$

will have opposite signs; so  $Q$  is *indefinite*.

**Example 16.4** Consider

$$A = \begin{pmatrix} 2 & 3 \\ 3 & 7 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 2 & 4 \\ 4 & 7 \end{pmatrix}.$$

Since  $|A_1| = 2$  and  $|A_2| = 5$ ,  $A$  is positive definite. Since  $|B_1| = 2$  and  $|B_2| = -2$ ,  $B$  is indefinite.

**Example 16.5** Consider

$$C = \begin{pmatrix} 0 & 0 \\ 0 & c \end{pmatrix}.$$

Note that  $|C_1| = 0$  and  $|C_2| = 0$ . The definiteness of  $C$  depends completely on  $c$ ;  $C$  is positive semidefinite if  $c \geq 0$  and negative semidefinite if  $c \leq 0$ . This is especially obvious if one looks at the corresponding quadratic form  $Q_C(x_1, x_2) = cx_2^2$ .

## EXERCISES

**16.1** Determine the definiteness of the following symmetric matrices:

$$\begin{array}{llll} \text{a)} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} & \text{b)} \begin{pmatrix} -3 & 4 \\ 4 & -5 \end{pmatrix} & \text{c)} \begin{pmatrix} -3 & 4 \\ 4 & -6 \end{pmatrix} & \text{d)} \begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix} \\ \text{e)} \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 5 \\ 0 & 5 & 6 \end{pmatrix} & \text{f)} \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix} & \text{g)} \begin{pmatrix} 1 & 0 & 3 & 0 \\ 0 & 2 & 0 & 5 \\ 3 & 0 & 4 & 0 \\ 0 & 5 & 0 & 6 \end{pmatrix} \end{array}$$

**16.2** Let  $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$  be a quadratic form on  $\mathbf{R}^n$ . By evaluating  $Q$  on each of the coordinate axes in  $\mathbf{R}^n$ , prove that a necessary condition for a symmetric matrix to be positive definite (positive semidefinite) is that all the diagonal entries be positive (nonnegative). State and prove the corresponding result for negative and negative semidefinite matrices. Give an example to show that this necessary condition is not sufficient.

**16.3** Using the method of the previous exercise, sketch a proof that if  $A$  is positive (or negative) definite, then every principal submatrix of  $A$  is also positive (or negative) definite.

**16.4** How many  $k$ th order principal minors will an  $n \times n$  matrix have for each  $k \leq n$ ?

**16.5** Mimic the computation in (8) to prove Theorem 16.1 for a general symmetric  $3 \times 3$  matrix. [Hint: After "completing the square" twice, you should find that

$$\begin{aligned} (x_1 \ x_2 \ x_3) \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \\ |A_1| \left( x_1 + \frac{a_{12}}{a_{11}} x_2 + \frac{a_{13}}{a_{11}} x_3 \right)^2 - \frac{|A_2|}{|A_1|} \left( x_2 + \frac{a_{11}a_{23} - a_{12}a_{13}}{|A_2|} x_3 \right)^2 + \frac{|A_3|}{|A_2|} (x_3)^2. \end{aligned}$$

## 16.3 LINEAR CONSTRAINTS AND BORDERED MATRICES

## Definiteness and Optimality

Keep in mind the fact that determining the definiteness of a quadratic form  $Q$  is equivalent to determining whether  $\mathbf{x} = \mathbf{0}$  is a max, a min, or neither for the real-valued function  $Q$ . For example,  $\mathbf{x} = \mathbf{0}$  is the unique *global minimum* of quadratic form  $Q$  if and only if  $Q$  is positive definite, by the very definition of positive definiteness. Similarly,  $\mathbf{x} = \mathbf{0}$  is the unique *global maximum* of  $Q$  if and only if  $Q$  is negative definite.

The characterization of definiteness in Theorem 16.1 works only if there are no constraints in the problem under consideration, that is, if  $\mathbf{x}$  can take on any value in  $\mathbf{R}^n$ . If there are constraints, the analysis becomes more delicate.

**Example 16.6** The quadratic form  $Q(x_1, x_2) = x_1^2 - x_2^2$  on  $\mathbf{R}^2$  is indefinite; the origin is neither a max nor a min. But, if we restrict our attention to the  $x_1$ -axis, that is, if we impose the constraint  $x_2 = 0$ , then  $Q(x_1, 0) = x_1^2$  has a strict global minimum at  $x_1 = 0$ , and therefore  $Q$  is positive definite on the constraint set  $\{x_2 = 0\}$ . Alternatively, if we impose the constraint  $x_1 = 0$  and consider  $Q$  only on the  $x_2$ -axis, then  $x_2 = 0$  is a global max of  $Q(0, x_2) = -x_2^2$  and  $Q$  is negative definite on the subspace  $\{x_1 = 0\}$ . On the line  $x_1 - 2x_2 = 0$ ,  $Q(2x_2, x_2) = (2x_2)^2 - x_2^2 = 3x_2^2$  is positive definite.

As will be shown in Chapter 19, the second order condition which distinguishes maxima from minima in a *constrained* optimization problem is a condition on the definiteness of a quadratic form which is restricted to a linear subspace. Since most optimization problems in economics involve constraints on the variables under study, the rest of this chapter will discuss the definiteness of quadratic forms which are restricted to linear subspaces of  $\mathbf{R}^n$ .

Let us look in detail at the simplest such problem: the problem of determining the definiteness of, or of optimizing, a general quadratic form of two variables:

$$Q(x_1, x_2) = ax_1^2 + 2bx_1x_2 + cx_2^2 = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad (9)$$

on the general linear subspace

$$Ax_1 + Bx_2 = 0. \quad (10)$$

In Example 16.6, we worked with  $A = 0$ , then with  $B = 0$ , and finally with  $A = 1$  and  $B = -2$ .

Since our focus is on the matrix and not on the quadratic form itself, we have multiplied the coefficient of  $x_1x_2$  in (9) by 2 so that we do not have to deal with  $1/2$ s in the corresponding matrix. The simplest approach to this problem is to solve (10) for  $x_1$  in terms of  $x_2$ ; obtain  $x_1 = -Bx_2/A$ , and then substitute this expression for  $x_1$  in the objective function (9):

$$\begin{aligned} Q\left(-\frac{Bx_2}{A}, x_2\right) &= a\left(-\frac{Bx_2}{A}\right)^2 + 2b\left(-\frac{Bx_2}{A}\right)x_2 + cx_2^2 \\ &= \frac{aB^2}{A^2}x_2^2 - \frac{2bB}{A}x_2^2 + cx_2^2 \\ &= \frac{aB^2 - 2bAB + cA^2}{A^2}x_2^2. \end{aligned} \quad (11)$$

We conclude from (11) that  $Q$  is positive definite on the constraint set (10) if and only if  $aB^2 - 2bAB + cA^2 > 0$  and negative definite on (10) if and only if  $aB^2 - 2bAB + cA^2 < 0$ . There is a convenient way of writing this expression:

$$aB^2 - 2bAB + cA^2 = -\det \begin{pmatrix} 0 & A & B \\ A & a & b \\ B & b & c \end{pmatrix}. \quad (12)$$

The matrix in (12) is obtained by “bordering” the  $2 \times 2$  matrix (9) of the quadratic  $Q$  on the top and left by the coefficients of the linear constraint (10). The following theorem summarizes these calculations.

**Theorem 16.3** The quadratic form  $Q(x_1, x_2) = ax_1^2 + 2bx_1x_2 + cx_2^2$  is positive (respectively, negative) definite on the constraint set  $Ax_1 + Bx_2 = 0$  if and only if

$$\det \begin{pmatrix} 0 & A & B \\ A & a & b \\ B & b & c \end{pmatrix}$$

is negative (respectively, positive).

This same result holds for the general problem of determining the definiteness of

$$Q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} = (x_1 \quad \cdots \quad x_n) \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{12} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad (13)$$

on the linear constraint set

$$\begin{pmatrix} B_{11} & B_{12} & \cdots & B_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ B_{m1} & B_{m2} & \cdots & B_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \quad (14)$$

Border the matrix (13) of the quadratic form on the top and on the left by the matrix (14) of the linear constraints:

$$H = \left( \begin{array}{ccc|ccc} 0 & \cdots & 0 & B_{11} & \cdots & B_{1n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & B_{m1} & \cdots & B_{mn} \\ \hline B_{11} & \cdots & B_{m1} & a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ B_{1n} & \cdots & B_{mn} & a_{1n} & \cdots & a_{nn} \end{array} \right). \quad (15)$$

We first need to figure out which submatrices of  $H$  to consider. In studying the quadratic (9) on the constraint set (10), we had only one condition to check in Theorem 16.3 because our single constraint in  $\mathbf{R}^2$  resulted in a *one*-dimensional problem. The above problem (13, 14) has  $m$  linear equations of  $n$  variables. We therefore expect that this problem is really  $n - m$  dimensional and therefore that we will have  $n - m$  conditions to check for the matrix  $H$  in (15). Furthermore, from our experience in Section 16.2, we expect that we will look for leading principal minors of the *same* sign to check for positive definiteness and for leading principal minors of *alternating* signs to check for negative definiteness. The following theorem indicates that these expectations are correct, and it makes precise the exact sign patterns that we need to verify.

**Theorem 16.4** To determine the definiteness of a quadratic form (13) of  $n$  variables,  $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ , when restricted to a constraint set (14) given by  $m$  linear equations  $B\mathbf{x} = \mathbf{0}$ , construct the  $(n + m) \times (n + m)$  symmetric matrix  $H$  by bordering the matrix  $A$  above and to the left by the coefficients  $B$  of the linear constraints:

$$H = \begin{pmatrix} \mathbf{0} & B \\ B^T & A \end{pmatrix}.$$

Check the signs of the *last*  $n - m$  leading principal minors of  $H$ , starting with the determinant of  $H$  itself.

- (a) If  $\det H$  has the same sign as  $(-1)^n$  and if these last  $n - m$  leading principal minors *alternate* in sign, then  $Q$  is *negative definite* on the constraint set  $B\mathbf{x} = \mathbf{0}$ , and  $\mathbf{x} = \mathbf{0}$  is a strict global max of  $Q$  on this constraint set.
- (b) If  $\det H$  and these last  $n - m$  leading principal minors all have the *same* sign as  $(-1)^m$ , then  $Q$  is *positive definite* on the constraint set  $B\mathbf{x} = \mathbf{0}$ , and  $\mathbf{x} = \mathbf{0}$  is a strict global min of  $Q$  on this constraint set.
- (c) If both of these conditions a) and b) are violated by *nonzero* leading principal minors, then  $Q$  is *indefinite* on the constraint set  $B\mathbf{x} = \mathbf{0}$ , and  $\mathbf{x} = \mathbf{0}$  is neither a max nor a min of  $Q$  on this constraint set.

We will not present the rather intricate proof of this theorem here. Notice that its conclusions are consistent with the conclusions of Theorem 16.3 for  $n = 2$  and  $m = 1$ , where for the case of two variables and one constraint, we only needed to compute  $n - m = 2 - 1 = 1$  determinant. Exercise 16.9 looks at some important special cases of Theorem 16.4.

**Example 16.7** To check the definiteness of

$$Q(x_1, x_2, x_3, x_4) = x_1^2 - x_2^2 + x_3^2 + x_4^2 + 4x_2x_3 - 2x_1x_4$$



on the constraint set

$$x_2 + x_3 + x_4 = 0, \quad x_1 - 9x_2 + x_4 = 0,$$

form the bordered matrix

$$H_6 = \left( \begin{array}{cc|ccc} 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & -9 & 0 & 1 \\ \hline 0 & 1 & 1 & 0 & 0 & -1 \\ 1 & -9 & 0 & -1 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 \end{array} \right).$$

Since this problem has  $n = 4$  variables and  $m = 2$  constraints, we need to check the largest  $n - m = 2$  leading principal submatrices of  $H_6$ :  $H_6$  itself and

$$H_5 = \left( \begin{array}{cc|ccc} 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & -9 & 0 & 1 \\ \hline 0 & 1 & 1 & 0 & 0 & -1 \\ 1 & -9 & 0 & -1 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{array} \right).$$

Since  $m = 2$  and  $(-1)^2 = +1$ , we need  $\det H_6 > 0$  and  $\det H_5 > 0$  to verify positive definiteness. Since  $n = 4$  and  $(-1)^4 = +1$ , we need  $\det H_6 > 0$  and  $\det H_5 < 0$  to verify negative definiteness. In fact,  $\det H_6 = 24$  and  $\det H_5 = 77$ ; so  $Q$  is positive definite on the constraint set, and  $\mathbf{x} = \mathbf{0}$  is a min of  $Q$  restricted to the constraint set.

**Remark** If the test for constrained definiteness of Theorem 16.4 fails only because one or more of the last  $n - m$  leading principal minors is zero, then we would like a test for semidefiniteness, analogous to the statement of Theorem 16.2 for the unconstrained problem. Unfortunately, tests for constrained semidefiniteness are much more tedious to state than the criteria described in Theorem 16.2. Fortunately, such tests are rarely required in applications.

## One Constraint

Constrained maximization problems with just one effective constraint are common in economic theory. For the problem of checking the definiteness of a quadratic  $Q$  subject to a single constraint  $A_1x_1 + \dots + A_nx_n = 0$ , Theorem 16.4 states that one needs to check the last  $n - 1$  leading principal minors of

$$H_{n+1} = \left( \begin{array}{cccc} 0 & A_1 & \cdots & A_n \\ A_1 & a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ A_n & a_{n1} & \cdots & a_{nn} \end{array} \right). \quad (16)$$

The only omitted leading principal submatrices are:

$$H_1 = (0) \quad \text{and} \quad H_2 = \begin{pmatrix} 0 & A_1 \\ A_1 & a_{11} \end{pmatrix}.$$

Let's suppose that  $A_1 \neq 0$ . (One of the  $A_i$ 's must be nonzero.) Then,  $\det H_2 = -A_1^2 < 0$ . Since  $m = 1$  and  $(-1)^1 = -1$ , the criterion for constrained positive definiteness is that the last  $n - 1$  leading principal minors of (16) are negative. Since  $\det H_2 < 0$ , this criterion is equivalent to the statement that the last  $n$  leading principal minors have the same sign. The criterion for constrained negative definiteness is that  $\det H_{n+1}$  have the sign of  $(-1)^n$  and that  $\det H_3, \dots, \det H_{n+1}$  alternate in sign. This means, in this case, that  $\det H_3$  must be positive. It follows that the condition for constrained negative definiteness is equivalent to the condition that the last  $n$  leading principal minors of  $H_{n+1}$  alternate in sign. The following theorem summarizes this discussion for  $m = 1$  and yields an easier-to-remember approach for the problem of determining definiteness when there is only one linear constraint.

**Theorem 16.5** To determine the definiteness of a quadratic  $Q(x_1, \dots, x_n)$  subject to *one* linear constraint, form the usual  $(n + 1) \times (n + 1)$  bordered matrix  $H$ , as in (16). Suppose that  $A_1 \neq 0$ . If the last  $n$  leading principal minors of  $H_{n+1}$  have the same sign,  $Q$  is positive definite on the constraint set (and  $\mathbf{x} = \mathbf{0}$  is a constrained min of  $Q$ ). If the last  $n$  leading principal minors of  $H_{n+1}$  alternate in sign,  $Q$  is negative definite on the constraint set (and  $\mathbf{x} = \mathbf{0}$  is a constrained max of  $Q$ ).

## Other Approaches

For the sake of completeness, we mention two alternative approaches to the problem of determining the definiteness of a quadratic form of  $n$  variables subject to  $m$  linear equations. The statement of Theorem 16.4 focuses on the sign of the largest submatrix  $H_{m+n}$  as the cornerstone of the algorithm. Some presentations focus instead on the smallest of the last  $n - m$  leading principal submatrices:  $H_{2m+1}$ , the  $(2m + 1)$ th order leading principal submatrix. Theorem 16.4 implies the following alternative checks:

- (A) To verify positive definiteness, check that  $\det H_{2m+1}$  has the same sign as  $(-1)^m$  and that all the larger leading principal minors have this sign too.
- (B) To verify negative definiteness, check that  $\det H_{2m+1}$  has the sign of  $(-1)^{m+1}$  and that the leading principal minors of larger order alternate in sign.

Some texts prefer to construct the bordered matrix  $H$  by bordering the matrix  $A$  of the quadratic form  $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$  *below* and to the *right* by the matrix  $B$  for

the linear equations  $B\mathbf{x} = \mathbf{0}$ :

$$H_{m-n} = \begin{pmatrix} A & B^T \\ B & \mathbf{0} \end{pmatrix}.$$

In this situation, one must still check  $n - m$  principal minors. However, the corresponding principal submatrices are no longer leading ones, but “border-preserving” ones. One removes from  $H_{m+n}$ , one at a time, the  $n - m - 1$  rows and columns which contain the last  $n - m - 1$  rows and columns of the matrix  $A$ , that is, rows and columns  $n, n - 1, \dots, m + 2$  of  $H_{m+n}$ .

**Example 16.8** To use this approach for the problem in Example 16.7, form the bordered matrix

$$\hat{H} = \left( \begin{array}{cccc|cc} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & -1 & 2 & 0 & 1 & -9 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 & 1 & 1 \\ \hline 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & -9 & 0 & 1 & 0 & 0 \end{array} \right)$$

and then form the submatrix  $\hat{H}_5$  by removing row 4 and column 4 from  $\hat{H}$ , the row and column just before the border of  $\hat{H}$ :

$$\hat{H}_5 = \left( \begin{array}{cccc|cc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & -1 & 2 & 1 & 1 & -9 \\ 0 & 2 & 1 & 1 & 1 & 0 \\ \hline 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & -9 & 0 & 0 & 0 & 0 \end{array} \right).$$

Note that  $\det \hat{H} = 24$  and  $\det \hat{H}_5 = 77$ , just as we found for the corresponding minors in Example 16.7.

## EXERCISES

**16.6** Determine the definiteness of the following constrained quadratics:

- $Q(x_1, x_2) = x_1^2 + 2x_1x_2 - x_2^2$ , subject to  $x_1 + x_2 = 0$ .
- $Q(x_1, x_2) = 4x_1^2 + 2x_1x_2 - x_2^2$ , subject to  $x_1 + x_2 = 0$ .
- $Q(x_1, x_2, x_3) = x_1^2 + x_2^2 - x_3^2 + 4x_1x_3 - 2x_1x_2$ , subject to  $x_1 + x_2 + x_3 = 0$  and  $x_1 + x_2 - x_3 = 0$ .
- $Q(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 + 4x_1x_3 - 2x_1x_2$ , subject to  $x_1 + x_2 + x_3 = 0$  and  $x_1 + x_2 - x_3 = 0$ .
- $Q(x_1, x_2, x_3) = x_1^2 - x_3^2 + 4x_1x_2 - 6x_2x_3$ , subject to  $x_1 + x_2 - x_3 = 0$ .

- 16.7** Prove that statements  $A$  and  $B$  above are equivalent to the statement of Theorem 16.4.
- 16.8** Use the theory of determinants to show why the corresponding minors in Examples 16.7 and 16.8 have the same values.
- 16.9** Use the techniques of Theorem 16.3 to verify Theorem 16.4 for the general problem with:  $a)$  three variables and one constraint,  $b)$  three variables and two constraints.

## 16.4 APPENDIX

This section presents the proof of Theorem 16.1. This proof has two major ingredients: the principle of induction and the theory of partitioned matrices as developed in Section 8.7. We will prove Theorem 16.1 for positive definite matrices and leave the proof for negative definite matrices as an exercise. First, we need two simple lemmas.

**Lemma 16.1** If  $A$  is a positive or negative definite matrix, then  $A$  is nonsingular.

*Proof* Suppose that such an  $A$  is singular. Then, there exists a nonzero vector  $\mathbf{x}$  such that  $A\mathbf{x} = \mathbf{0}$ . But then,

$$\mathbf{x}^T \cdot A\mathbf{x} = \mathbf{x}^T \cdot \mathbf{0} = 0,$$

a contradiction to the definiteness of  $A$ . ■

**Lemma 16.2** Suppose that  $A$  is a symmetric matrix and that  $Q$  is a nonsingular matrix. Then,  $Q^T A Q$  is a symmetric matrix, and  $A$  is positive (negative) definite if and only if  $Q^T A Q$  is positive (negative) definite.

*Proof* To see that  $Q^T A Q$  is symmetric, one checks directly that it equals its own transpose:

$$(Q^T A Q)^T = Q^T A^T (Q^T)^T = Q^T A^T Q = Q^T A Q.$$

Suppose that  $Q^T A Q$  is positive definite. Let  $\mathbf{x} \neq \mathbf{0}$  be an arbitrary nonzero vector in  $\mathbf{R}^n$ . Since  $Q$  is nonsingular, there exists a nonzero vector  $\mathbf{y}$  such that  $\mathbf{x} = Q\mathbf{y}$ . Then

$$\mathbf{x}^T A \mathbf{x} = (Q\mathbf{y})^T A (Q\mathbf{y}) = \mathbf{y}^T Q^T A Q \mathbf{y} = \mathbf{y}^T (Q^T A Q) \mathbf{y},$$

which is positive, since  $Q^T A Q$  is positive definite. Therefore,  $A$  is positive definite.

On the other hand, if  $A$  is positive definite and  $\mathbf{z}$  is an arbitrary nonzero vector, then  $Q\mathbf{z}$  will be nonzero too, since  $Q$  is nonsingular. Therefore,

$$0 < (Q\mathbf{z})^T A (Q\mathbf{z}) = \mathbf{z}^T Q^T A Q \mathbf{z} = \mathbf{z}^T (Q^T A Q) \mathbf{z}.$$

and  $Q^T A Q$  is positive definite. ■

**Theorem 16.1** Let  $A$  be a symmetric matrix. Then,  $A$  is positive definite if and only if all its leading principal minors are positive.

*Proof* We will prove this result by using induction on the size  $n$  of  $A$ . The result is trivially true for  $1 \times 1$  matrices. We proved it for  $2 \times 2$  matrices in Section 16.2. We suppose that the theorem is true for  $n \times n$  matrices and prove it true for  $(n+1) \times (n+1)$  matrices.

Let  $A$  be an  $(n+1) \times (n+1)$  symmetric matrix. Write  $A_j$  for the  $j \times j$  leading principal submatrix of  $A$  for  $j = 1, \dots, n+1$ .

We first prove that if all the  $A_j$ 's have positive determinants, then  $A$  is positive definite. The leading principal submatrices of  $A_n$  are  $A_1, \dots, A_n$ , which are positive definite by hypothesis, since they are the first  $n$  leading principal submatrices of  $A$ . By the inductive hypothesis that the theorem is true for  $n \times n$  matrices, the  $n \times n$  symmetric matrix  $A_n$  is positive definite. By Lemma 16.1 above,  $A_n$  is invertible. Partition  $A$  as

$$A = \left( \begin{array}{c|c} A_n & \mathbf{a} \\ \hline \mathbf{a}^T & a_{n+1, n+1} \end{array} \right), \quad (17)$$

where  $\mathbf{a}$  denotes the  $n \times 1$  column matrix

$$\mathbf{a} = \begin{pmatrix} a_{1, n+1} \\ \vdots \\ a_{n, n+1} \end{pmatrix}.$$

Let  $d = a_{n+1, n+1} - \mathbf{a}^T (A_n)^{-1} \mathbf{a}$ , let  $I_n$  denote the  $n \times n$  identity matrix, and let  $\mathbf{0}_n$  denote the  $n \times 1$  column matrix of all 0s. Then, the matrix  $A$  in (17) can be written as

$$\begin{aligned} A &= \left( \begin{array}{c|c} I_n & \mathbf{0}_n \\ \hline (\mathbf{A}_n^{-1} \mathbf{a})^T & 1 \end{array} \right) \left( \begin{array}{c|c} A_n & \mathbf{0}_n \\ \hline \mathbf{0}_n^T & d \end{array} \right) \left( \begin{array}{c|c} I_n & \mathbf{A}_n^{-1} \mathbf{a} \\ \hline \mathbf{0}_n^T & 1 \end{array} \right) \\ &\equiv Q^T B Q. \end{aligned} \quad (18)$$

(Exercise.) By properties of the determinant,

$$\det Q = \det Q^T = 1 \quad \text{and} \quad \det B = d \cdot \det A_n.$$

$$\text{Therefore,} \quad \det A = d \cdot \det A_n. \quad (19)$$

Since  $\det A > 0$  and  $\det A_n > 0$ , then  $d > 0$ .

Let  $\mathbf{X}$  be an arbitrary  $(n+1)$ -vector. Write  $\mathbf{X}$  as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x} \\ x_{n+1} \end{pmatrix},$$

where  $\mathbf{x}$  is an  $n$ -vector. Then,

$$\begin{aligned}\mathbf{X}^T B \mathbf{X} &= (\mathbf{x}^T \quad x_{n+1}) \left( \frac{A_n \quad | \quad \mathbf{0}_n}{\mathbf{0}_n^T \quad | \quad d} \right) \begin{pmatrix} \mathbf{x} \\ x_{n+1} \end{pmatrix} \\ &= \mathbf{x}^T A_n \mathbf{x} + dx_{n+1}^2.\end{aligned}\tag{20}$$

Since  $A_n$  is positive definite by inductive hypothesis and  $d > 0$ , this last expression is strictly positive. Therefore,  $B = Q^T A Q$  is positive definite. By Lemma 16.2,  $A$  is positive definite.

To prove the converse —  $A$  positive definite implies that all the  $|A_j|$ 's are positive — we use induction once more. We have seen that this result is true for  $1 \times 1$  and  $2 \times 2$  matrices. Assume that it is true for  $n \times n$  symmetric matrices, and let  $A$  be an  $(n+1) \times (n+1)$  positive definite symmetric matrix. We first show that all the  $A_j$ 's are positive definite. Let  $\mathbf{x}_j$  be a nonzero  $j$ -vector, and let  $\mathbf{0}^*$  be the zero  $(n+1) - j$  vector. Then

$$0 < (\mathbf{x}_j^T \quad \mathbf{0}^*) A \begin{pmatrix} \mathbf{x}_j \\ \mathbf{0}^* \end{pmatrix} = \mathbf{x}_j^T A_j \mathbf{x}_j;$$

and  $A_j$  is positive definite.

In particular, since  $A_n$  is positive definite, the inductive hypothesis tells us that  $A_1, \dots, A_n$  all have positive determinants. We need only prove that the determinant of  $A$  itself is positive. Since  $A_n$  is invertible, we can once again write  $A$  as  $Q^T B Q$  as in (18) and conclude that (19) still holds. Since  $A$  is positive definite,  $B$  is positive definite by Lemma 16.2. Choose  $\mathbf{X}$  in (20) so that  $\mathbf{x} = \mathbf{0}$  and  $x_{n+1} = 1$ . Then,

$$0 < \mathbf{X}^T B \mathbf{X} = d.$$

Since  $\det A_n > 0$  and  $d > 0$ ,  $\det A > 0$ . ■

## EXERCISES

- 16.10** Show that the block decomposition (18) is correct.  
**16.11** Prove the corresponding theorem for negative definite matrices.

# Unconstrained Optimization

Since optimization plays such a major role in economic theory, this chapter on unconstrained optimization and the next three chapters on constrained optimization can be considered the core of this book. This chapter turns from the matrix criteria that specify the conditions for optimizing a quadratic form to the first and second order derivative conditions that characterize the optima of a general differentiable function. Just as techniques of calculus play a major role in optimization problems for functions of one variable, they play an equally important role for functions of several variables. The main results for multivariable functions are analogous to the one-dimensional results:

- (1) a necessary condition for  $\mathbf{x}_0$  to be an interior max of  $z = F(\mathbf{x})$  is that the first derivatives of  $F$  at  $\mathbf{x}_0$  be zero, and
- (2) if we include an appropriate condition on the second derivatives of  $F$ , this necessary condition becomes a sufficient condition.

## 17.1 DEFINITIONS

The definitions of a maximum and minimum for a function of several variables are the same as the corresponding definitions for a function of one variable. Let  $F: U \rightarrow \mathbf{R}^1$  be a real-valued function of  $n$  variables, whose domain  $U$  is a subset of  $\mathbf{R}^n$ .

- (1) A point  $\mathbf{x}^* \in U$  is a **max** of  $F$  on  $U$  if  $F(\mathbf{x}^*) \geq F(\mathbf{x})$  for all  $\mathbf{x} \in U$ .
- (2)  $\mathbf{x}^* \in U$  is a **strict max** if  $\mathbf{x}^*$  is a max and  $F(\mathbf{x}^*) > F(\mathbf{x})$  for all  $\mathbf{x} \neq \mathbf{x}^*$  in  $U$ .
- (3)  $\mathbf{x}^* \in U$  is a **local (or relative) max** of  $F$  if there is a ball  $B_r(\mathbf{x}^*)$  about  $\mathbf{x}^*$  such that  $F(\mathbf{x}^*) \geq F(\mathbf{x})$  for all  $\mathbf{x} \in B_r(\mathbf{x}^*) \cap U$ .
- (4)  $\mathbf{x}^* \in U$  is a **strict local max** of  $F$  if there is a ball  $B_r(\mathbf{x}^*)$  about  $\mathbf{x}^*$  such that  $F(\mathbf{x}^*) > F(\mathbf{x})$  for all  $\mathbf{x} \neq \mathbf{x}^*$  in  $B_r(\mathbf{x}^*) \cap U$ .

In other words, a point  $\mathbf{x}^*$  is a local max if there are no *nearby* points at which  $F$  takes on a larger value. Of course, a max is always a local max. If we want to

emphasize that a point  $\mathbf{x}^*$  is a max of  $F$  on the whole domain  $U$ , not just a local max, we call  $\mathbf{x}^*$  a **global max** or **absolute max** of  $F$  on  $U$ .

To be precise, we should say, for example in (1), that  $\mathbf{x}^*$  is a **maximizer** or **maximum point** of  $F$ , or that  $F$  has its **maximum value** at  $\mathbf{x}^*$ . The word “max” is a convenient shortcut.

Reversing the inequalities in the above four definitions leads to the definitions of a global min, a strict global min, a local min, and a strict local min, respectively.

## 17.2 FIRST ORDER CONDITIONS

The first order condition for a point  $\mathbf{x}^*$  to be a max or min of a function  $f$  of one variable is that  $f'(\mathbf{x}^*) = 0$ , that is, that  $\mathbf{x}^*$  be a **critical point** of  $f$ . This condition requires that  $\mathbf{x}^*$  not be an endpoint of the interval under consideration, in other words, that  $\mathbf{x}^*$  lie in the interior of the domain of  $f$ . The same first order condition works for a function  $F$  of  $n$  variables. However, a function of  $n$  variables has  $n$  first derivatives: the partials  $\partial F / \partial x_i$ . The  $n$ -dimensional analogue of  $f'(\mathbf{x}^*) = 0$  is that each  $\partial F / \partial x_i = 0$  at  $\mathbf{x}^*$ . In this case,  $\mathbf{x}^*$  is an **interior point** of the domain of  $F$  if there is a whole ball  $B_r(\mathbf{x}^*)$  about  $\mathbf{x}^*$  in the domain of  $F$ .

**Theorem 17.1** Let  $F: U \rightarrow \mathbf{R}^1$  be a  $C^1$  function defined on a subset  $U$  of  $\mathbf{R}^n$ . If  $\mathbf{x}^*$  is a local max or min of  $F$  in  $U$  and if  $\mathbf{x}^*$  is an interior point of  $U$ , then

$$\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0 \quad \text{for } i = 1, \dots, n. \quad (1)$$

*Proof* We will work with the local max case; the same proof works for the min case. Let  $B = B_r(\mathbf{x}^*)$  be a ball about  $\mathbf{x}^*$  in  $U$  with the property that  $F(\mathbf{x}^*) \geq F(\mathbf{x})$  for all  $\mathbf{x} \in B$ . Since  $\mathbf{x}^*$  maximizes  $F$  on  $B$ , along each *line segment* through  $\mathbf{x}^*$  that lies in  $B$  and that is parallel to one of the axes,  $F$  takes on its maximum value at  $\mathbf{x}^*$ . In other words,  $x_i^*$  maximizes the function of *one* variable:

$$x_i \mapsto F(x_1^*, \dots, x_{i-1}^*, x_i, x_{i+1}^*, \dots, x_n^*)$$

for  $x_i \in (x_i^* - r, x_i^* + r)$ . Apply the one-variable maximization criterion of Theorem 3.3 to each of these  $n$  one-dimensional problems to conclude that

$$\frac{\partial F}{\partial x_1}(\mathbf{x}^*) = 0, \dots, \frac{\partial F}{\partial x_n}(\mathbf{x}^*) = 0. \quad \blacksquare$$

**Example 17.1** To find the local maxs and mins of  $F(x, y) = x^3 - y^3 + 9xy$ , one computes the first order partial derivatives and sets them equal to zero:

$$\frac{\partial F}{\partial x} = 3x^2 + 9y = 0 \quad \text{and} \quad \frac{\partial F}{\partial y} = -3y^2 + 9x = 0. \quad (2)$$



The first equation yields:  $y = -\frac{1}{3}x^2$ . Substitute this into the second equation:

$$0 = -3y^2 + 9x = -3\left(-\frac{1}{3}x^2\right)^2 + 9x = -\frac{1}{3}x^4 + 9x.$$

This equation can be written as:  $27x - x^4 = x(27 - x^3) = 0$ , whose solutions are  $x = 0$  and  $x = 3$ . Substitute these solutions into  $y = -\frac{1}{3}x^2$  to conclude that the solutions to (2) are the two points  $(0, 0)$  and  $(3, -3)$ . At this stage, we can conclude that the only candidates for a local max or min of  $F$  are these two points:  $(0, 0)$  and  $(3, -3)$ . We are unable to say whether either of these two is a max or a min.

## 17.3 SECOND ORDER CONDITIONS

**Definition** As we did for functions of one variable, we say that the  $n$ -vector  $\mathbf{x}^*$  is a **critical point** of a function  $F(x_1, \dots, x_n)$  if  $\mathbf{x}^*$  satisfies

$$\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0 \quad \text{for } i = 1, \dots, n. \quad (3)$$

The critical points of  $F(x, y) = x^3 - y^3 + 9xy$  in Example 17.1 are  $(0, 0)$  and  $(3, -3)$ . To determine whether either of these critical points is a max or a min, we need to use a condition on the second derivatives of  $F$ , as we did for functions of one variable. As noted in Chapter 14, a  $C^2$  function of  $n$  variables has  $n^2$  second order partial derivatives at each point in its domain and it is natural to combine them into an  $n \times n$  matrix, called the **Hessian** of  $F$ :

$$D^2F(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial^2 F}{\partial x_1^2}(\mathbf{x}^*) & \cdots & \frac{\partial^2 F}{\partial x_n \partial x_1}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 F}{\partial x_1 \partial x_n}(\mathbf{x}^*) & \cdots & \frac{\partial^2 F}{\partial x_n^2}(\mathbf{x}^*) \end{pmatrix}. \quad (4)$$

Since cross-partials are equal for a  $C^2$  function (Theorem 14.5),  $D^2F(\mathbf{x}^*)$  is a symmetric matrix.

### Sufficient Conditions

The second order condition for a critical point  $\mathbf{x}^*$  of a function  $f$  on  $\mathbf{R}^1$  to be a max is that the second derivative  $f''(x^*)$  be negative. The corresponding condition for a function  $F$  of  $n$  variables is that the second derivative  $D^2F(\mathbf{x}^*)$  be *negative definite* as a symmetric matrix at the critical point  $\mathbf{x}^*$ . Similarly, the second order

sufficient condition for a critical point of a function  $f$  of one variable to be a local min is that  $f''(x^*)$  be positive; the analogous second order condition for an  $n$ -dimensional critical point  $\mathbf{x}^*$  to be a local min is that the Hessian of  $F$  at  $\mathbf{x}^*$ ,  $D^2F(\mathbf{x}^*)$ , be *positive definite*.

**Theorem 17.2** Let  $F: U \rightarrow \mathbf{R}^1$  be a  $C^2$  function whose domain is an open set  $U$  in  $\mathbf{R}^n$ . Suppose that  $\mathbf{x}^*$  is a critical point of  $F$  in that it satisfies (3).

- (1) If the Hessian  $D^2F(\mathbf{x}^*)$  is a negative definite symmetric matrix, then  $\mathbf{x}^*$  is a strict local max of  $F$ ;
- (2) If the Hessian  $D^2F(\mathbf{x}^*)$  is a positive definite symmetric matrix, then  $\mathbf{x}^*$  is a strict local min of  $F$ .
- (3) If  $D^2F(\mathbf{x}^*)$  is indefinite, then  $\mathbf{x}^*$  is neither a local max nor a local min of  $F$ .

**Definition** A critical point  $\mathbf{x}^*$  of  $F$  for which the Hessian  $D^2F(\mathbf{x}^*)$  is indefinite is called a **saddle point** of  $F$ . A saddle point  $\mathbf{x}^*$  is a min of  $F$  in some directions and a max of  $F$  in other directions. Therefore, its graph, like that of  $F(x_1, x_2) = x_1^2 - x_2^2$  in Figure 16.4, is saddle-shaped.

We will present a careful proof of Theorem 17.2 in Chapter 30 after we develop the theory of Taylor polynomial approximation. The idea behind the proof is a simple one. Let  $\mathbf{x}^*$  be a critical point of  $F$ , as in the statement of Theorem 17.2. The presentation in Chapter 30 shows how to approximate a  $C^2$  function by its Taylor polynomial of order two about  $\mathbf{x}^*$ :

$$F(\mathbf{x}^* + \mathbf{h}) = F(\mathbf{x}^*) + DF(\mathbf{x}^*)\mathbf{h} + \frac{1}{2}\mathbf{h}^T D^2F(\mathbf{x}^*)\mathbf{h} + R(\mathbf{h}), \quad (5)$$

where  $DF(\mathbf{x}^*)$  is the Jacobian matrix of  $F$  at  $\mathbf{x}^*$ ,  $D^2F(\mathbf{x}^*)$  is the Hessian matrix (4) of second order partial derivatives of  $F$  at  $\mathbf{x}^*$ , and  $R(\mathbf{h})$  is the remainder term that goes to 0 very quickly as  $\mathbf{h} \rightarrow \mathbf{0}$ . In this proof sketch, we will ignore the negligibly small remainder term  $R(\mathbf{h})$  and, as in the hypothesis of Theorem 17.2, set  $DF(\mathbf{x}^*)$  equal to zero. We can then write equation (5) as

$$F(\mathbf{x}^* + \mathbf{h}) - F(\mathbf{x}^*) \approx \frac{1}{2}\mathbf{h}^T D^2F(\mathbf{x}^*)\mathbf{h}. \quad (6)$$

If  $D^2F(\mathbf{x}^*)$  is negative definite, then for all (small enough)  $\mathbf{h} \neq \mathbf{0}$ , the right-hand side of (6) is negative. Then, the left hand-side is negative too:

$$F(\mathbf{x}^* + \mathbf{h}) - F(\mathbf{x}^*) < 0 \quad \text{or} \quad F(\mathbf{x}^* + \mathbf{h}) < F(\mathbf{x}^*),$$

for all small enough  $\mathbf{h} \neq \mathbf{0}$ . In other words,  $\mathbf{x}^*$  is a strict local max of  $F$ . The complete proof in Chapter 30 works out the details of the Taylor polynomial

representation (5) and shows how to handle the remainder term  $R$  in the above argument.

The presentation in Chapter 16 developed an analytic characterization of positive definite and negative definite matrices. In terms of that formulation, Theorem 17.2 can be restated as the following theorems.

**Theorem 17.3** Let  $F: U \rightarrow \mathbf{R}^1$  be a  $C^2$  function whose domain is an open set  $U$  in  $\mathbf{R}^1$ . Suppose that

$$\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0 \quad \text{for } i = 1, \dots, n$$

and that the  $n$  leading principal minors of  $D^2F(\mathbf{x}^*)$  alternate in sign

$$|F_{x_1x_1}| < 0, \quad \begin{vmatrix} F_{x_1x_1} & F_{x_2x_1} \\ F_{x_1x_2} & F_{x_2x_2} \end{vmatrix} > 0, \quad \begin{vmatrix} F_{x_1x_1} & F_{x_2x_1} & F_{x_3x_1} \\ F_{x_1x_2} & F_{x_2x_2} & F_{x_3x_2} \\ F_{x_1x_3} & F_{x_2x_3} & F_{x_3x_3} \end{vmatrix} < 0, \dots$$

at  $\mathbf{x}^*$ . Then,  $\mathbf{x}^*$  is a strict local max of  $F$ .

**Theorem 17.4** Let  $F: U \rightarrow \mathbf{R}^1$  be a  $C^2$  function whose domain is an open set  $U$  in  $\mathbf{R}^n$ . Suppose that

$$\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0 \quad \text{for } i = 1, \dots, n$$

and that the  $n$  leading principal minors of  $D^2F(\mathbf{x}^*)$  are all positive:

$$|F_{x_1x_1}| > 0, \quad \begin{vmatrix} F_{x_1x_1} & F_{x_2x_1} \\ F_{x_1x_2} & F_{x_2x_2} \end{vmatrix} > 0, \quad \begin{vmatrix} F_{x_1x_1} & F_{x_2x_1} & F_{x_3x_1} \\ F_{x_1x_2} & F_{x_2x_2} & F_{x_3x_2} \\ F_{x_1x_3} & F_{x_2x_3} & F_{x_3x_3} \end{vmatrix} > 0, \dots$$

at  $\mathbf{x}^*$ . Then,  $\mathbf{x}^*$  is a strict local min of  $F$ .

**Theorem 17.5** Let  $F: U \rightarrow \mathbf{R}^1$  be a  $C^2$  function whose domain is an open set  $U$  in  $\mathbf{R}^n$ . Suppose that

$$\frac{\partial F}{\partial x_i}(\mathbf{x}^*) = 0 \quad \text{for } i = 1, \dots, n$$

and that some nonzero leading principal minors of  $D^2F(\mathbf{x}^*)$  violate the sign patterns in the hypotheses of Theorems 17.3 and 17.4. Then,  $\mathbf{x}^*$  is a saddle point of  $F$ ; it is neither a local max nor a local min.

## Necessary Conditions

The second order *necessary* condition for a max or min of a function of one variable is weaker than the second order *sufficient* condition. The weak inequality of the necessary condition, namely, that  $f''(x^*) \leq 0$  at a local max and that  $f''(x^*) \geq 0$  at a local min, replaces the strict inequality of the sufficient condition. The functions  $f_1(x) = x^4$  and  $f_2(x) = -x^4$  provide the simplest illustrations of the difference between the necessary and sufficient conditions on  $\mathbf{R}^1$ . For both functions,  $f'_i(0) = 0$  and  $f''_i(0) = 0$ ; yet  $f_1$  has a strict global *min* at  $x = 0$  and  $f_2$  has a strict global *max* at  $x = 0$ .

A similar weakening occurs for functions of several variables. In this case, one replaces the negative definite and positive definite conditions on the Hessian of  $F$  in the sufficient conditions of Theorems 17.3 and 17.4 by requirements that the Hessian must be negative *semidefinite* at a local max and positive *semidefinite* at a local min.

**Theorem 17.6** Let  $F: U \rightarrow \mathbf{R}^1$  be a  $C^2$  function whose domain  $U$  is in  $\mathbf{R}^n$ . Suppose that  $\mathbf{x}^*$  is an interior point of  $U$  and that  $\mathbf{x}^*$  is a local max (respectively, min) of  $F$ . Then,  $DF(\mathbf{x}^*) = \mathbf{0}$  and  $D^2F(\mathbf{x}^*)$  is negative semidefinite (respectively, positive semidefinite).

The discussion in Chapter 16 showed that a symmetric function is positive semidefinite if and only if all of its  $2^n - 1$  principal minors are  $\geq 0$  and that a symmetric matrix is negative semidefinite if and only if all of its principal minors of odd order are  $\leq 0$  and all of its principal minors of even order are  $\geq 0$ . With this analytical test for semidefiniteness in mind, we rewrite Theorem 17.6 as follows.

**Theorem 17.7** Let  $F: U \rightarrow \mathbf{R}^1$  be a  $C^2$  function of  $n$  variables. Suppose that  $\mathbf{x}^*$  is an interior point of  $U$ .

- (a) If  $\mathbf{x}^*$  is a local min of  $F$ , then  $(\partial F / \partial x_i)(\mathbf{x}^*) = 0$  for  $i = 1, \dots, n$  and all the principal minors of the Hessian  $D^2F(\mathbf{x}^*)$  are  $\geq 0$ .
- (b) If  $\mathbf{x}^*$  is a local max of  $F$ , then  $(\partial F / \partial x_i)(\mathbf{x}^*) = 0$  for  $i = 1, \dots, n$  and all the principal minors of the Hessian  $D^2F(\mathbf{x}^*)$  of odd order are  $\leq 0$  and all the principal minors of the Hessian  $D^2F(\mathbf{x}^*)$  of even order are  $\geq 0$ .

The proof of Theorem 17.6 is in Chapter 30. Now, let's apply all our second order conditions to the cubic function in Example 17.1.

**Example 17.2** In Example 17.1, we computed that the critical points of  $F(x, y) = x^3 - y^3 + 9xy$  are  $(0, 0)$  and  $(3, -3)$ . By differentiating the first derivatives in expression (2), we compute that the Hessian of  $F$  is

$$\begin{pmatrix} F_{xx} & F_{xy} \\ F_{yx} & F_{yy} \end{pmatrix} = \begin{pmatrix} 6x & 9 \\ 9 & -6y \end{pmatrix}.$$

The first order leading principal minor is  $F_{xx} = 6x$  and the second order leading principal minor is  $\det D^2F(\mathbf{x}) = -36xy - 81$ . At  $(0, 0)$ , these two minors are 0 and  $-81$ , respectively. Since the second order leading principal minor is negative,  $(0, 0)$  is a saddle of  $F$  — neither a max nor a min. At  $(3, -3)$ , these two minors are 18 and 243. Since these two numbers are positive,  $D^2F(3, -3)$  is positive definite and  $(3, -3)$  is a strict local min of  $F$ .

Notice that  $(3, -3)$  is not a *global* min, because at the point  $(0, n)$ ,  $F(0, n) = -n^3$ , which goes to  $-\infty$  as  $n \rightarrow \infty$ .

## EXERCISES

- 17.1** For each of the following functions defined on  $\mathbf{R}^2$ , find the critical points and classify these as local max, local min, saddle point, or “can’t tell”:

$$\begin{array}{ll} a) x^4 + x^2 - 6xy + 3y^2, & b) x^2 - 6xy + 2y^2 + 10x + 2y - 5, \\ c) xy^2 + x^3y - xy, & d) 3x^4 + 3x^2y - y^3. \end{array}$$

- 17.2** For each of the following functions defined on  $\mathbf{R}^3$ , find the critical points and classify them as local max, local min, saddle point, or “can’t tell”:

$$\begin{array}{l} a) x^2 + 6xy + y^2 - 3yz + 4z^2 - 10x - 5y - 21z, \\ b) (x^2 + 2y^2 + 3z^2)e^{-(x^2+y^2+z^2)}. \end{array}$$

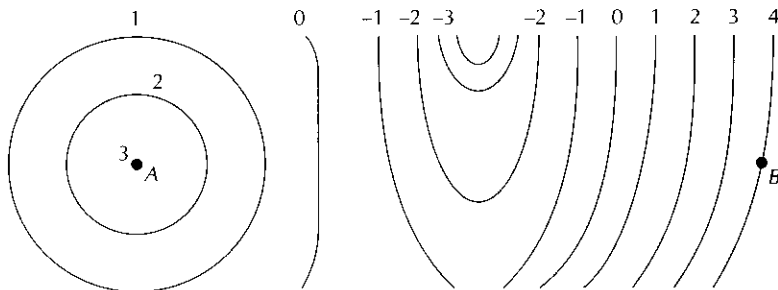
## 17.4 GLOBAL MAXIMA AND MINIMA

The first and second order sufficient conditions of the last section will find all the local maxima and minima of a differentiable function whose domain is an open set in  $\mathbf{R}^n$ . As Example 17.2 illustrates, these conditions say nothing about whether or not any of these local extrema is a *global* max or min. In this section, we will discuss sufficient conditions for global maxima and minima of a real-valued function on  $\mathbf{R}^n$ .

The study of one-dimensional optimization problems in Section 3.5 put forth two conditions for a critical point  $x^*$  of  $f$  to be a global max (or min), when  $f$  is a  $C^2$  function defined on a connected interval  $I$  of  $\mathbf{R}^1$ :

- (1)  $x^*$  is a local max (or min) and it’s the only critical point of  $f$  in  $I$ ; or
- (2)  $f'' \leq 0$  on all of  $I$  (or  $f'' \geq 0$  on  $I$  for a min), that is,  $f$  is a concave function on  $I$  (or  $f$  is a convex function for a min).

Condition 1 does not work in higher dimensions, as the function  $F$  whose level sets are pictured in Figure 17.1 illustrates. The point A in Figure 17.1 is a local max of  $F$  in the open set  $U$ . Even though A is the only critical point of  $F$  in  $U$ , the function  $F$  takes on a higher value at point B.



Level sets of a function whose unique local max is not a global max.

Figure  
17.1

## Global Maxima of Concave Functions

Condition 2 *does* work in higher dimensions. As we have seen, the  $n$ -dimensional version of  $f''(x) \leq 0$  states that the Hessian  $D^2F(\mathbf{x})$  is negative semidefinite. We will see in Chapter 20 that this condition is exactly the second derivative criterion for  $F$  to be a concave function. The following theorem summarizes the first and second order criteria for a function on  $\mathbf{R}^n$  to be a concave or convex function and relates these properties to global extrema.

**Theorem 17.8** Let  $F: U \rightarrow \mathbf{R}^1$  be a  $C^2$  function whose domain is a convex open subset  $U$  of  $\mathbf{R}^n$ .

- (a) The following three conditions are equivalent:
  - (i)  $F$  is a concave function on  $U$ ; and
  - (ii)  $F(\mathbf{y}) - F(\mathbf{x}) \leq DF(\mathbf{x})(\mathbf{y} - \mathbf{x})$  for all  $\mathbf{x}, \mathbf{y} \in U$ ; and
  - (iii)  $D^2F(\mathbf{x})$  is negative semidefinite for all  $\mathbf{x} \in U$ .
- (b) The following three conditions are equivalent:
  - (i)  $F$  is a convex function on  $U$ ; and
  - (ii)  $F(\mathbf{y}) - F(\mathbf{x}) \geq DF(\mathbf{x})(\mathbf{y} - \mathbf{x})$  for all  $\mathbf{x}, \mathbf{y} \in U$ ; and
  - (iii)  $D^2F(\mathbf{x})$  is positive semidefinite for all  $\mathbf{x} \in U$ .
- (c) If  $F$  is a concave function on  $U$  and  $DF(\mathbf{x}^*) = \mathbf{0}$  for some  $\mathbf{x}^* \in U$ , then  $\mathbf{x}^*$  is a *global max* of  $F$  on  $U$ .
- (d) If  $F$  is a convex function on  $U$  and  $DF(\mathbf{x}^*) = \mathbf{0}$  for some  $\mathbf{x}^* \in U$ , then  $\mathbf{x}^*$  is a *global min* of  $F$  on  $U$ .

We will present a careful proof of these results in Chapter 20, where we give a complete introduction to the study of smooth concave and convex functions. To get a flavor of that proof, we sketch the proof that  $ai \implies aii \implies c$  for functions of one variable here.

**Proof Sketch on  $\mathbf{R}^1$ :** By definition, a function  $F$  is concave on  $U$  if and only if

$$tF(y) + (1 - t)F(x) \leq F(ty + (1 - t)x) \quad (7)$$

for all  $x, y \in U$  and all  $t \in [0, 1]$ . Write (7) as

$$F(y) - F(x) \leq \frac{F(x + t(y - x)) - F(x)}{t},$$

or, letting  $h = t(y - x)$ , as

$$F(y) - F(x) \leq \frac{F(x + h) - F(x)}{h} (y - x). \quad (8)$$

Taking limits as  $h \rightarrow 0$ , (8) becomes

$$F(y) - F(x) \leq F'(x)(y - x). \quad (9)$$

This shows that *ai* implies *aii*.

To prove *c*, suppose that  $F$  is concave and that  $F'(x^*) = 0$ . Then, by inequality (9) above, for all  $y \in U$ ,

$$F(y) - F(x^*) \leq 0 \cdot (y - x^*) = 0;$$

that is,  $F(y) \leq F(x^*)$  for all  $y \in U$ . Therefore,  $x^*$  is a global max of  $F$  in  $U$ . ■

There is an interesting comparison between Theorem 17.8 and Theorem 17.2. In order to guarantee that a critical point  $\mathbf{x}^*$  of a  $C^2$  function  $F$  is a local max, we need to show that  $D^2F(\mathbf{x}^*)$  is negative definite; showing  $D^2F(\mathbf{y})$  negative semidefinite is not strong enough. However, if we can show that  $D^2F(\mathbf{y})$  is negative semidefinite not just at  $\mathbf{x}^*$  but for all  $\mathbf{y}$  in a whole ball about  $\mathbf{x}^*$ , then by Theorem 17.8, we can conclude that  $\mathbf{x}^*$  is a max of  $F$ . For example, evaluating  $f_1''(0)$  cannot distinguish the critical point 0 as a min of  $f_1(x) = x^4$  from 0 as a max of  $f_2(x) = -x^4$ . However, since  $f_2''(x) = -12x^2 \leq 0$  for all  $x \in \mathbf{R}^1$ , we know by Theorem 17.8 that 0 is a max of  $f_2$  on  $\mathbf{R}^1$ .

## EXERCISES

- 17.3 Which of the maxima and minima in Exercise 17.1 are global maxima or global minima?

## 17.5 ECONOMIC APPLICATIONS

Most of the principles of economic theory are derived from the first order conditions — and sometimes the second order conditions — of an underlying optimization problem. We have seen some examples of this phenomenon in our discussion of the economic applications of functions of *one* variable in Section 3.6.

## Profit-Maximizing Firm

The simplest multivariable example occurs in the study of the profit-maximizing firm. Suppose a firm uses  $n$  inputs to produce a single product. If  $\mathbf{x} \in \mathbf{R}_+^n$  represents an input bundle, if  $y = G(\mathbf{x})$  is the firm's  $C^1$  production function, and if  $p$  is the selling price of its product, then the firm's revenue  $R(\mathbf{x}) = pG(\mathbf{x})$ . If  $C(\mathbf{x})$  represents the cost of input bundle  $\mathbf{x}$ , the firm's profit  $F(\mathbf{x})$  from optimal use of input bundle  $\mathbf{x}$  is

$$F(\mathbf{x}) = R(\mathbf{x}) - C(\mathbf{x}).$$

Assume that  $R$  and  $C$  are such that the profit-maximizing firm uses a positive amount of each input so that the profit-maximizing  $\mathbf{x}$  occurs in the interior of the positive orthant  $\mathbf{R}_+^n$ . Then, by Theorem 17.1, the partial derivatives of  $F$  must be zero at the profit-maximizing  $\mathbf{x}^*$ :

$$0 = \frac{\partial F}{\partial x_i}(\mathbf{x}^*) = \frac{\partial R}{\partial x_i}(\mathbf{x}^*) - \frac{\partial C}{\partial x_i}(\mathbf{x}^*). \quad (10)$$

In particular, the *marginal revenue product* from using one more unit of input  $i$  must just balance the marginal cost of purchasing another unit of input  $i$ .

Suppose that each input  $i$  has constant unit cost  $w_i$ , so that

$$C(\mathbf{x}) = w_1x_1 + \dots + w_nx_n = \mathbf{w} \cdot \mathbf{x}.$$

The first order conditions (10) now become

$$\frac{\partial R}{\partial x_i}(\mathbf{x}^*) = p \frac{\partial G}{\partial x_i}(\mathbf{x}^*) = w_i \quad \text{or} \quad \frac{\partial G}{\partial x_i}(\mathbf{x}^*) = \frac{w_i}{p}.$$

for  $i = 1, \dots, n$ . The second order necessary condition in Theorem 17.7 requires that  $D^2F(\mathbf{x}^*)$  be negative semidefinite. In the case of constant marginal cost, this second order condition implies that the Hessian of the production function  $D^2G(\mathbf{x}^*)$  must be negative semidefinite at the optimal input bundle. In particular, this implies that each  $\partial^2 G / \partial x_i^2$  must be  $\leq 0$  at  $\mathbf{x}^*$ . Another way of saying this is that the profit-maximizing input vectors occur only in those regions of the input space where all the  $\partial^2 G / \partial x_i^2$ 's are  $\leq 0$ . If for each  $\mathbf{x} \in \mathbf{R}_+^n$ , there is an index  $i$  such that  $(\partial^2 G / \partial x_i^2)(\mathbf{x}) > 0$ , then the firm under study cannot have a profit-maximizing output in the interior of  $\mathbf{R}_+^n$ .

## Discriminating Monopolist

The model of a discriminating monopolist is another example of a maximization problem to which the results of this chapter apply. Suppose that a monopolist faces two distinct and separated markets, for example, a domestic market and a foreign



market, each with its own demand function. Let  $Q_i$  be the amount supplied to market  $i$  and let  $P_i = G_i(Q_i)$  be the inverse demand function for market  $i$ , so that the revenue from supplying  $Q_i$  units of output to market  $i$  is  $Q_i G_i(Q_i)$ . Suppose that production costs  $C$  depend only on the total produced  $Q = Q_1 + Q_2$ ; that is, the firm's cost function can be written as  $C(Q_1 + Q_2)$ . Its profit is then

$$F(Q_1, Q_2) = Q_1 \cdot G_1(Q_1) + Q_2 \cdot G_2(Q_2) - C(Q_1 + Q_2).$$

If we know that the firm will produce a positive amount for each market, then our problem is to compute the maxima of the profit function  $F$  in the interior of the positive quadrant. These maximizers satisfy

$$0 = \frac{\partial F}{\partial Q_1} = \frac{d(Q_1 G_1(Q_1))}{dQ_1} - C'(Q_1 + Q_2)$$

$$0 = \frac{\partial F}{\partial Q_2} = \frac{d(Q_2 G_2(Q_2))}{dQ_2} - C'(Q_1 + Q_2)$$

or 
$$\frac{d(Q_1 G_1(Q_1))}{dQ_1} = \frac{d(Q_2 G_2(Q_2))}{dQ_2} = C'(Q_1 + Q_2).$$

The marginal revenue in *each* market should just equal the marginal cost of the total output.

**Example 17.3** A monopolist producing a single output has two types of customers. If it produces  $Q_1$  units for customers of type 1, then these customers are willing to pay a price of  $50 - 5Q_1$  dollars per unit. If it produces  $Q_2$  units for customers of type 2, then these customers are willing to pay a price of  $100 - 10Q_2$  dollars per unit. The monopolist's cost of manufacturing  $Q$  units of output is  $90 + 20Q$  dollars. In order to maximize profits, how much should the monopolist produce for each market?

This discriminating monopolist's profit function is

$$F(Q_1, Q_2) = Q_1(50 - 5Q_1) + Q_2(100 - 10Q_2) - (90 + 20(Q_1 + Q_2)).$$

The critical points of  $F$  satisfy

$$\frac{\partial F}{\partial Q_1} = 50 - 10Q_1 - 20 = 0, \quad \text{or} \quad Q_1 = 3,$$

$$\frac{\partial F}{\partial Q_2} = 100 - 20Q_2 - 20 = 0, \quad \text{or} \quad Q_2 = 4.$$

Now check the second order conditions. Since

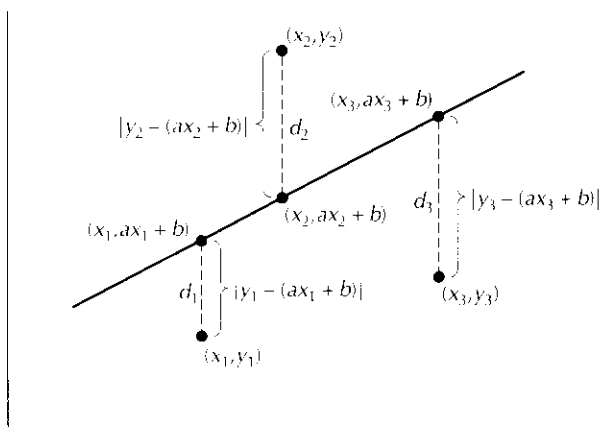
$$F_{Q_1 Q_1} = -10, \quad F_{Q_2 Q_2} = -20, \quad F_{Q_1 Q_2} = F_{Q_2 Q_1} = 0,$$

the first order leading principal minor of  $D^2F(3, 4)$  is  $-10$  and the second order leading principal minor is  $200$ . Therefore,  $F$  is a concave function and the point  $(3, 4)$  is the profit-maximizing supply plan over all plans in the positive orthant.

## Least Squares Analysis

Scientists studying the data from some observations or experiments are often interested in discovering whether the variables under study may be linearly related, or at least in finding the linear graph which best fits the data points. Suppose that we are studying the relationship between two variables, so that each observation can be represented by a point  $(x_i, y_i)$  in the plane. If we have only two points, then by the method of Section 2.2, we can find the equation of the unique straight line that passes through these two points. However, if there are three or more data points, it is unlikely that they will line up on a single line. In this case, we'd like to find the line that best fits these data points, both to shed light on any underlying patterns and possibly to predict other points.

Suppose that the data points are:  $(x_1, y_1), \dots, (x_n, y_n)$ ,  $n \geq 3$ , as pictured in Figure 17.2. For any given line  $y = mx + b$ , we can measure the vertical distance from each of these  $n$  points to the line. These distances are represented by the lengths of the dotted vertical line segments  $d_1, \dots, d_n$  in Figure 17.2. The  $x$ -coordinate of the point on the line corresponding to  $(x_i, y_i)$  is  $x_i$  itself, since we are taking vertical distances. Since the equation of the line is  $y = mx + b$ , the corresponding  $y$ -value of this point on the line is  $y_i = mx_i + b$ , as we have noted in Figure 17.2. The vertical distance from the data point  $(x_i, y_i)$  to the point on the line  $(x_i, mx_i + b)$  is  $|mx_i + b - y_i|$ . Since we are working with calculus and the absolute value function is not differentiable, we replace  $|mx_i + b - y_i|$  by  $(mx_i + b - y_i)^2$ . We use the square since we want the distance to be a positive number; otherwise, a point which is, say, 100 units above the line and a point which is 100 units below



*The aggregate distance from three data points to a line.*

**Figure**  
**17.2**

the line will cancel each other out in this process. In summary, if we are given  $n$  data points in the plane:  $(x_1, y_1), \dots, (x_n, y_n)$ , and a straight line whose equation is  $y = mx + b$ , then we can write the *aggregate distance* from the  $n$  points to the line as

$$S = (mx_1 + b - y_1)^2 + (mx_2 + b - y_2)^2 + \cdots + (mx_n + b - y_n)^2. \quad (11)$$

Consider calculating  $S$  for a number of different lines, so that  $S$  becomes a function of  $m$  and  $b$ :

$$S(m, b) = \sum_{i=1}^n (mx_i + b - y_i)^2.$$

The line which best fits the  $n$  data points should be the line whose parameters  $m^*$  and  $b^*$  minimize the total discrepancy  $S$ . Since  $m$  and  $b$  can be any real numbers, to find the minima of  $S$ , we first find its critical points by solving

$$\begin{aligned} \frac{\partial S}{\partial m} &= \sum_{i=1}^n 2(mx_i + b - y_i) \cdot x_i = 0, \\ \frac{\partial S}{\partial b} &= \sum_{i=1}^n 2(mx_i + b - y_i) \cdot 1 = 0, \end{aligned} \quad (12)$$

using the Chain Rule on each summand in (11). [Remember: the  $x_i$ 's and  $y_i$ 's are knowns here; the  $m$ 's and the  $b$ 's are the variables.] Divide the equations in (12) through by 2 and redistribute the summation signs to obtain the system of two equations in two unknowns:

$$\begin{aligned} \sum_i mx_i^2 + \sum_i bx_i - \sum_i x_i y_i &= 0 \\ \sum_i mx_i + \sum_i b - \sum_i y_i &= 0, \end{aligned}$$

or

$$\begin{aligned} \left( \sum_i x_i^2 \right) m + \left( \sum_i x_i \right) b &= \sum_i x_i y_i \\ \left( \sum_i x_i \right) m + n \cdot b &= \sum_i y_i \end{aligned} \quad (13)$$

We can solve (13) using Cramer's rule:

$$m^* = \frac{\begin{vmatrix} \sum_i x_i y_i & \sum_i x_i \\ \sum_i y_i & n \end{vmatrix}}{\begin{vmatrix} \sum_i x_i^2 & \sum_i x_i \\ \sum_i x_i & n \end{vmatrix}} = \frac{n \sum_i x_i y_i - (\sum_i x_i)(\sum_i y_i)}{n \sum_i x_i^2 - (\sum_i x_i)^2} \quad (14)$$

$$b^* = \frac{\begin{vmatrix} \sum x_i^2 & \sum x_i y_i \\ \sum x_i & \sum y_i \end{vmatrix}}{\begin{vmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{vmatrix}} = \frac{(\sum x_i^2)(\sum y_i) - (\sum x_i)(\sum x_i y_i)}{n \sum x_i^2 - (\sum x_i)^2}. \quad (15)$$

**Example 17.4** To find the straight line which best fits the points (0, 4), (3, 3), (4, 2), (3, 1), and (5, 0), construct a table:

| $i$      | $x_i$ | $y_i$ | $x_i y_i$ | $x_i^2$ |
|----------|-------|-------|-----------|---------|
| 1        | 0     | 4     | 0         | 0       |
| 2        | 3     | 3     | 9         | 9       |
| 3        | 4     | 2     | 8         | 16      |
| 4        | 3     | 1     | 3         | 9       |
| 5        | 5     | 0     | 0         | 25      |
| $\Sigma$ | 15    | 10    | 20        | 59      |

Then, by (14) and (15),

$$m^* = \frac{5 \cdot 20 - 15 \cdot 10}{5 \cdot 59 - 15 \cdot 15} = -\frac{5}{7}$$

$$b^* = \frac{59 \cdot 10 - 15 \cdot 20}{5 \cdot 59 - 15 \cdot 15} = \frac{29}{7}.$$

The line which best fits these five points has the equation

$$y = -\frac{5}{7}x + \frac{29}{7} \quad \text{or} \quad 5x + 7y = 29.$$

**Vocabulary** The technique we have been discussing is called the **method of least squares** or **regression analysis**. In this case, the line  $y = m^*x + b^*$  which best fits the data points is called the **regression line**.

## EXERCISES

- 17.4** A firm uses two inputs to produce a single product. If its production function is  $Q = x^{1/4}y^{1/4}$  and if it sells its output for a dollar a unit and buys each input for \$4 dollars a unit, find its profit-maximizing input bundle. (Check the second order conditions.)
- 17.5** More generally, suppose that a firm has a Cobb-Douglas production function  $Q = x^a y^b$  and that it faces output price  $p$  and input prices  $w$  and  $r$ , respectively. Solve the first order conditions for a profit-maximizing input bundle. Use the second order conditions to determine the values of the parameters  $a$ ,  $b$ ,  $p$ ,  $w$ , and  $r$  for which this solution is a global max.

- 17.6** Dingbat Airlines has regular flights between Ypsilanti and Kalamazoo. It can treat business and pleasure travelers as separate markets by demanding advance purchase and Saturday night stay-over for pleasure travelers. Suppose that it notes a demand function of  $Q = 16 - p$  for business travelers and a demand function  $Q = 10 - p$  for pleasure travelers and that it has a cost function for all travelers of  $C(Q) = 10 + Q^2$ . How much should it charge in each market to maximize its profit?
- 17.7** For the discriminating monopolist in Example 17.3, compute the demand function for the market as a whole, without price discrimination. Compute the firm's profit-maximizing output for this situation and compare the profit to the computation in Example 17.3.
- 17.8** Find the equation of the line which best fits the data points: (1, 2), (3, 4), (5, 3), and (6, 6).
- 17.9** a) Prove that  $2ab \leq a^2 + b^2$  for all numbers  $a, b$ .  
b) Use this result to show that

$$\begin{aligned}(x_1 + \cdots + x_n)^2 &= x_1^2 + \cdots + x_n^2 + \sum_{i < j} 2x_i x_j \\ &\leq x_1^2 + \cdots + x_n^2 + (n-1)(x_1^2 + \cdots + x_n^2) \\ &= n(x_1^2 + \cdots + x_n^2).\end{aligned}$$

- c) Conclude that the point  $(m^*, b^*)$  in (14) and (15) is a global minimizer of the function  $S$  in (11).
- 17.10** Use parts *a* and *b* of the previous exercise to determine the geometric interpretation of a zero determinant in the numerators of (14) and (15).
- 17.11** Given a collection of points  $\{(x_i, y_i, z_i)\}_{i=1}^n$  in  $\mathbf{R}^3$ , find an equation for the coefficients  $A, B$ , and  $C$  of the plane  $z = Ax + By + C$  whose graph best fits these points.
- 17.12** Divide the equations (14) and (15) through by  $n^3$  to express the least-squares coefficients  $m^*$  and  $b^*$  in terms of *average values* of the  $x_i$ 's,  $y_i$ 's, and so on.
-

# Constrained Optimization I: First Order Conditions

Economics is often defined as the study of the optimal allocation of scarce resources. The word “optimal” implies that we are dealing with some sort of an optimization problem. The word “scarce” implies that the objects in this optimization problem are not free to take on any value but are constrained. A household’s consumption is constrained by its available income; a firm’s production is constrained by the cost and availability of its inputs. Thus, constrained optimization problems lie at the heart of economic theory. The central mathematical problem here is that of maximizing (or in the case of cost, expenditure, or risk, minimizing) a function of several variables, where these variables are bound by some constraining equations. The prototype problem is

$$\text{maximize } f(x_1, \dots, x_n)$$

where  $(x_1, \dots, x_n) \in \mathbf{R}^n$  must satisfy

$$\begin{aligned} g_1(x_1, \dots, x_n) &\leq b_1, \dots, g_k(x_1, \dots, x_n) \leq b_k, \\ h_1(x_1, \dots, x_n) &= c_1, \dots, h_m(x_1, \dots, x_n) = c_m. \end{aligned} \tag{1}$$

The function  $f$  is called the **objective function**, while  $g_1, \dots, g_k$  and  $h_1, \dots, h_m$  are called **constraint functions**. The  $g_j$ ’s define **inequality constraints**, and the  $h_i$ ’s define **equality constraints**. In applications, the most common inequality constraints are **nonnegativity constraints**:  $x_1 \geq 0, \dots, x_n \geq 0$ . Equality constraints often arise as definitions of one variable in terms of others. This chapter begins the treatment of constrained maximization problems. We bring all the mathematical background of the previous chapters to bear on this central topic of economic theory.

## 18.1 EXAMPLES

We begin by discussing some important economic examples of constrained optimization problems.

**Example 18.1 (Utility Maximization Problem)** In this most basic problem,  $x_i$  represents the amount of commodity  $i$  and  $f(x_1, \dots, x_n)$ , usually written as  $U(x_1, \dots, x_n)$ , measures the individual's level of utility or satisfaction with consuming  $x_1$  units of good 1,  $x_2$  units of good 2, and so on. Let  $p_1, \dots, p_n$  denote the prices of the commodities and let  $I$  denote the individual's income. The consumer wants to

$$\begin{aligned} &\text{maximize} && U(x_1, \dots, x_n) \\ &\text{subject to} && p_1x_1 + p_2x_2 + \cdots + p_nx_n \leq I, \\ &&& x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0. \end{aligned}$$

To be consistent with the general format in (1), the nonnegativity constraints  $x_i \geq 0$  should be written as  $-x_i \leq 0$  so that all inequality constraints are written with  $\leq$  signs. However, when we first formulate a problem mathematically, we will write the constraints in their most natural form. Later, when we begin working with the constrained problem, we will worry about the proper mathematical conventions.

**Example 18.2 (Utility Maximization with Labor/Leisure Choice)** Let  $U, x_1, \dots, x_n, p_1, \dots, p_n$  be as in the preceding example. In addition, let  $w$  denote the wage rate,  $I'$  the consumer's nonwage income,  $\ell_0$  hours of labor, and  $\ell_1$  hours of leisure. The consumer has  $I' + w\ell_0$  dollars to spend and wants to

$$\begin{aligned} &\text{maximize} && U(x_1, \dots, x_n, \ell_1) \\ &\text{subject to} && p_1x_1 + \cdots + p_nx_n \leq I' + w\ell_0, \\ &&& \ell_0 + \ell_1 = 24, \\ &&& x_1 \geq 0, \dots, x_n \geq 0, \ell_0 \geq 0, \ell_1 \geq 0. \end{aligned}$$

**Example 18.3 (Profit Maximization of a Competitive Firm)** Suppose that a firm in a competitive industry uses  $n$  inputs to manufacture its product. Let  $y$  denote the amount of its output, and let  $x_1, \dots, x_n$  denote the amounts of its inputs — all flow concepts. Let  $y = f(x_1, \dots, x_n)$  denote the firm's production function, describing the maximal amount of output that can be produced from the input bundle  $(x_1, \dots, x_n)$ . Let  $p$  be the unit price of the output and let  $w_i$  denote the cost of input  $i$ . The firm's goal is to choose  $(x_1, \dots, x_n)$  to maximize its **profit**

$$\Pi(x_1, \dots, x_n) = pf(x_1, \dots, x_n) - \sum_{i=1}^n w_i x_i$$

under the constraints

$$\begin{aligned} pf(x_1, \dots, x_n) - \sum_1^n w_i x_i &\geq 0, \\ g_1(\mathbf{x}) &\leq b_1, \dots, g_k(\mathbf{x}) \leq b_k, \\ x_1 &\geq 0, \dots, x_n \geq 0. \end{aligned}$$

The first inequality constraint reflects the requirement that the firm make a nonnegative profit. The  $g_j$ -constraints represent constraints on the availability of the inputs.

## 18.2 EQUALITY CONSTRAINTS

### Two Variables and One Equality Constraint

We begin with the simplest constrained maximization problem, that of maximizing a function  $f(x_1, x_2)$  of two variables subject to a single equality constraint:  $h(x_1, x_2) = c$ . The student of economics first encounters such a problem in the utility maximization problem in an intermediate microeconomics course:

$$\begin{aligned} &\text{maximize} && f(x_1, x_2) \\ &\text{subject to} && p_1 x_1 + p_2 x_2 = I. \end{aligned}$$

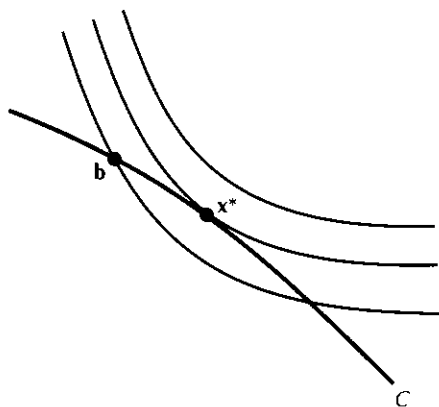
We ignore for a moment the requirement that  $x_1$  and  $x_2$  should be nonnegative and the possibility that not all income  $I$  may be spent.

With this model of consumer choice in mind, let us examine the usual geometric solution to such a problem. First, draw the constraint set  $C$  in the  $x_1 x_2$ -plane — the thick line in Figure 18.1. Then, draw a representative sample of the level curves of the objective function  $f$ . Geometrically, our goal is to find the highest-valued level curve of  $f$  which meets the constraint set. The highest level curve of  $f$  cannot cross the constraint curve  $C$ ; if it did, nearby *higher* level sets would cross too, as occurs at point **b** in Figure 18.1. This highest level set of  $f$  must touch  $C$  (so that the constraint is satisfied) but must otherwise lie on one side of  $C$  (since it cannot cross over  $C$ ). Another way of saying this is that the highest level curve of  $f$  to touch the constraint set  $C$  must be *tangent* to  $C$  at the constrained max. This situation occurs at point  $\mathbf{x}^*$  in Figure 18.1.

The fact that the level curve of  $f$  is tangent to the constraint set  $C$  at the constrained maximizer  $\mathbf{x}^*$  means that the slope of the level set of  $f$  equals the slope of the constraint curve  $C$  at  $\mathbf{x}^*$ . Recall from Section 15.2 that the slope of the level set of  $f$  at  $\mathbf{x}^*$  is

$$-\frac{\partial f}{\partial x_1}(\mathbf{x}^*) \bigg/ \frac{\partial f}{\partial x_2}(\mathbf{x}^*)$$





**Figure 18.1** *At the constrained max  $\mathbf{x}^*$ , the highest level curve of  $f$  is tangent to the constraint set  $C$ .*

and the slope of the constraint set  $\{h(x_1, x_2) = c\}$  at  $\mathbf{x}^*$  is

$$-\frac{\partial h}{\partial x_1}(\mathbf{x}^*) \bigg/ \frac{\partial h}{\partial x_2}(\mathbf{x}^*).$$

The fact that these two slopes are equal at  $\mathbf{x}^*$  means

$$\frac{\frac{\partial f}{\partial x_1}(\mathbf{x}^*)}{\frac{\partial f}{\partial x_2}(\mathbf{x}^*)} = \frac{\frac{\partial h}{\partial x_1}(\mathbf{x}^*)}{\frac{\partial h}{\partial x_2}(\mathbf{x}^*)}. \quad (2)$$

As we will see shortly, it is handy to rewrite this equation as

$$\frac{\frac{\partial f}{\partial x_1}(\mathbf{x}^*)}{\frac{\partial h}{\partial x_1}(\mathbf{x}^*)} = \frac{\frac{\partial f}{\partial x_2}(\mathbf{x}^*)}{\frac{\partial h}{\partial x_2}(\mathbf{x}^*)}. \quad (3)$$

To avoid working with (possibly) zero denominators, let  $\mu$  denote the common value of the two quotients in (3):

$$\frac{\frac{\partial f}{\partial x_1}(\mathbf{x}^*)}{\frac{\partial h}{\partial x_1}(\mathbf{x}^*)} = \frac{\frac{\partial f}{\partial x_2}(\mathbf{x}^*)}{\frac{\partial h}{\partial x_2}(\mathbf{x}^*)} = \mu. \quad (4)$$

Rewrite (4) as the two equations

$$\begin{aligned}\frac{\partial f}{\partial x_1}(\mathbf{x}^*) - \mu \frac{\partial h}{\partial x_1}(\mathbf{x}^*) &= 0, \\ \frac{\partial f}{\partial x_2}(\mathbf{x}^*) - \mu \frac{\partial h}{\partial x_2}(\mathbf{x}^*) &= 0.\end{aligned}\tag{5}$$

Since we have to solve for the three unknowns  $(x_1, x_2, \mu)$ , we need three equations—one more than is listed in (5). But we do have a third equation: the constraint equation  $h(x_1, x_2) = c$ . Including the constraint equation with the two equations in (5) yields a system of three equations in three unknowns:

$$\begin{aligned}\frac{\partial f}{\partial x_1}(\mathbf{x}) - \mu \frac{\partial h}{\partial x_1}(\mathbf{x}) &= 0 \\ \frac{\partial f}{\partial x_2}(\mathbf{x}) - \mu \frac{\partial h}{\partial x_2}(\mathbf{x}) &= 0 \\ h(x_1, x_2) - c &= 0.\end{aligned}\tag{6}$$

There is a convenient way of writing this system (6). Form the **Lagrangian function**

$$L(x_1, x_2, \mu) \equiv f(x_1, x_2) - \mu(h(x_1, x_2) - c).$$

Find the critical points of the Lagrangian  $L$  by computing  $\partial L/\partial x_1$ ,  $\partial L/\partial x_2$ , and  $\partial L/\partial \mu$  and setting each equal to zero. As can easily be checked, the result of this process is precisely system (6). Note that since  $\mu$  just multiplies the constraint in the definition of  $L$ , the equation  $\partial L/\partial \mu = 0$  is equivalent to the constraint equation  $c - h(x_1, x_2) = 0$ . This new variable  $\mu$  which multiplies the constraint is called a **Lagrange multiplier**.

This process is truly magical. When we want to maximize a function in an unconstrained problem, we simply solve for its critical points by setting its first order partial derivatives equal to zero. However, by the introduction of the Lagrange multiplier  $\mu$  into the constrained problem, we have transformed a two-variable constrained problem to the problem of finding the *critical points* of a function  $L(x_1, x_2, \mu)$  of one more variable. In a sense, we have reduced a *constrained* optimization problem in two variables to an *unconstrained* problem in three variables. The penalty for this reduction is the inclusion of a new and somewhat artificial variable  $\mu$ . As we will see in Chapter 19, this new variable  $\mu$  is loaded with economic meaning; it will give us a new measure of value of the scarce resources in the problem under consideration.

A word of caution is in order here. This reduction would not have worked if both  $\partial h/\partial x_1$  and  $\partial h/\partial x_2$  were zero at the maximizer  $\mathbf{x}^*$  in equation (3). For this reason, we will need to make the assumption that  $\partial h/\partial x_1$  or  $\partial h/\partial x_2$  (or both) is

not zero at the constrained maximizer. Since this is a (slight) restriction on the constraint set, it is called a **constraint qualification**. If the constraint is linear, as it is in the utility maximization problems of Examples 18.1 and 18.2, this constraint qualification will automatically be satisfied.

We can now summarize our geometric analysis of the problem of maximizing (or minimizing) a function of two variables restricted by a single equality constraint.

**Theorem 18.1** Let  $f$  and  $h$  be  $C^1$  functions of two variables. Suppose that  $\mathbf{x}^* = (x_1^*, x_2^*)$  is a solution of the problem

$$\begin{aligned} &\text{maximize} && f(x_1, x_2) \\ &\text{subject to} && h(x_1, x_2) = c \end{aligned}$$

Suppose further that  $(x_1^*, x_2^*)$  is not a critical point of  $h$ . Then, there is a real number  $\mu^*$  such that  $(x_1^*, x_2^*, \mu^*)$  is a critical point of the Lagrangian function

$$L(x_1, x_2, \mu) \equiv f(x_1, x_2) - \mu[h(x_1, x_2) - c].$$

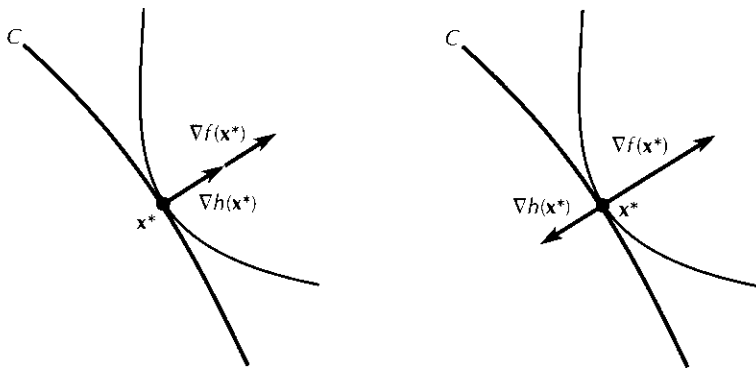
In other words, at  $(x_1^*, x_2^*, \mu^*)$

$$\frac{\partial L}{\partial x_1} = 0, \quad \frac{\partial L}{\partial x_2} = 0, \quad \text{and} \quad \frac{\partial L}{\partial \mu} = 0.$$

**Remark** Our geometric proof of Theorem 18.1 was based on the fact that the level curves of  $f$  and of  $h$  are tangent at  $(x_1^*, x_2^*)$  and therefore have equal slope. We now present another version of this proof based on the fact that the gradient vectors

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix} \quad \text{and} \quad \nabla h(\mathbf{x}) = \begin{pmatrix} \frac{\partial h}{\partial x_1} \\ \frac{\partial h}{\partial x_2} \end{pmatrix},$$

considered as displacement vectors or arrows at the point  $\mathbf{x}$ , are perpendicular to the level sets of  $f$  and  $h$  respectively. Since the level sets of  $f$  and  $h$  have the same slope at  $\mathbf{x}^*$ , the gradient vectors  $\nabla f(\mathbf{x}^*)$  and  $\nabla h(\mathbf{x}^*)$  must line up at  $\mathbf{x}^*$ ; they point in the same direction as in the left side of Figure 18.2 or in opposite directions as in the right side of Figure 18.2. In either case, the gradients are scalar multiples of each other. If we write this *multiplier* as  $\mu^*$ , then we find  $\nabla f(\mathbf{x}^*) = \mu^* \nabla h(\mathbf{x}^*)$ ;



$\nabla f(\mathbf{x}^*)$  and  $\nabla h(\mathbf{x}^*)$  line up at the constrained max or min  $\mathbf{x}^*$ .

**Figure 18.2**

that is,

$$\begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix} = \mu^* \begin{pmatrix} \frac{\partial h}{\partial x_1} \\ \frac{\partial h}{\partial x_2} \end{pmatrix}.$$

This immediately translates into system (5).

We present an analytic proof of Theorem 18.1 in the last section of Chapter 19.

**Remark** If we were minimizing  $f$  instead of maximizing  $f$  on the constraint set  $C_h$ , we would have used the same arguments that we used in the geometric proof of Theorem 18.1. In other words, the conclusion of Theorem 18.1 holds whether we are maximizing  $f$  or minimizing  $f$  on  $C_h$ . In Section 19.3, we will describe a second order condition that distinguishes maxima from minima.

**Example 18.4** Let's use Theorem 18.1 to solve a simple utility maximization problem:

$$\begin{aligned} &\text{maximize} && f(x_1, x_2) = x_1 x_2 \\ &\text{subject to} && h(x_1, x_2) \equiv x_1 + 4x_2 = 16. \end{aligned} \tag{7}$$

Since the gradient of  $h$  is  $(1, 4)$ ,  $h$  has no critical points and the constraint qualification is satisfied. Form the Lagrangian

$$L(x_1, x_2, \mu) = x_1 x_2 - \mu(x_1 + 4x_2 - 16),$$

and set its partial derivatives equal to zero:

$$\begin{aligned}\frac{\partial L}{\partial x_1} &= x_2 - \mu = 0 \\ \frac{\partial L}{\partial x_2} &= x_1 - 4\mu = 0 \\ \frac{\partial L}{\partial \mu} &= -(x_1 + 4x_2 - 16) = 0.\end{aligned}\tag{8}$$

Note that, as expected, the equation  $\partial L / \partial \mu = 0$  just repeats the equality constraint. We want to solve this system (8) of three equations in three unknowns. In this simple example, system (8) is linear and we can use the methods of Chapter 7. Since (8) is simple, we will use substitution.

From the first two equations in (8),

$$\mu = x_2 = \frac{1}{4}x_1,\tag{9}$$

$$\text{and therefore} \quad x_1 = 4x_2\tag{10}$$

Substitute (10) into the third equation in (8):

$$(4x_2) + 4x_2 = 16, \quad \text{or} \quad x_2 = 2.$$

From (9) and (10), we conclude the solution of system (8) is

$$x_1 = 8, \quad x_2 = 2, \quad \mu = 2.$$

Theorem 18.1 states that the only candidate for a solution to problem (7) is  $x_1 = 8, x_2 = 2$ .

*Example 18.5* Let's work out a more complex example:

$$\begin{aligned}\text{maximize} \quad & f(x_1, x_2) = x_1^2 x_2 \\ \text{subject to} \quad & (x_1, x_2) \text{ in the constraint set} \\ & C_h = \{(x_1, x_2) : 2x_1^2 + x_2^2 = 3\}.\end{aligned}$$

To check the constraint qualification, we compute the critical points of  $h(x_1, x_2) = 2x_1^2 + x_2^2$ . The only such critical point occurs at  $(x_1, x_2) = (0, 0)$  — a point which is *not* in the constraint set  $C_h$ . Therefore, the constraint qualification for this problem is satisfied. Now, form the Lagrangian

$$L(x_1, x_2, \mu) = x_1^2 x_2 - \mu(2x_1^2 + x_2^2 - 3),$$

compute its partial derivatives, and set them equal to 0:

$$\begin{aligned}\frac{\partial L}{\partial x_1} &= 2x_1x_2 - 4\mu x_1 = 2x_1(x_2 - 2\mu) = 0 \\ \frac{\partial L}{\partial x_2} &= x_1^2 - 2\mu x_2 = 0 \\ \frac{\partial L}{\partial \mu} &= -2x_1^2 - x_2^2 + 3 = 0.\end{aligned}\tag{11}$$

Once again, the equation  $\partial L/\partial \mu = 0$  just repeats the equality constraint. We want to solve this *nonlinear* system (11) of three equations in three unknowns. The usual method is to solve one of the first two equations for  $\mu$  and plug this expression into the other to eliminate  $\mu$  as a variable in the computation.

The first equation in (11) yields  $x_1 = 0$  or  $x_2 = 2\mu$ . We work with each of these two cases separately. If  $x_1 = 0$ , then  $x_2 = \pm\sqrt{3}$  from the third equation and  $\mu = 0$  from the second. Therefore,

$$(0, \sqrt{3}, 0) \quad \text{and} \quad (0, -\sqrt{3}, 0)$$

are two solutions of system (11).

If  $x_1 \neq 0$  in (11), then  $x_2 = 2\mu$  or  $\mu = x_2/2$  by the first equation in (11). Plug  $\mu = x_2/2$  into the second equation of (11) to find  $x_1^2 = x_2^2$ . Plug this expression into the third equation to obtain  $3x_1^2 = 3$  or  $x_1 = \pm 1$ . Since  $x_1^2 = x_2^2$ ,  $x_1 = \pm 1$  implies that  $x_2 = \pm 1$  and, via the first equation, that  $\mu = 0.5$  when  $x_2 = +1$ , and  $\mu = -0.5$  when  $x_2 = -1$ . Consequently, we obtain four more solutions of system (11) of first order conditions:

$$(1, 1, 0.5), \quad (-1, -1, -0.5), \quad (1, -1, -0.5), \quad \text{and} \quad (-1, 1, 0.5).$$

As we will see in Chapter 30, the fact the constraint set  $C_h$  is a compact set means that a constrained max exists for this problem. (Theorem 30.1.) Since we don't have a second order test and we know by Theorem 18.1 that the constrained max is one of these six candidates, we can plug each of the six candidates into  $f(x_1, x_2)$  and see which gives the highest value. Since

$$\begin{aligned}f(1, 1) &= f(-1, 1) = 1, \\ f(1, -1) &= f(-1, -1) = -1, \\ f(0, \sqrt{3}) &= f(0, -\sqrt{3}) = 0,\end{aligned}$$

the max occurs at  $(1, 1)$  and  $(-1, 1)$ . Note that  $(1, -1)$  and  $(-1, -1)$  *minimize*  $f$  on  $C_h$ . See Exercise 18.1.

The computations in Examples 18.4 and 18.5 illustrate how one applies Theorem 18.1. First, check the constraint qualification by calculating the critical points of the constraint function  $h$ , that is, the solutions of  $\partial h / \partial x_1 = 0$  and  $\partial h / \partial x_2 = 0$ . If none of these lie in the constraint set, we can write out the Lagrangian, set its partial derivatives equal to zero, and solve the resulting system of equations. If the constraint set does contain a critical point of the constraint function, then we include this point among our candidates for a solution to the original constrained maximization problem, along with the critical points of the Lagrangian. We will say more about constraint qualifications next chapter.

### Several Equality Constraints

We consider next the problem of maximizing a function  $f(x_1, \dots, x_n)$  of  $n$  variables constrained by more than one, say  $m$ , *equality* constraints. Let the functions  $h_1(\mathbf{x}), \dots, h_m(\mathbf{x})$  define the constraint set. In other words, we want to

$$\begin{aligned} &\text{maximize or minimize} && f(x_1, \dots, x_n) \\ &\text{subject to} && C_h = \{\mathbf{x} = (x_1, \dots, x_n) \mid h_1(\mathbf{x}) = a_1, \dots, h_m(\mathbf{x}) = a_m\}. \end{aligned}$$

We will need to generalize to  $m$  functions the constraint qualification that we used for one function of two variables:

$$\left( \frac{\partial h}{\partial x_1}(\mathbf{x}^*), \frac{\partial h}{\partial x_2}(\mathbf{x}^*) \right) \neq (0, 0). \quad (12)$$

If we have just one constraint,  $h(x_1, \dots, x_n) = a$ , then the natural generalization of (12) is that some first order partial derivative of  $h$  is not zero at the optimal  $\mathbf{x}^*$ :

$$\left( \frac{\partial h}{\partial x_1}(\mathbf{x}^*), \frac{\partial h}{\partial x_2}(\mathbf{x}^*), \dots, \frac{\partial h}{\partial x_n}(\mathbf{x}^*) \right) \neq (0, 0, \dots, 0). \quad (13)$$

If we are dealing with  $m$  constraint functions,  $m > 1$ , the natural generalization of (12) and (13) involves the Jacobian derivative

$$D\mathbf{h}(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial h_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial h_1}{\partial x_n}(\mathbf{x}^*) \\ \frac{\partial h_2}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial h_2}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial h_m}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial h_m}{\partial x_n}(\mathbf{x}^*) \end{pmatrix}$$

of the constraint functions. In general, a point  $\mathbf{x}^*$  is called a **critical point** of  $\mathbf{h} = (h_1, \dots, h_m)$  if the rank of the matrix  $D\mathbf{h}(\mathbf{x}^*)$  is  $< m$ . Therefore, the natural

generalization of the constraint qualification (13) is that the rank of  $D\mathbf{h}(\mathbf{x})$  be  $m$ —as large as it can be. More formally, we say that  $(h_1, \dots, h_m)$  satisfies the **nondegenerate constraint qualification (NDCQ)** at  $\mathbf{x}^*$  if the rank of the Jacobian matrix  $D\mathbf{h}(\mathbf{x}^*)$  at  $\mathbf{x}^*$  is  $m$ . The NDCQ is a regularity condition, like the definition of a regular curve in Section 14.5. It implies that the constraint set has a well-defined  $n - m$ -dimensional tangent plane everywhere.

The statement of the general theorem for maximizing a function of  $n$  variables constrained by  $m$  equality constraints is a straightforward generalization of Theorem 18.1. However, we can't use our two-dimensional geometric proof. We will present a complete proof in Section 19.6.

**Theorem 18.2** Let  $f, h_1, \dots, h_m$  be  $C^1$  functions of  $n$  variables. Consider the problem of maximizing (or minimizing)  $f(\mathbf{x})$  on the constraint set

$$C_h \equiv \{\mathbf{x} = (x_1, \dots, x_n) : h_1(\mathbf{x}) = a_1, \dots, h_m(\mathbf{x}) = a_m\}.$$

Suppose that  $\mathbf{x}^* \in C_h$  and that  $\mathbf{x}^*$  is a (local) max or min of  $f$  on  $C_h$ . Suppose further that  $\mathbf{x}^*$  satisfies condition NDCQ above. Then, there exist  $\mu_1^*, \dots, \mu_m^*$  such that  $(x_1^*, \dots, x_n^*, \mu_1^*, \dots, \mu_m^*) \equiv (\mathbf{x}^*, \mu^*)$  is a critical point of the Lagrangian

$$L(\mathbf{x}, \mu) \equiv f(\mathbf{x}) - \mu_1[h_1(\mathbf{x}) - a_1] - \mu_2[h_2(\mathbf{x}) - a_2] - \dots - \mu_m[h_m(\mathbf{x}) - a_m].$$

In other words,

$$\begin{aligned} \frac{\partial L}{\partial x_1}(\mathbf{x}^*, \mu^*) = 0, \dots, \frac{\partial L}{\partial x_n}(\mathbf{x}^*, \mu^*) &= 0, \\ \frac{\partial L}{\partial \mu_1}(\mathbf{x}^*, \mu^*) = 0, \dots, \frac{\partial L}{\partial \mu_m}(\mathbf{x}^*, \mu^*) &= 0. \end{aligned} \tag{14}$$

**Example 18.6** Consider the problem of maximizing  $f(x, y, z) = xyz$  on the constraint set defined by

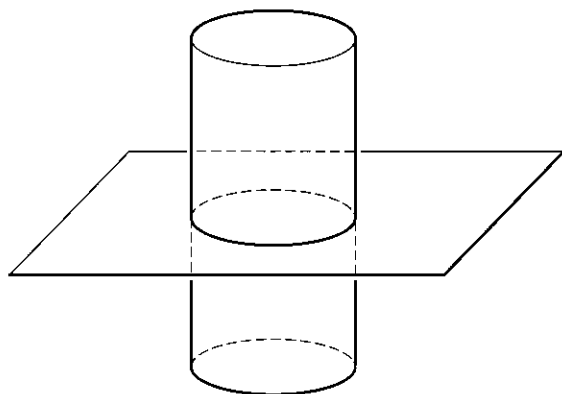
$$h_1(x, y, z) \equiv x^2 + y^2 = 1 \quad \text{and} \quad h_2(x, y, z) \equiv x + z = 1.$$

Looking at this problem geometrically first, we see that the set defined by  $h_1$  is the cylinder  $C_1$  parallel to the  $z$ -axis, as drawn in Figure 18.3. The set defined by  $h_2$  is the plane  $C_2$  parallel to the  $y$ -axis, as drawn in Figure 18.3. The actual constraint set  $C_h$  is the intersection of  $C_1$  and  $C_2$ .

Now let's tackle this problem analytically. First, compute the Jacobian matrix of the constraint functions

$$D\mathbf{h}(x, y, z) = \begin{pmatrix} 2x & 2y & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$





**Figure  
18.3**

*The constraint set in Example 18.6.*

Its rank is less than 2 if and only if  $x = y = 0$ . Since any point with  $x = y = 0$  would violate the first constraint, all points in the constraint set satisfy NDCQ. Next, form the Lagrangian

$$L(x, y, z, \mu_1, \mu_2) = xyz - \mu_1(x^2 + y^2 - 1) - \mu_2(x + z - 1),$$

and set its first partial derivatives equal to 0:

$$\frac{\partial L}{\partial x} = yz - 2\mu_1 x - \mu_2 = 0$$

$$\frac{\partial L}{\partial y} = xz - 2\mu_1 y = 0$$

$$\frac{\partial L}{\partial z} = xy - \mu_2 = 0$$

$$\frac{\partial L}{\partial \mu_1} = 1 - x^2 - y^2 = 0$$

$$\frac{\partial L}{\partial \mu_2} = 1 - x - z = 0.$$

Solve the second and third equations for  $\mu_1$  and  $\mu_2$  in terms of  $x$ ,  $y$ , and  $z$  and plug these into the first equation to obtain

$$yz - 2\left(\frac{xz}{2y}\right)x - xy = 0,$$

or

$$y^2z - x^2z - xy^2 = 0. \quad (15)$$

Then, solve the fourth equation for  $y^2$  in terms of  $x^2$  and the last equation for  $z$  in terms of  $x$  and plug these into (15) to obtain

$$(1 - x^2)(1 - x) - x^2(1 - x) - x(1 - x^2) = 0.$$

The result is a cubic equation with  $x = 1$  as one root. One can divide the cubic through by  $x - 1$  and solve the resulting quadratic to obtain  $x = \frac{1}{6}(-1 \pm \sqrt{13})$ , approximately  $-0.7676$  and  $0.4343$ . Plugging these numbers into the constraints, we obtain the four solution candidates

$$x \approx 0.4343, \quad y \approx \pm 0.9008, \quad z \approx 0.5657;$$

$$x \approx -0.7676, \quad y \approx \pm 0.6409, \quad z \approx 1.7676.$$

Evaluating the objective function at these four points, we find that the maximizer is

$$x \approx -0.7676, \quad y \approx -0.6409, \quad z \approx 1.7676.$$

As Example 18.6 illustrates, we use Theorem 18.2 just as we used Theorem 18.1. The candidates for the constrained maximizer are the critical points of the constraint functions *and* the critical points of the Lagrangian.

### EXERCISES

- 18.1 For the problem of Example 18.5, draw  $C_h$  and some level sets of  $f$ , as in Figure 18.1, to get a geometric interpretation of the solutions. Use this picture to determine whether the points  $(0, \pm\sqrt{3})$  are local max, local min, or neither.
- 18.2 Find the maximum and minimum distance from the origin to the ellipse  $x^2 + xy + y^2 = 3$ . [Hint: Use  $x^2 + y^2$  as your objective function.]
- 18.3 Find the point on the parabola  $y = x^2$  that is closest to the point  $(2, 1)$ . (Estimate the solution of the cubic equation which results.)
- 18.4 Find the general expression (in terms of all the parameters) for the commodity bundle  $(x_1, x_2)$  which maximizes the Cobb-Douglas utility function  $U(x_1, x_2) = kx_1^\alpha x_2^{1-\alpha}$  on the budget set  $p_1x_1 + p_2x_2 = I$ .
- 18.5 Find the point closest to the origin in  $\mathbf{R}^3$  that is on both the planes  $3x + y + z = 5$  and  $x + y + z = 1$ .
- 18.6 Find the max and min of  $f(x, y, z) = x + y + z^2$  subject to  $x^2 + y^2 + z^2 = 1$  and  $y = 0$ .
- 18.7 Maximize  $f(x, y, z) = yz - xz$  subject to  $y^2 + z^2 = 1$  and  $xz = 3$ .
- 18.8 Show that the NDCQ implies that  $m \leq n$ .
- 18.9 Maximize  $x^2y^2z^2$  subject to  $x^2 + y^2 + z^2 = c^2$ , where  $c$  is some fixed positive constant. What is the maximum value of the objective function on the constraint set? Show that for all  $x, y, z$ ,

$$x^2y^2z^2 \leq \left(\frac{1}{3}(x^2 + y^2 + z^2)\right)^3, \quad \text{or} \quad (x^2y^2z^2)^{1/3} \leq \frac{x^2 + y^2 + z^2}{3}.$$

This equality states that the **geometric mean** of three positive numbers is always  $\leq$  the **arithmetic mean** or **average** of these three numbers. Furthermore, these means are equal only when the three numbers are equal. Of course, the same proof works for all sets of  $n$  positive numbers:

$$(x_1^2 \cdot x_2^2 \cdots x_n^2)^{\frac{1}{n}} \leq \frac{x_1^2 + x_2^2 + \cdots + x_n^2}{n}$$

with equality if and only if  $x_1^2 = x_2^2 = \cdots = x_n^2$ .

### 18.3 INEQUALITY CONSTRAINTS

In the previous section, we worked solely with constraint sets defined by *equality* constraints:

$$h_1(x_1, \dots, x_n) = c_1, \dots, h_m(x_1, \dots, x_n) = c_m.$$

To find the maximum or minimum of a function on such a constraint set, we simply constructed the Lagrangian, set its  $(m + n)$  first partial derivatives equal to zero, and then solved these  $(m + n)$  equations in  $(m + n)$  unknowns — worrying a little about constraint qualifications along the way. However, the vast majority of constrained optimization problems that arise in economics have their constraints defined by *inequalities*:

$$g_1(x_1, \dots, x_n) \leq b_1, \dots, g_k(x_1, \dots, x_n) \leq b_k.$$

Unfortunately, the method for finding the constrained maxima in problems with inequality constraints is a bit more complex than the method we used for equality constraints. The first order conditions involve both equalities and inequalities and their solution entails the investigation of a number of cases.

As a matter of bookkeeping, for the rest of this chapter we will continue to use  $h$ 's to denote functions which define equality constraints and we will use  $g$ 's to denote functions which define inequality constraints. We will use  $\mu$ 's for the multipliers of the  $h$ 's and  $\lambda$ 's for the multipliers of the  $g$ 's. This convention will be especially helpful when we work with the general case which includes both equality and inequality constraints.

#### One Inequality Constraint

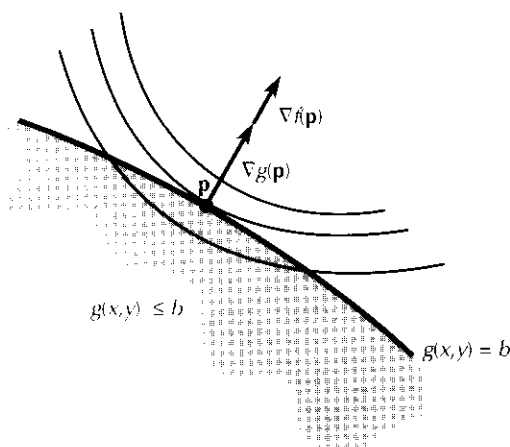
Let's begin again by looking at the simplest case — two variables and one *inequality* constraint:

$$\begin{array}{ll} \text{maximize} & f(x, y) \\ \text{subject to} & g(x, y) \leq b. \end{array}$$

In Figure 18.4, the thicker curve is the curve  $g(x, y) = b$ ; the region to the left and below this curve is the constraint set  $g(x, y) \leq b$ . The thinner lines are the level sets of the objective function  $f$ . By studying Figure 18.4, one notes that the highest level curve of  $f$  which meets the constraint set meets it at the point  $\mathbf{p}$ . Since  $\mathbf{p}$  lies on the boundary of the constraint set where  $g(x, y) = b$ , we say that the constraint is **binding** (or is active, effective, or tight) at  $\mathbf{p}$ . As we noted in Figure 18.1, the level set of  $f$  and the level set of  $g$  are tangent to each other at  $\mathbf{p}$ . As in Figure 18.2, this means that  $\nabla f(\mathbf{p})$  and  $\nabla g(\mathbf{p})$  line up — point in the same direction or in opposite directions — and therefore that  $\nabla f(\mathbf{p})$  is a multiple of  $\nabla g(\mathbf{p})$ . If we let  $\lambda$  denote the multiplier, then  $\nabla f(\mathbf{p}) = \lambda \nabla g(\mathbf{p})$ , or

$$\nabla f(\mathbf{p}) - \lambda \nabla g(\mathbf{p}) = \mathbf{0}. \quad (16)$$

This time, however, the sign of the multiplier is important! Recall from Section 14.6 that  $\nabla f(\mathbf{p})$  points in the direction in which  $f$  *increases* most rapidly at  $\mathbf{p}$ . In particular,  $\nabla g(\mathbf{p})$  points to the set  $g(x, y) \geq b$ , and not to the set  $g(x, y) \leq b$ . Since  $\mathbf{p}$  maximizes  $f$  on the set  $g(x, y) \leq b$ , the gradient of  $f$  cannot point to the constraint set. If it did, we could increase  $f$  and still keep  $g(x, y) \leq b$ . So,  $\nabla f(\mathbf{p})$  must point to the region where  $g(x, y) \geq b$ . This means that  $\nabla f(\mathbf{p})$  and  $\nabla g(\mathbf{p})$  *point in the same direction*. Thus, if  $\nabla f(\mathbf{p}) = \lambda \nabla g(\mathbf{p})$ , the multiplier  $\lambda$  must be  $\geq 0$ . It is worthwhile to take the time to understand clearly this difference between equality and inequality constraints.



$\nabla f$  and  $\nabla g$  point in the same direction at the maximizer  $\mathbf{p}$ .

**Figure**  
**18.4**

Consequently, for the situation depicted in Figure 18.4, we still form the Lagrangian function

$$L(x, y, \lambda) = f(x, y) - \lambda[g(x, y) - b]$$

and, as shown in (16), set

$$\frac{\partial L}{\partial x} = \frac{\partial f}{\partial x} - \lambda \frac{\partial g}{\partial x} \quad \text{and} \quad \frac{\partial L}{\partial y} = \frac{\partial f}{\partial y} - \lambda \frac{\partial g}{\partial y}$$

both equal to zero. We require, as we did in the previous section, the constraint qualification that the maximizer not be a critical point of the constraint function  $g$ .

Before we consider  $\partial L / \partial \lambda$ , there is one more situation we need to examine. Suppose that the maximum of  $f$  on the constraint set  $g(x, y) \leq b$  occurs not where  $g(x, y) = b$  but at a point where  $g(x, y) < b$ . Figure 18.5 depicts this scenario, where we still use a thicker line to represent the boundary  $g(x, y) = b$  of the constraint set. The full constraint set  $g(x, y) \leq b$  lies to the left and below this curve. However, this time the maximum of  $f$  occurs at the point  $\mathbf{q}$  in the interior of the constraint set. There is a point  $\mathbf{r}$  on the level set  $g(x, y) = b$  where this level set is tangent to a level set of  $f$ , but  $\nabla f$  and  $\nabla g$  point in opposite directions at  $\mathbf{r}$ . In fact, one can increase the value of  $f$  by moving further *into* the constraint set from  $\mathbf{r}$  until one reaches  $\mathbf{q}$ . Since  $g(x, y)$  is strictly less than  $b$  at  $\mathbf{q}$ , in other words  $\mathbf{q}$  is in the interior of the constraint set, we say that the constraint is **not binding** (inactive, ineffective, loose) at  $\mathbf{q}$ . Note that the point  $\mathbf{q}$  must be a local max of  $f$ , that is, a local *unconstrained* max. By Theorem 17.1, this means that

$$\frac{\partial f}{\partial x}(\mathbf{q}) = 0 \quad \text{and} \quad \frac{\partial f}{\partial y}(\mathbf{q}) = 0.$$

The derivatives of  $g$  do not enter the criterion or the calculations at  $\mathbf{q}$ . We can still use our Lagrangian

$$L(x, y, \lambda) = f(x, y) - \lambda[g(x, y) - b]$$

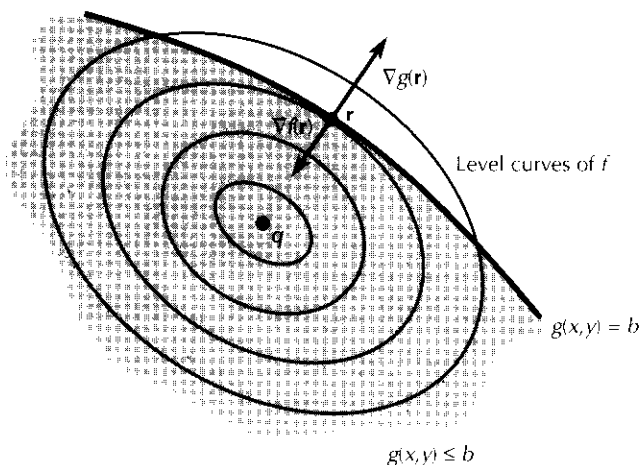


Figure  
18.5

The situation in which the constraint is not binding.

and set  $\partial L/\partial x$  and  $\partial L/\partial y$  equal to zero, *provided* that we set  $\lambda$  equal to zero. Setting  $\lambda = 0$  causes the constraint function to drop out of the analysis; this is just what we want when the constraint is not binding at the max.

In summary, either the constraint is binding, that is,  $g(x, y) - b = 0$  as in Figure 18.4, in which case the multiplier  $\lambda$  must be  $\geq 0$ , or the constraint is not binding as in Figure 18.5, in which case the multiplier  $\lambda$  must be zero. Such a condition in which one of two inequalities must be binding is called a **complementary slackness condition**. A convenient way to summarize such a condition that one of two numbers be zero is to say that their *product* must be zero. Therefore, we will summarize our criterion that either  $g(x, y) - b = 0$  or  $\lambda = 0$  by requiring that

$$\lambda \cdot [g(x, y) - b] = 0. \quad (17)$$

Since we do not know *a priori* whether or not the constraint will be binding at the maximizer, we cannot use the condition  $\partial L/\partial \mu = 0$  that we used with equality constraints since this condition is equivalent to  $g(x, y) - b = 0$ . We will replace this statement by condition (17), which says that *either* the constraint is binding or its multiplier is zero (or, in rare circumstances, both). Putting these two cases together, we can summarize our observations in this section by the following theorem.

**Theorem 18.3** Suppose that  $f$  and  $g$  are  $C^1$  functions on  $\mathbf{R}^2$  and that  $(x^*, y^*)$  maximizes  $f$  on the constraint set  $g(x, y) \leq b$ . If  $g(x^*, y^*) = b$ , suppose that

$$\frac{\partial g}{\partial x}(x^*, y^*) \neq 0 \quad \text{or} \quad \frac{\partial g}{\partial y}(x^*, y^*) \neq 0.$$

In any case, form the Lagrangian function

$$L(x, y, \lambda) = f(x, y) - \lambda \cdot [g(x, y) - b].$$

Then, there is a multiplier  $\lambda^*$  such that:

- (a)  $\frac{\partial L}{\partial x}(x^*, y^*, \lambda^*) = 0,$
- (b)  $\frac{\partial L}{\partial y}(x^*, y^*, \lambda^*) = 0,$
- (c)  $\lambda^* \cdot [g(x^*, y^*) - b] = 0,$
- (d)  $\lambda^* \geq 0,$
- (e)  $g(x^*, y^*) \leq b.$

**Remark** In some texts, the constraint is written as  $g(x, y) \geq b$  instead of as  $g(x, y) \leq b$  and the Lagrangian is then written as  $L(x, y, \lambda) = f(x, y) + \lambda \cdot [g(x, y) - b]$ . These two changes cancel each other so that the conclusion of Theorem 18.3 still holds at a constrained max.

Notice the similarities and differences between the statement of Theorem 18.2 which treats equality constraints and the statement of Theorem 18.3 which covers inequality constraints:

- (1) Both use the same Lagrangian  $L$  and both require that the derivatives of  $L$  with respect to the  $x_i$ 's be zero.
- (2) The condition that  $\partial L / \partial \mu = h(x, y) - c = 0$  for equality constraints may no longer hold for inequality constraints since the constraint need not be binding at the maximizer in the inequality constraint case. It is replaced by two conditions:

$$\lambda \cdot [g(x, y) - b] = 0 \quad \text{and} \quad \frac{\partial L}{\partial \lambda} = g(x, y) - b \leq 0.$$

The second of these two conditions is simply a repetition of the inequality constraint itself.

- (3) Both situations require that we check a constraint qualification. However, we need only check the constraint qualification for an inequality constraint if that constraint is binding at the solution candidate.
- (4) There were no restrictions on the sign of the multiplier in the equality constraint situation; however, the multiplier for inequality constraints must be nonnegative.
- (5) For equality constraints (and for problems with no constraints), the same first order conditions that work for maximization problems also hold for minimization problems. However, the argument, summarized in Figure 18.4, that  $\nabla f(\mathbf{p})$  and  $\nabla g(\mathbf{p})$  point in the *same* direction for inequality constraints holds only for the maximization problem. The same line of reasoning concludes that  $\nabla f(\mathbf{p})$  and  $\nabla g(\mathbf{p})$  must point in *opposite* directions in a constrained minimization problem. We will say more about the distinction between minimization problems and maximization problems in Section 18.5.

**Example 18.7** Consider the problem of maximizing  $f(x, y) = xy$  on the constraint set  $g(x, y) = x^2 + y^2 \leq 1$ . The only critical point of  $g$  occurs at the origin — far away from the boundary of the constraint set  $x^2 + y^2 = 1$ . So, the constraint qualification will be satisfied at any candidate for a solution. Form the Lagrangian

$$L(x, y, \lambda) = xy - \lambda(x^2 + y^2 - 1),$$

and write out the first order conditions described in Theorem 18.3:

$$\begin{aligned} \frac{\partial L}{\partial x} = y - 2\lambda x &= 0, & \frac{\partial L}{\partial y} = x - 2\lambda y &= 0, \\ \lambda(x^2 + y^2 - 1) &= 0, & x^2 + y^2 &\leq 1, & \lambda &\geq 0. \end{aligned}$$

The first two equations yield

$$\lambda = \frac{y}{2x} = \frac{x}{2y}, \quad \text{or} \quad x^2 = y^2. \quad (18)$$

If  $\lambda = 0$ , then  $x = y = 0$ . This combination satisfies all the first order conditions, so it is a candidate for a solution. If  $\lambda \neq 0$ , then the third equation becomes  $x^2 + y^2 - 1 = 0$ . Combining this with (18), we find that  $x^2 = y^2 = 1/2$ , or  $x = \pm 1/\sqrt{2}$ ,  $y = \pm 1/\sqrt{2}$ . Combining these with the equation for  $\lambda$  in (18), we find the following four candidates:

$$x = +\frac{1}{\sqrt{2}}, \quad y = +\frac{1}{\sqrt{2}}, \quad \lambda = +\frac{1}{2};$$

$$x = -\frac{1}{\sqrt{2}}, \quad y = -\frac{1}{\sqrt{2}}, \quad \lambda = +\frac{1}{2};$$

$$x = +\frac{1}{\sqrt{2}}, \quad y = -\frac{1}{\sqrt{2}}, \quad \lambda = -\frac{1}{2};$$

$$x = -\frac{1}{\sqrt{2}}, \quad y = +\frac{1}{\sqrt{2}}, \quad \lambda = -\frac{1}{2}.$$

We disregard the last two candidates since they involve a negative multiplier. So, including  $(0, 0, 0)$ , there are three candidates which satisfy all five first order conditions. Plugging these three into the objective function, we find that

$$x = \frac{1}{\sqrt{2}}, \quad y = \frac{1}{\sqrt{2}} \quad \text{and} \quad x = -\frac{1}{\sqrt{2}}, \quad y = -\frac{1}{\sqrt{2}}$$

are the solutions of our original problem.

The two points with the negative multipliers are the solutions of the problem of minimizing  $xy$  on the constraint set  $x^2 + y^2 \leq 1$ .

One way to think of condition  $c$  in Theorem 18.3 is that if  $\lambda > 0$ , we know the constraint will be binding and we can treat it as an equality constraint instead of as an inequality constraint — a much simpler criterion to work with. In some economics problems this type of analysis can give us useful information about the phenomenon under study, as the following example illustrates.

**Example 18.8** Consider once again the standard utility maximization problem of Example 18.1. We continue to ignore the nonnegativity constraints but now do not force the budget constraint to be binding in the statement of the problem. We will see that the tightness of the budget constraint — the conclusion that the consumer spends all the available income — is a consequence of a natural monotonicity assumption on the utility function.



Our goal is to maximize a  $C^1$  utility function  $U(x_1, x_2)$  subject to the budget constraint  $p_1x_1 + p_2x_2 \leq I$ , where  $p_1$  and  $p_2$  represent positive unit prices. We assume that for each commodity bundle  $(x_1, x_2)$ ,

$$\frac{\partial U}{\partial x_1}(x_1, x_2) > 0 \quad \text{or} \quad \frac{\partial U}{\partial x_2}(x_1, x_2) > 0.$$

This is a version of the usual monotonicity or nonsatiation assumption. It states that the commodities under study are *goods* in that increasing consumption increases utility. Since the usual constraint qualification is satisfied, we can form the Lagrangian

$$L(x_1, x_2, \lambda) = U(x_1, x_2) - \lambda(p_1x_1 + p_2x_2 - I)$$

and compute its  $x_1$ - and  $x_2$ -critical points:

$$\begin{aligned} \frac{\partial L}{\partial x_1}(x_1, x_2) &= \frac{\partial U}{\partial x_1}(x_1, x_2) - \lambda p_1 = 0, \\ \frac{\partial L}{\partial x_2}(x_1, x_2) &= \frac{\partial U}{\partial x_2}(x_1, x_2) - \lambda p_2 = 0. \end{aligned} \tag{19}$$

At the maximizer, the multiplier  $\lambda$  cannot be zero; otherwise both  $\partial U / \partial x_1$  and  $\partial U / \partial x_2$  would be zero in (19)—a contradiction to our monotonicity assumption. Since

$$\lambda > 0 \quad \text{and} \quad \lambda(p_1x_1 + p_2x_2 - I) = 0,$$

it follows that  $p_1x_1 + p_2x_2 = I$ ; the consumer will spend all available income and we can treat the budget constraint as an equality constraint.

## Several Inequality Constraints

The generalization of Theorem 18.3 to more variables and more constraints is straightforward. We will present its statement here and then work through some concrete problems using it, but delay its proof until Section 19.6. Recall that a constraint  $g(\mathbf{x}) \leq b$  is **binding** (active, effective) at a solution candidate  $\mathbf{x}^*$  if  $g(\mathbf{x}^*) = b$ . If  $g(\mathbf{x}^*) < b$ , we say that the constraint is **not binding** (inactive, ineffective) at  $\mathbf{x}^*$ .

**Theorem 18.4** Suppose that  $f, g_1, \dots, g_k$  are  $C^1$  functions of  $n$  variables. Suppose that  $\mathbf{x}^* \in \mathbf{R}^n$  is a local maximizer of  $f$  on the constraint set defined by the  $k$  inequalities

$$g_1(x_1, \dots, x_n) \leq b_1, \dots, g_k(x_1, \dots, x_n) \leq b_k.$$

For ease of notation, assume that the first  $k_0$  constraints are binding at  $\mathbf{x}^*$  and that the last  $k - k_0$  constraints are not binding. Suppose that the following nondegenerate constraint qualification is satisfied at  $\mathbf{x}^*$ .

The rank at  $\mathbf{x}^*$  of the Jacobian matrix of the *binding* constraints

$$\begin{pmatrix} \frac{\partial g_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial g_1}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_{k_0}}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial g_{k_0}}{\partial x_n}(\mathbf{x}^*) \end{pmatrix}$$

is  $k_0$  — as large as it can be.

Form the Lagrangian

$$L(x_1, \dots, x_n, \lambda_1, \dots, \lambda_k) \equiv f(\mathbf{x}) - \lambda_1[g_1(\mathbf{x}) - b_1] - \cdots - \lambda_k[g_k(\mathbf{x}) - b_k].$$

Then, there exist multipliers  $\lambda_1^*, \dots, \lambda_k^*$  such that:

- (a)  $\frac{\partial L}{\partial x_1}(\mathbf{x}^*, \lambda^*) = 0, \dots, \frac{\partial L}{\partial x_n}(\mathbf{x}^*, \lambda^*) = 0,$
- (b)  $\lambda_1^*[g_1(\mathbf{x}^*) - b_1] = 0, \dots, \lambda_k^*[g_k(\mathbf{x}^*) - b_k] = 0,$
- (c)  $\lambda_1^* \geq 0, \dots, \lambda_k^* \geq 0,$
- (d)  $g_1(\mathbf{x}^*) \leq b_1, \dots, g_k(\mathbf{x}^*) \leq b_k.$

**Remark** The constraint qualification in the statement of Theorem 18.4 is the natural generalization of the constraint qualifications in Theorems 18.2 and 18.3. This condition involves only the *binding* constraints since the nonbinding constraints should play no role in the first order conditions. Then, we treat the binding constraints just as we did the equality constraints in Theorem 18.2, by assuming that their Jacobian has maximal rank. We will still abbreviate this version of the nondegenerate constraint qualifications as NDCQ.

**Example 18.9** Consider the problem of maximizing  $f(x, y, z) = xyz$  on the constraint set defined by the inequalities

$$x + y + z \leq 1, \quad x \geq 0, \quad y \geq 0, \quad \text{and} \quad z \geq 0.$$

This is the typical example of a utility maximization problem in a three-dimensional commodity space. Since we need to write all our inequality constraints consistently — with a  $\leq$  — we write the three nonnegativity constraints as

$$-x \leq 0, \quad -y \leq 0, \quad \text{and} \quad -z \leq 0.$$

The Jacobian of the constraint functions is

$$\begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Since its columns are linearly independent, it has rank three. Since at most three of the four constraints can be binding at any one time, the NDCQ holds at any solution candidate. Form the Lagrangian

$$\begin{aligned} L(x, y, z, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = & xyz - \lambda_1(x + y + z - 1) \\ & - \lambda_2(-x) - \lambda_3(-y) - \lambda_4(-z). \end{aligned}$$

Because of the double minus signs in the last three terms of this Lagrangian, we can rewrite it more aesthetically as

$$L(x, y, z, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = xyz - \lambda_1(x + y + z - 1) + \lambda_2x + \lambda_3y + \lambda_4z.$$

From now on, we will treat nonnegativity constraints this way, by including them in the Lagrangian as  $+\lambda_i x_i$  rather than as  $-\lambda_i(-x_i)$ . We now write out the complete set of first order conditions, according to Theorem 18.4:

$$(1) \quad \frac{\partial L}{\partial x} = yz - \lambda_1 + \lambda_2 = 0,$$

$$(2) \quad \frac{\partial L}{\partial y} = xz - \lambda_1 + \lambda_3 = 0,$$

$$(3) \quad \frac{\partial L}{\partial z} = xy - \lambda_1 + \lambda_4 = 0,$$

$$(4) \quad \lambda_1(x + y + z - 1) = 0, \quad (5) \quad \lambda_2x = 0,$$

$$(6) \quad \lambda_3y = 0, \quad (7) \quad \lambda_4z = 0,$$

$$(8) \quad \lambda_1 \geq 0, \quad (9) \quad \lambda_2 \geq 0,$$

$$(10) \quad \lambda_3 \geq 0, \quad (11) \quad \lambda_4 \geq 0,$$

$$(12) \quad x + y + z \leq 1, \quad (13) \quad x \geq 0,$$

$$(14) \quad y \geq 0, \quad (15) \quad z \geq 0.$$

Rewrite conditions 1, 2, and 3, without minus signs, as

$$\lambda_1 = yz + \lambda_2 = xz + \lambda_3 = xy + \lambda_4. \quad (20)$$

We will look at two cases:  $\lambda_1 = 0$  and  $\lambda_1 > 0$ .

If  $\lambda_1 = 0$  in equation (20), then because every variable in equation (20) is nonnegative,

$$yz = xz = xy = 0 \quad \text{and} \quad \lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0. \quad (21)$$

Equations (21) lead to the (infinite) set of solution candidates in which two of the variables equal zero and the third is any number in the interval  $[0, 1]$ . In particular, the objective function equals zero for all  $(x, y, z)$  which satisfy equations (21).

Next, look at the case  $\lambda_1 > 0$ . By condition 4,  $x + y + z = 1$ ; at least one of  $x, y, z$  must be nonzero. Suppose for a moment that  $x = 0$ . Then, using equations (20) and the assumption that  $\lambda_1 > 0$ , we see that  $\lambda_3 = \lambda_4 = \lambda_1 > 0$ . But then, conditions 6 and 7 imply that  $y = z = 0$ —a contradiction to  $x + y + z = 1$ . Since the assumption that  $x = 0$  leads to a contradiction, we conclude that  $x > 0$ . Similar arguments show that  $y$  and  $z$  are positive too. Then, conditions 5, 6, and 7 imply that  $\lambda_2 = \lambda_3 = \lambda_4 = 0$  and equations (20) become simply

$$yz = xz = xy.$$

It follows now that

$$x = y = z = \frac{1}{3} \quad (22)$$

and, by using equation (20) once more, that  $\lambda_1 = 1/9$ . Since

$$f\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) = \frac{1}{27} > 0,$$

(22) is the solution of the constrained maximization problem.

As this example shows, the solution of a constrained maximization problem usually involves breaking the first order conditions into a number of cases. It is often easiest to start with the nonnegativity constraints or the signs of the multipliers. In Example 18.9, we first worked with the case  $\lambda_1 = 0$ . Each case needs to be carried out until either a complete candidate for a solution is computed, including values for the multipliers, or a contradiction to one of the first order conditions is reached. While working with any given case, one might have to break that case into two subcases depending on whether or not a second inequality constraint is binding or not. In Example 18.9, while studying the case  $\lambda_1 > 0$ , we had to examine two subcases depending on the sign of  $x$ .

In economic theory, however, one rarely needs to compute the maxima or minima of a specific problem. One is usually more interested in studying the first order conditions which arise in a specific *type* of problem, since these can lead to interesting relationships between the variables of the problem or even to

a general economic principle. We have provided a number of concrete examples and exercises in this chapter to make it easier to understand and work with the first order conditions of Theorem 18.4. In this vein, one more concrete example with inequality constraints is presented in the last section of this chapter.

### EXERCISES

- 18.10** Find the maximizer of  $f(x, y) = x^2 + y^2$ , subject to the constraints  $2x + y \leq 2$ ,  $x \geq 0$ ,  $y \geq 0$ .
- 18.11** Find the maximizer of  $f(x, y) = 2y^2 - x$ , subject to the constraints  $x^2 + y^2 \leq 1$ ,  $x \geq 0$ ,  $y \geq 0$ .
- 18.12** Consider the problem of maximizing  $f(x, y, z) = xyz + z$ , subject to the constraints  $x^2 + y^2 + z \leq 6$ ,  $x \geq 0$ ,  $y \geq 0$ ,  $z \geq 0$ .
- Write out a complete set of first order conditions for this problem.
  - Determine whether or not the constraint  $x^2 + y^2 + z \leq 6$  is binding at any solution.
  - Find a solution of the first order conditions that includes  $x = 0$ .
  - Find three equations in the three unknowns  $x, y, z$  that must be satisfied if  $x \neq 0$  at the solution.
  - Show that  $x = 1, y = 1, z = 4$  satisfies these equations.
- 18.13** Show that the budget inequality constraint is binding in Example 18.8 even in the presence of the nonnegativity constraints  $x_1 \geq 0$ ,  $x_2 \geq 0$ . In the process, check the NDCQ for this more general problem.
- 18.14** Allow more general prices and income in Example 18.9; in other words, change the constraint  $x + y + z \leq 1$  to the constraint  $p_1x + p_2y + p_3z \leq I$ .

### 18.4 MIXED CONSTRAINTS

As we noted in Example 18.8, some maximization problems involve both equality and inequality constraints. It is straightforward to combine the statements of Theorems 18.2 and 18.4 into one result which handles the general case.

**Theorem 18.5** Suppose that  $f, g_1, \dots, g_k, h_1, \dots, h_m$  are  $C^1$  functions of  $n$  variables. Suppose that  $\mathbf{x}^* \in \mathbf{R}^n$  is a local maximizer of  $f$  on the constraint set defined by the  $k$  inequalities and  $m$  equalities:

$$g_1(x_1, \dots, x_n) \leq b_1, \dots, g_k(x_1, \dots, x_n) \leq b_k,$$

$$h_1(x_1, \dots, x_n) = c_1, \dots, h_m(x_1, \dots, x_n) = c_m.$$

Without loss of generality, we can assume that the first  $k_0$  inequality constraints are binding at  $\mathbf{x}^*$  and that the other  $k - k_0$  inequality constraints are not binding. Suppose that the following nondegenerate constraint qualification is satisfied at  $\mathbf{x}^*$ :

The rank at  $\mathbf{x}^*$  of the Jacobian matrix of the equality constraints and the binding inequality constraints

$$\begin{pmatrix} \frac{\partial g_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial g_1}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_{k_0}}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial g_{k_0}}{\partial x_n}(\mathbf{x}^*) \\ \frac{\partial h_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial h_1}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial h_m}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial h_m}{\partial x_n}(\mathbf{x}^*) \end{pmatrix}$$

is  $k_0 + m$ —as large as it can be.

Form the Lagrangian

$$\begin{aligned} L(x_1, \dots, x_n, \lambda_1, \dots, \lambda_k, \mu_1, \dots, \mu_m) \\ = f(\mathbf{x}) - \lambda_1[g_1(\mathbf{x}) - b_1] - \cdots - \lambda_k[g_k(\mathbf{x}) - b_k] \\ - \mu_1[h_1(\mathbf{x}) - c_1] - \cdots - \mu_m[h_m(\mathbf{x}) - c_m]. \end{aligned}$$

Then, there exist multipliers  $\lambda_1^*, \dots, \lambda_k^*, \mu_1^*, \dots, \mu_m^*$  such that:

- (a)  $\frac{\partial L}{\partial x_1}(\mathbf{x}^*, \lambda^*) = 0, \dots, \frac{\partial L}{\partial x_n}(\mathbf{x}^*, \lambda^*) = 0,$
- (b)  $\lambda_1^*[g_1(\mathbf{x}^*) - b_1] = 0, \dots, \lambda_k^*[g_k(\mathbf{x}^*) - b_k] = 0,$
- (c)  $h_1(\mathbf{x}^*) = c_1, \dots, h_m(\mathbf{x}^*) = c_m,$
- (d)  $\lambda_1^* \geq 0, \dots, \lambda_k^* \geq 0,$
- (e)  $g_1(\mathbf{x}^*) \leq b_1, \dots, g_k(\mathbf{x}^*) \leq b_k.$

**Example 18.10** Consider the problem of maximizing  $x - y^2$  on the constraint set  $x^2 + y^2 = 4$ ,  $x \geq 0$ ,  $y \geq 0$ . Checking the NDCQ, first note that the gradient of  $x^2 + y^2$  is zero only at the origin, a point which is not in the constraint set. If either nonnegativity constraint is binding, then the solution candidate is  $(2, 0)$  or  $(0, 2)$ . In both cases, the corresponding  $2 \times 2$  Jacobian matrix of constraints has rank two. Therefore, the NDCQ will automatically be satisfied. Form the Lagrangian

$$L = x - y^2 - \mu(x^2 + y^2 - 4) + \lambda_1 x + \lambda_2 y.$$

The first order conditions become:

$$(1) \quad \frac{\partial L}{\partial x} = 1 - 2\mu x + \lambda_1 = 0,$$

$$(2) \quad \frac{\partial L}{\partial y} = -2y - 2\mu y + \lambda_2 = 0,$$

$$(3) \quad x^2 + y^2 - 4 = 0,$$

$$(4) \quad \lambda_1 x = 0, \quad (5) \quad \lambda_2 y = 0,$$

$$(6) \quad \lambda_1 \geq 0, \quad (7) \quad \lambda_2 \geq 0,$$

$$(8) \quad x \geq 0, \quad (9) \quad y \geq 0.$$

Write 1 without minus signs as  $1 + \lambda_1 = 2\mu x$ . Since  $\lambda_1 \geq 0$ ,  $1 + \lambda_1 > 0$ . Therefore,  $\mu > 0$  and  $x > 0$  (and  $\lambda_1 = 0$ ). Write 2 as  $2y(1 + \mu) = \lambda_2$ . Since  $1 + \mu > 0$ , either both  $y$  and  $\lambda_2$  are zero or both are positive. By 5, both cannot be positive. Therefore,  $\lambda_2 = y = 0$ . Now,  $x = 2$  by 3 and 8,  $\lambda_1 = 0$  by 4, and  $\mu = 1/4$  by 1. This leads to the solution

$$(x, y, \mu, \lambda_1, \lambda_2) = (2, 0, 1/4, 0, 0).$$

## EXERCISES

- 18.15** Maximize  $3xy - x^3$  subject to the constraints  $2x - y = -5$ ,  $5x + 2y \geq 37$ ,  $x \geq 0$ ,  $y \geq 0$ .

## 18.5 CONSTRAINED MINIMIZATION PROBLEMS

For the equality constrained problems that we considered in Section 18.2, the first order conditions for a minimization problem are the same as the first order conditions for a maximization problem. However, in Section 18.3 on *inequality* constrained problems, our discussion about the sign of the multiplier applied only to maximization problems. The same argument would show that if we wanted to minimize  $f$  instead of maximize it over the same constraint set, we would write the Lagrangian the same way but would require that the multipliers be  $\leq 0$ . This fact is illustrated in Example 18.7 in which the minimizing points correspond to negative multipliers.

However, there is a more common way of treating constrained minimization problems. The inequality constraints in a minimization problem are usually presented as  $g(\mathbf{x}) \geq b$ , instead of as  $g(\mathbf{x}) \leq b$  as they are in maximization problems. Therefore, we will use a formulation for minimization problems which takes advantage of this situation and which is a more natural analogue to our approach for maximization problems, especially for studying duality. We present this formulation as a theorem — the analogue of Theorem 18.5 for maximization problems.

**Theorem 18.6** Suppose that  $f, g_1, \dots, g_k, h_1, \dots, h_m$  are  $C^1$  functions of  $n$  variables. Suppose that  $\mathbf{x}^* \in \mathbf{R}^n$  is a local **minimizer** of  $f$  on the constraint set defined by the  $k$  inequalities and  $m$  equalities:

$$g_1(x_1, \dots, x_n) \geq b_1, \dots, g_k(x_1, \dots, x_n) \geq b_k,$$

$$h_1(x_1, \dots, x_n) = c_1, \dots, h_m(x_1, \dots, x_n) = c_m.$$

Without loss of generality, we can assume that the first  $k_0$  inequality constraints are binding at  $\mathbf{x}^*$  and that the other  $k - k_0$  inequality constraints are not binding. Suppose that the following nondegenerate constraint qualification is satisfied at  $\mathbf{x}^*$ :

The rank at  $\mathbf{x}^*$  of the Jacobian matrix of the equality constraints and the *binding* inequality constraints

$$\begin{pmatrix} \frac{\partial g_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial g_1}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_{k_0}}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial g_{k_0}}{\partial x_n}(\mathbf{x}^*) \\ \frac{\partial h_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial h_1}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial h_m}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial h_m}{\partial x_n}(\mathbf{x}^*) \end{pmatrix}$$

is  $k_0 + m$  — as large as it can be.

Form the Lagrangian

$$\begin{aligned} L(x_1, \dots, x_n, \lambda_1, \dots, \lambda_k, \mu_1, \dots, \mu_m) \\ \equiv f(\mathbf{x}) - \lambda_1[g_1(\mathbf{x}) - b_1] - \cdots - \lambda_k[g_k(\mathbf{x}) - b_k] \\ - \mu_1[h_1(\mathbf{x}) - c_1] - \cdots - \mu_m[h_m(\mathbf{x}) - c_m]. \end{aligned}$$

Then, there exist multipliers  $\lambda_1^*, \dots, \lambda_k^*, \mu_1^*, \dots, \mu_m^*$  such that:

- (a)  $\frac{\partial L}{\partial x_1}(\mathbf{x}^*, \lambda^*) = 0, \dots, \frac{\partial L}{\partial x_n}(\mathbf{x}^*, \lambda^*) = 0,$
- (b)  $\lambda_1^*[g_1(\mathbf{x}^*) - b_1] = 0, \dots, \lambda_k^*[g_k(\mathbf{x}^*) - b_k] = 0,$
- (c)  $h_1(\mathbf{x}^*) = c_1, \dots, h_m(\mathbf{x}^*) = c_m,$
- (d)  $\lambda_1^* \geq 0, \dots, \lambda_k^* \geq 0,$  and
- (e)  $g_1(\mathbf{x}^*) \geq b_1, \dots, g_k(\mathbf{x}^*) \geq b_k.$



To use this formulation, we need to write all the inequality constraints in the form  $g(\mathbf{x}) \geq b$ . For example, we would write the constraint  $x^2 + y^2 \leq 5$  as the constraint  $-(x^2 + y^2) \geq -5$ . Note that in this case, the nonnegativity constraints are already in the right form; now, they enter the Lagrangian as  $-\lambda_i x_i$ .

Other texts may take other approaches to constrained min problems (and even to constrained max problems). Other possible formulations for constrained minimization problems are:

- (1) Replace  $f$  by  $-f$ , because minimizing  $f$  is equivalent to maximizing  $-f$ . Otherwise, keep everything else the same as in the constrained max formulation, including the form of the inequality constraints.
- (2) Put the multipliers in the Lagrangian with plus signs instead of with minus signs, keeping the constraints as they had been for the constrained maximization problem.

*Example 18.11* Consider the problem

$$\begin{aligned} &\text{minimize} && f(x, y) = 2y - x^2 \\ &\text{subject to} && x^2 + y^2 \leq 1, \quad x \geq 0, \quad y \geq 0. \end{aligned} \tag{23}$$

Before writing the Lagrangian, we write the first constraint as  $-x^2 - y^2 \geq -1$ . Then, the Lagrangian is

$$L(x, y, \lambda_1, \lambda_2, \lambda_3) = 2y - x^2 - \lambda_1(-x^2 - y^2 + 1) - \lambda_2 x - \lambda_3 y.$$

The first order conditions are

$$\frac{\partial L}{\partial x} = -2x + 2\lambda_1 x - \lambda_2 = 0, \tag{24}$$

$$\frac{\partial L}{\partial y} = 2 + 2\lambda_1 y - \lambda_3 = 0, \tag{25}$$

$$\lambda_1(-x^2 - y^2 + 1) = 0, \tag{26}$$

$$\lambda_2 x = 0, \quad \lambda_3 y = 0, \tag{27}$$

$$\lambda_1, \lambda_2, \lambda_3 \geq 0, \tag{28}$$

along with the original constraints (23).

An effective way to start the solution of such a system of equations and inequalities is to rewrite the  $\partial L / \partial x_i = 0$  equations without minus signs. In this case, equations (24) and (25) become

$$2x + \lambda_2 = 2\lambda_1 x, \tag{29}$$

$$2 + 2\lambda_1 y = \lambda_3. \tag{30}$$

Since every variable in (29) and (30) is nonnegative, we can read immediately from (30) that  $\lambda_3 \geq 2 > 0$ . We conclude from (27) that  $y = 0$  and from (30) that  $\lambda_3 = 2$ .

Next, we examine two cases, depending on the sign of  $x$ . If  $x = 0$ , then  $\lambda_1 = 0$  by (26) and the fact that  $y = 0$ , and  $\lambda_2 = 0$  by (24). Thus,  $(x, y) = (0, 0)$  satisfies the first order conditions (24) to (28), and  $f(0, 0) = 0$ .

If  $x > 0$ , then  $\lambda_2 = 0$  by (27),  $\lambda_1 = 1$  by (29), and  $x^2 + y^2 = 1$  by (26). Since  $y = 0$ ,  $x^2 = 1$ ; since  $x \geq 0$ ,  $x = 1$ . The solution  $(x, y) = (1, 0)$  of the first order conditions leads to the value  $f(1, 0) = -1$  of the objective function. We conclude that  $(x, y) = (1, 0)$  minimizes  $f(x, y) = 2y - x^2$  on the constraint set (23).

## EXERCISES

**18.16** Check that the NDCQ are satisfied in Example 18.11.

**18.17** Minimize  $x^2 - 2y$  subject to the constraints  $x^2 + y^2 \leq 1$ ,  $x \geq 0$ ,  $y \geq 0$ .

**18.18** Minimize  $2x^2 + 2y^2 - 2xy - 9y$  on the constraint set

$$4x + 3y \leq 10, \quad y - 4x^2 \geq -2, \quad x \geq 0, \quad y \geq 0.$$

This is a rather complex problem, so we provide a few hints for its solutions. 1) As usual, first write out the Lagrangian and the complete set of first order conditions. 2) Rewrite the first two equations of the first order conditions without minus signs. 3) Examine each of the following three cases:  $x = 0$ ,  $y = 0$ ;  $x > 0$ ,  $y = 0$ ; and  $x = 0$ ,  $y > 0$ , separately and show that each leads to a violation of the first order conditions. 4) Conclude that  $x > 0$  and  $y > 0$  and that each of their multipliers  $\lambda_3$  and  $\lambda_4$  are zero. 5) Look at four cases according to the positivity of  $\lambda_1$  and  $\lambda_2$ . Show that each of the three cases:  $\lambda_1 = 0$ ,  $\lambda_2 = 0$ ;  $\lambda_1 = 0$ ,  $\lambda_2 > 0$ ; and  $\lambda_1 > 0$ ,  $\lambda_2 > 0$  leads to a contradiction of the first order conditions. 6) Compute the solution that corresponds to  $\lambda_1 > 0$  and  $\lambda_2 = 0$ .

**18.19** Present a geometric proof that in the problem of minimizing  $f(x, y)$  on the constraint set  $g(x, y) \geq b$ , the gradient of  $f$  and the gradient of  $g$  point in the *same* direction at a minimizer for which the constraint is binding.

## 18.6 KUHN-TUCKER FORMULATION

The most common constrained maximization problems in economics involve only inequality constraints and a complete set of nonnegativity constraints:

$$\begin{aligned} &\text{maximize} && f(x_1, \dots, x_n) \\ &\text{subject to} && g_1(x_1, \dots, x_n) \leq b_1, \dots, g_k(x_1, \dots, x_n) \leq b_k, \\ &&& x_1 \geq 0, \dots, x_n \geq 0. \end{aligned} \tag{31}$$

Notice that we have separated the nonnegativity constraints from the rest of the inequality constraints. This situation is so common that a special Lagrangian has been devised to handle it. Since the approach we are about to describe goes back to the pioneering work of Harold Kuhn and A. W. Tucker on inequality constrained maximization problems, it is known as the **Kuhn-Tucker formulation**.

We will assume throughout this section that the usual NDCQ holds at the solution  $\mathbf{x}^*$  of problem (31). If we use the techniques of Section 18.3 to solve problem (31), we would write the Lagrangian as

$$\begin{aligned} L(\mathbf{x}, \lambda_1, \dots, \lambda_k, \nu_1, \dots, \nu_n) \\ = f(\mathbf{x}) - \lambda_1 \cdot [g_1(\mathbf{x}) - b_1] - \dots - \lambda_k \cdot [g_k(\mathbf{x}) - b_k] + \nu_1 x_1 + \dots + \nu_n x_n. \end{aligned}$$

The corresponding first order conditions are

$$\begin{aligned} \frac{\partial L}{\partial x_1} &= \frac{\partial f}{\partial x_1} - \lambda_1 \frac{\partial g_1}{\partial x_1} - \dots - \lambda_k \frac{\partial g_k}{\partial x_1} + \nu_1 = 0, \\ &\vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \\ \frac{\partial L}{\partial x_n} &= \frac{\partial f}{\partial x_n} - \lambda_n \frac{\partial g_1}{\partial x_n} - \dots - \lambda_k \frac{\partial g_k}{\partial x_n} + \nu_n = 0, \end{aligned} \tag{32}$$

$$\begin{aligned} \lambda_1 (g_1(\mathbf{x}) - b_1) &= -\lambda_1 \frac{\partial L}{\partial \lambda_1} = 0 \\ &\vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \\ \lambda_k (g_k(\mathbf{x}) - b_k) &= -\lambda_k \frac{\partial L}{\partial \lambda_k} = 0, \end{aligned} \tag{33}$$

$$\begin{aligned} \nu_1 x_1 &= 0 \\ &\vdots \qquad \qquad \qquad \vdots \\ \nu_n x_n &= 0, \end{aligned} \tag{34}$$

$$\lambda_1, \dots, \lambda_k, \nu_1, \dots, \nu_n \geq 0,$$

plus the inequalities in (31).

Kuhn and Tucker worked with a Lagrangian  $\tilde{L}$  which did *not* include the nonnegativity constraints:

$$\tilde{L}(\mathbf{x}, \lambda_1, \dots, \lambda_k) \equiv f(\mathbf{x}) - \lambda_1 \cdot [g_1(\mathbf{x}) - b_1] - \dots - \lambda_k \cdot [g_k(\mathbf{x}) - b_k]. \tag{35}$$

We will call  $\tilde{L}$  the **Kuhn-Tucker Lagrangian**. Note that

$$L(\mathbf{x}, \lambda_1, \dots, \lambda_k, \nu_1, \dots, \nu_n) = \tilde{L}(\mathbf{x}, \lambda_1, \dots, \lambda_k) + \nu_1 x_1 + \dots + \nu_n x_n.$$

For  $j = 1, \dots, n$ , rewrite (32) as

$$\frac{\partial \tilde{L}}{\partial x_j} = \frac{\partial \tilde{L}}{\partial x_j} + \nu_j = 0, \quad (36)$$

$$\text{or} \quad \frac{\partial \tilde{L}}{\partial x_j} = -\nu_j \quad (37)$$

at the solution for each  $j$ . By (34), (37), and the equations  $\nu_j \geq 0$ ,

$$\frac{\partial \tilde{L}}{\partial x_j} \leq 0 \quad \text{and} \quad x_j \frac{\partial \tilde{L}}{\partial x_j} = 0. \quad (38)$$

On the other hand, for any  $\mathbf{x}$ ,

$$\frac{\partial \tilde{L}}{\partial \lambda_j} = \frac{\partial L}{\partial \lambda_j} = b_j - g_j(\mathbf{x}) \geq 0. \quad (39)$$

Combining (34), (38), and (39), we find that the first order conditions in terms of the Kuhn-Tucker Lagrangian are

$$\begin{aligned} \frac{\partial \tilde{L}}{\partial x_1} \leq 0, \dots, \quad \frac{\partial \tilde{L}}{\partial x_n} \leq 0, \quad \frac{\partial \tilde{L}}{\partial \lambda_1} \geq 0, \dots, \quad \frac{\partial \tilde{L}}{\partial \lambda_k} \geq 0, \\ x_1 \frac{\partial \tilde{L}}{\partial x_1} = 0, \dots, x_n \frac{\partial \tilde{L}}{\partial x_n} = 0, \quad \lambda_1 \frac{\partial \tilde{L}}{\partial \lambda_1} = 0, \dots, \lambda_k \frac{\partial \tilde{L}}{\partial \lambda_k} = 0. \end{aligned} \quad (40)$$

This formulation has some advantages over the formulation of Section 18.3. First, it involves  $n + k$  equations in  $n + k$  unknowns, compared with the  $2n + k$  equations in  $2n + k$  unknowns in system (32) to (34). However, in solving a specific numerical problem, as in the exercises at the end of Section 18.3, it is often easier to work with the  $\nu_j$ 's in the original formulation. A more important advantage of (40) is the symmetric way that the  $x_i$ 's and  $\lambda_j$ 's enter the first order conditions. This approach leads naturally to considerations of the dual problem to (31) — the minimization problem in which the  $\lambda_j$ 's become the primary variables. The following theorem summarizes the observations of this section.

**Theorem 18.7** Consider the constrained maximization problem (31) with no equality constraints and with a complete set of nonnegativity constraints. Form the Kuhn-Tucker Lagrangian  $\tilde{L}$  as in (35). Suppose that  $\mathbf{x}^*$  is a solution of (31) and that the matrix  $(\partial g_i / \partial x_j)$  has maximal rank at  $\mathbf{x}^*$ , where the  $i$ 's vary over the indices of the  $g_i$ -constraints that are binding at  $\mathbf{x}^*$  and the  $j$ 's range over the indices  $j$  for which  $x_j^* > 0$ . Then, there exist nonnegative multipliers  $\lambda_1^*, \dots, \lambda_k^*$  such that  $x_1^*, \dots, x_n^*, \lambda_1^*, \dots, \lambda_k^*$  satisfies the system of equalities and inequalities (40).

**Example 18.12** In this framework, the Kuhn-Tucker Lagrangian for the usual utility maximization problem of Example 18.1 would be

$$\tilde{L}(x_1, x_2, \lambda) = U(x_1, x_2) - \lambda(p_1x_1 + p_2x_2 - I),$$

and the first order conditions are:

$$\begin{aligned} \frac{\partial U}{\partial x_1} - \lambda p_1 &\leq 0, & \frac{\partial U}{\partial x_2} - \lambda p_2 &\leq 0, \\ x_1 \cdot \left( \frac{\partial U}{\partial x_1} - \lambda p_1 \right) &= 0, & x_2 \cdot \left( \frac{\partial U}{\partial x_2} - \lambda p_2 \right) &= 0, \\ \frac{\partial \tilde{L}}{\partial \lambda} = I - p_1x_1 - p_2x_2 &\geq 0, & \lambda \frac{\partial \tilde{L}}{\partial \lambda} = \lambda(I - p_1x_1 - p_2x_2) &= 0. \end{aligned}$$

## EXERCISES

- 18.20** Show that the usual NDCQ as applied to problem (31) leads to the constraint qualification in the statement of Theorem 18.7.
- 18.21** Write out the Kuhn-Tucker formulation for a constrained *minimization* problem.

## 18.7 EXAMPLES AND APPLICATIONS

### Application: A Sales-Maximizing Firm with Advertising

Programming principles are used to determine the effect of such items as advertising, sales maximization, taxes, and government constraints on the optimal behavior of a firm. We close this chapter with two examples of such studies.

Suppose that a firm has its policies determined by a manager whose objective function is to maximize sales, i.e., revenue, without letting profit drop below some fixed level. To spice up the problem, let's add an advertising cost  $a \in \mathbf{R}_+$ . Let  $R(y, a)$  denote the firm's revenue when the level of production is  $y \in \mathbf{R}_+$  and the advertising cost is  $a \in \mathbf{R}_+$ . Let  $C(y)$  denote the cost of manufacturing  $y$  units of output. We assume that  $C$  and  $R$  are  $C^1$  functions,  $C' > 0$ , and  $\partial R / \partial a > 0$ . Our programming problem is to maximize  $R(y, a)$  subject to the constraints

$$\Pi \equiv R(y, a) - C(y) - a \geq m, \quad y \geq 0, \quad a \geq 0.$$

Assume that  $(y^*, a^*)$  is an optimal solution with  $y^* > 0$ . Form the Lagrangian

$$L(y, a; \lambda_1, \lambda_2) \equiv R(y, a) + \lambda_1 a - \lambda_2(m - R(y, a) + C(y) + a).$$

Under the assumption that the NDCQ holds at  $(y^*, a^*)$ , the first order conditions are

$$\frac{\partial L}{\partial y} = (1 + \lambda_2) \frac{\partial R}{\partial y} - \lambda_2 C'(y) = 0, \quad (41)$$

$$\frac{\partial L}{\partial a} = (1 + \lambda_2) \frac{\partial R}{\partial a} + \lambda_1 - \lambda_2 = 0, \quad (42)$$

$$\lambda_1 a = 0, \quad \text{and} \quad \lambda_2(m - R(y, a) + C(y) + a) = 0. \quad (43)$$

Since  $\partial R/\partial a > 0$  and  $\lambda_2 \geq 0$  in (42),  $\lambda_1 - \lambda_2 < 0$ . Since  $\lambda_1 \geq 0$ ,  $\lambda_2$  must be strictly positive. Therefore,  $\Pi(y^*, a^*) = m$ ; *the profit realized is the minimal profit allowed.*

Since  $\lambda_2 > 0$  and  $C'(y^*) > 0$  in (41),  $\partial R/\partial y > 0$ , and the marginal revenue is positive at the optimal level. On the other hand, the marginal profit  $\partial \Pi/\partial y$  is negative at  $(y^*, a^*)$ , since by (41),

$$\begin{aligned} (1 + \lambda_2) \frac{\partial \Pi}{\partial y}(y^*, a^*) &= (1 + \lambda_2) \left( \frac{\partial R^*}{\partial y} - C'(y) \right) \\ &= \frac{\partial L}{\partial y}(y^*, a^*) - C'(y^*) \\ &= 0 - C'(y^*) \\ &< 0. \end{aligned}$$

Consequently, *output  $y^*$  is greater than the output in the profit-maximizing situation.*

### Application: The Averch-Johnson Effect

Consider the effect of a fair-rate-of-return regulatory constraint on the behavior of a monopolist firm that uses capital and labor to produce a single commodity. In this case, the price charged by the firm is subject to public control in that a regulatory agency ensures that, after the firm subtracts its operating expenses from its revenues, the remaining net revenue should just suffice to compensate the firm for its investment in plant and equipment. This example is due to H. Averch and L. Johnson (see chapter Notes), who applied their analysis to the Federal Communications Commission regulation of the telephone and telegraph industry.

Let  $y = f(x_1, x_2)$  denote the amount of output produced from  $x_1$  units of capital and  $x_2$  units of labor. Suppose that production requires positive amounts of both capital and labor in that  $y = 0$  if  $x_1 = 0$  or  $x_2 = 0$ . Let  $p(y)$  denote the inverse demand function and  $R(y) = p(y)y$  the revenue from the sales of  $y$  units of output. Let  $r_1$  denote the interest cost involved in holding one unit of plant and

equipment, and let  $c_1$  denote the acquisition cost per unit of plant and material. Let  $r_2$  denote the wages paid per unit of labor. For simplicity, choose units so that  $c_1 = 1$ .

Now  $c_1x_1$  is the value of capital used in production,  $r_2x_2$  is the total cost of labor, and  $(R(y) - r_2x_2)/c_1x_1$  is the rate of return on plant and equipment. The regulatory constraint is expressed by the fact that there is a constant  $s_1 > 0$  such that

$$\frac{R(y) - r_2x_2}{c_1x_1} \leq s_1. \quad (44)$$

Assuming that the firm's objective is to maximize profit, the firm faces the following optimization problem:

$$\begin{aligned} &\text{maximize} && \pi = R(f(x_1, x_2)) - r_1x_1 - r_2x_2 \\ &\text{subject to} && R(f(x_1, x_2)) - r_2x_2 - s_1x_1 \leq 0. \\ &&& x_1 \geq 0, \quad x_2 \geq 0 \end{aligned} \quad (45)$$

We write  $R^*(x_1, x_2)$  for  $R(f(x_1, x_2))$ , the firm's revenue as a function of its inputs. We will assume that  $R^*$  has sufficient regularity so that the NDCQ is satisfied at solutions of (45). The Lagrangian for problem (45) is

$$L = R^*(x_1, x_2) - r_1x_1 - r_2x_2 - \lambda(R^*(x_1, x_2) - r_2x_2 - s_1x_1) + \nu_1x_1 + \nu_2x_2.$$

Necessary conditions for optimization are that there exist  $\lambda$ ,  $\nu_1$ ,  $\nu_2$  all nonnegative such that

$$\frac{\partial L}{\partial x_1} = (1 - \lambda) \frac{\partial R^*}{\partial x_1} - r_1 + \lambda s_1 + \nu_1 = 0, \quad (46)$$

$$\frac{\partial L}{\partial x_2} = (1 - \lambda) \frac{\partial R^*}{\partial x_2} - (1 - \lambda)r_2 + \nu_2 = 0, \quad (47)$$

$$\lambda(R^*(x_1, x_2) - r_2x_2 - s_1x_1) = 0, \quad \nu_1x_1 = 0, \quad \text{and} \quad \nu_2x_2 = 0. \quad (48)$$

We first claim that the allowable rate of return  $s_1$  must be greater than the cost of capital  $r_1$  in order for the firm to realize any profit. Suppose  $s_1 \leq r_1$ . Then,

$$\begin{aligned} \pi &= R^*(x_1, x_2) - r_1x_1 - r_2x_2 \\ &= (R^*(x_1, x_2) - s_1x_1 - r_2x_2) + (s_1 - r_1)x_1 \\ &\leq 0 + (s_1 - r_1)x_1 \quad (\text{by (44)}) \\ &\leq 0. \end{aligned}$$

Thus, we can assume that  $s_1 > r_1$  and that  $x_1$  and  $x_2$  are positive, in which case,  $v_1 = v_2 = 0$ . This also means that  $\lambda \neq 1$  in (46). We claim that  $\lambda$  must be less than 1. To see this, choose  $s_1$  so large that the regulatory constraint is ineffective, that is,  $\lambda = 0$ . As  $s_1$  tends back toward  $r_1$ ,  $\lambda$  varies continuously, but it can never reach 1. Consequently,  $0 \leq \lambda < 1$ .

Since  $v_2 = 0$  and  $1 - \lambda \neq 0$  in (47),

$$\frac{\partial(p(y)y)}{\partial x_2} = r_2,$$

as in the unregulated situation. By (46),

$$\begin{aligned} \frac{\partial(p(y)y)}{\partial x_1} &\leq \frac{r_1 - \lambda s_1}{1 - \lambda} \\ &= r_1 - \frac{(1 - \lambda)}{(1 - \lambda)} r_1 + \frac{r_1 - \lambda s_1}{1 - \lambda} \\ &= r_1 - \frac{\lambda(s_1 - r_1)}{1 - \lambda}. \end{aligned}$$

If  $\lambda > 0$  so that the regulatory constraint is binding,  $\lambda/(1 - \lambda) > 0$ . Since  $s_1 - r_1 > 0$  too,

$$r_1 > \frac{\partial(p(y)y)}{\partial x_1}$$

at the maximizer. Therefore, input of capital is such that its marginal cost is greater than its marginal revenue product. The firm substitutes capital for labor and operates at a level of output where cost is not minimized. Thus, the regulatory constraint distorts the efficient relative proportions of capital and labor that would be used in the absence of the regulatory constraint.

### One More Worked Example

We close this chapter on first order conditions by working out the solution of a rather complex constrained maximization problem with one inequality constraint and two nonnegativity constraints.

**Example 18.13** Consider the problem of maximizing  $f(x, y) = x^2 + x + 4y^2$  subject to the inequality constraints

$$2x + 2y \leq 1, \quad x \geq 0, \quad \text{and} \quad y \geq 0.$$



The Jacobian of the constraint functions is

$$\begin{pmatrix} 2 & 2 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

At most two constraints can be binding at the same time, and any  $2 \times 2$  submatrix of this Jacobian has rank two. Therefore, the NDCQ will hold at any solution candidate. Form the Lagrangian

$$L(x, y, \lambda_1, \lambda_2, \lambda_3) = x^2 + x + 4y^2 - \lambda_1(2x + 2y - 1) + \lambda_2x + \lambda_3y,$$

where we treat the nonnegativity constraints as in the previous example. Next, write the first order conditions:

$$1) \quad \frac{\partial L}{\partial x} = 2x + 1 - 2\lambda_1 + \lambda_2 = 0,$$

$$2) \quad \frac{\partial L}{\partial y} = 8y - 2\lambda_1 + \lambda_3 = 0,$$

$$3) \quad \lambda_1(2x + 2y - 1) = 0, \quad 4) \quad \lambda_2x = 0, \quad 5) \quad \lambda_3y = 0,$$

$$6) \quad \lambda_1 \geq 0, \quad 7) \quad \lambda_2 \geq 0, \quad 8) \quad \lambda_3 \geq 0,$$

$$9) \quad 2x + 2y \leq 1, \quad 10) \quad x \geq 0, \quad 11) \quad y \geq 0.$$

Rewrite condition 1 without minus signs as  $2x + 1 + \lambda_2 = 2\lambda_1$ . Along with conditions 7 and 10, this equation implies that  $2\lambda_1 \geq 1 > 0$ , and this, by condition 3, implies that the first constraint is binding:

$$2x + 2y = 1. \quad (49)$$

Now we need to examine a few cases. Let's first look at the case  $\lambda_2 > 0$ . In this case, it follows from condition 4 that  $x = 0$ , from equation (49) that  $y = 0.5$ , and from condition 5 that  $\lambda_3 = 0$ . Plugging  $y = 0.5$  and  $\lambda_3 = 0$  into condition 2 yields  $\lambda_1 = 2$ . Plugging  $x = 0$  and  $\lambda_1 = 2$  into condition 1 yields  $\lambda_2 = 3$ . So, the assumption that  $\lambda_2 > 0$  leads to the candidate

$$(x, y, \lambda_1, \lambda_2, \lambda_3) = (0, 0.5, 2, 3, 0).$$

Now, let's try the opposite case:  $\lambda_2 = 0$ . Combining conditions 1 and 2, we find

$$2x + 1 + \lambda_2 = 8y + \lambda_3 = 2\lambda_1. \quad (50)$$

Substituting  $\lambda_2 = 0$  and  $2x = 1 - 2y$  from equation (49), we find

$$1 - 2y + 1 = 8y + \lambda_3, \quad \text{or} \quad 2 = 10y + \lambda_3.$$

Combining the latter equation with condition 5 leads to the conclusion that either  $y = 0$  or  $\lambda_3 = 0$ , but not both. If  $y = 0$ , then  $\lambda_3 = 2$ ; by equation (49),  $x = 0.5$ ; and by equation (50),  $\lambda_1 = 1$ . If, instead,  $\lambda_3 = 0$ , then  $y = 0.2$ ,  $x = 0.3$ , and  $\lambda_1 = 0.8$ . Consequently, the case  $\lambda_2 = 0$  leads to the two candidates

$$(x, y, \lambda_1, \lambda_2, \lambda_3) = \begin{cases} (0.5, 0, 1, 0, 2) & \text{or} \\ (0.3, 0.2, 0.8, 0, 0). \end{cases}$$

By evaluating the objective function at each of these three candidates, we find that the constrained maximum occurs at the point  $x = 0$ ,  $y = 0.5$ , where  $\lambda_1 = 2$ ,  $\lambda_2 = 3$ , and  $\lambda_3 = 0$ .

## NOTES

See C. Simon, "Scalar and Vector Maximization: Calculus Techniques with Economic Applications," in (S. Reiter, ed.) *Studies in Mathematical Economics*, (Washington, D.C.: Mathematical Association of America, 1986), pp. 62–159, for an earlier version of the material in this chapter and Chapters 19, 21, and 22. For further discussion on the sales maximization application in Section 18.7, see W. J. Baumol, *Economic Theory and Operations Research* (Englewood Cliffs, N.J.: Prentice Hall, 1961); and H. Kuhn, "Lectures on mathematical economics," in G. Dantzig and A. Veinott (eds.), *Mathematics of the Decision Sciences*, American Math. Soc. Lectures in Applied Math., vol. 12, Providence, RI, 1968, pp. 49–84. The Averch-Johnson analysis is presented in H. Averch and L. Johnson, "Behavior of the firm under regulatory constraint," *American Economic Review* 52 (1962), 1052–1069. See E. E. Zajac, "Lagrange Multiplier Values at Constrained Optima," *Journal of Economic Theory* 4 (1972), 125–131, for a more careful and detailed argument that  $0 \leq \lambda < 1$  in the Averch-Johnson example.

# Constrained Optimization II

This chapter continues our presentation of the central mathematical technique in economic theory: the solution of constrained optimization problems. The last chapter introduced the Lagrangian formulation of that solution and focused on its most important aspect: the first order conditions that form the basis of a large number of economic principles. This chapter looks at three other aspects of the Lagrangian approach:

- (1) the sensitivity of the optimal value of the objective function to changes in the parameters of the problem,
- (2) the second order conditions that distinguish maxima from minima, and
- (3) the constraint qualifications that are a subtle but necessary hypothesis in the Lagrangian approach.

The last section of this chapter contains careful proofs of the basic first order conditions of the last chapter.

## 19.1 THE MEANING OF THE MULTIPLIER

In solving constrained optimization problems, we seem to be deriving extraneous information in the values of the multipliers  $(\lambda_1^*, \dots, \lambda_m^*)$ . However, the multipliers play an important role in economic analysis — in some problems, a role at least as important as that of the maximizer itself. We will see in this section that the multipliers measure the sensitivity of the optimal value of the objective function to changes in the right-hand sides of the constraints and, as a result, they provide a natural measure of value for scarce resources in economics maximization problems.

## One Equality Constraint

We return to the simplest problem — two variables and one equality constraint:

$$\begin{aligned} &\text{maximize} && f(x, y) \\ &\text{subject to} && h(x, y) = a \end{aligned} \quad (1)$$

We will consider  $a$  as a parameter which varies from problem to problem. For any fixed value of  $a$ , write  $(x^*(a), y^*(a))$  for the solution to problem (1), and write  $\mu^*(a)$  for the multiplier which corresponds to this solution. Let  $f(x^*(a), y^*(a))$  be the corresponding *optimal value* of the objective function. We prove that under reasonable conditions which hold for nearly all constrained maximization problems,  $\mu^*(a)$  measures the rate of change of the optimal value of  $f$  with respect to the parameter  $a$ , or roughly speaking, the (infinitesimal) effect of a unit increase in  $a$  on  $f(x^*(a), y^*(a))$ .

**Theorem 19.1** Let  $f$  and  $h$  be  $C^1$  functions of two variables. For any fixed value of the parameter  $a$ , let  $(x^*(a), y^*(a))$  be the solution of problem (1) with corresponding multiplier  $\mu^*(a)$ . Suppose that  $x^*$ ,  $y^*$ , and  $\mu^*$  are  $C^1$  functions of  $a$  and that NDCQ holds at  $(x^*(a), y^*(a), \mu^*(a))$ . Then,

$$\mu^*(a) = \frac{d}{da} f(x^*(a), y^*(a)). \quad (2)$$

*Proof* The Lagrangian for problem (1) is

$$L(x, y, \mu; a) \equiv f(x, y) - \mu(h(x, y) - a), \quad (3)$$

with  $a$  entering as a parameter. By Theorem 18.1, the solution  $(x^*(a), y^*(a), \mu^*(a))$  of (1) satisfies

$$\begin{aligned} 0 &= \frac{\partial L}{\partial x}(x^*(a), y^*(a), \mu^*(a); a) \\ &= \frac{\partial f}{\partial x}(x^*(a), y^*(a), \mu^*(a)) - \mu^*(a) \frac{\partial h}{\partial x}(x^*(a), y^*(a), \mu^*(a)), \\ 0 &= \frac{\partial L}{\partial y}(x^*(a), y^*(a), \mu^*(a); a) \\ &= \frac{\partial f}{\partial y}(x^*(a), y^*(a), \mu^*(a)) - \mu^*(a) \frac{\partial h}{\partial y}(x^*(a), y^*(a), \mu^*(a)), \end{aligned} \quad (4)$$

for all  $a$ . Furthermore, since  $h(x^*(a), y^*(a)) = a$  for all  $a$ ,

$$\frac{\partial h}{\partial x}(x^*, y^*) \frac{dx^*}{da}(a) + \frac{\partial h}{\partial y}(x^*, y^*) \frac{dy^*}{da}(a) = 1 \quad (5)$$

for every  $a$ . Therefore, using the Chain Rule and equations (4) and (5),

$$\begin{aligned}
 \frac{d}{da} f(x^*(a), y^*(a)) &= \frac{\partial f}{\partial x}(x^*(a), y^*(a)) \frac{dx^*}{da}(a) + \frac{\partial f}{\partial y}(x^*(a), y^*(a)) \frac{dy^*}{da}(a) \\
 &= \mu^* \frac{\partial h}{\partial x}(x^*(a), y^*(a)) \frac{dx^*}{da}(a) + \mu^* \frac{\partial h}{\partial y}(x^*(a), y^*(a)) \frac{dy^*}{da}(a) \\
 &= \mu^* \left[ \frac{\partial h}{\partial x}(x^*(a), y^*(a)) \frac{dx^*}{da}(a) + \frac{\partial h}{\partial y}(x^*(a), y^*(a)) \frac{dy^*}{da}(a) \right] \\
 &= \mu^* \cdot 1. \quad \blacksquare
 \end{aligned}$$

**Example 19.1** In Example 18.5, we found that a maximizer of  $f(x_1, x_2) = x_1^2 x_2$  on the constraint set  $2x_1^2 + x_2^2 = 3$  is  $x_1 = 1, x_2 = 1$ , with multiplier  $\mu = 0.5$ . The maximum value of  $f$  is  $f^* = f(1, 1) = 1$ . Redo the problem, this time using the constraint  $2x_1^2 + x_2^2 = 3.3$ . The same computation as in Example 18.5 yields the solution  $x_1 = x_2 = \sqrt{1.1}$ , with maximum value  $f^* = (1.1)^{3/2} \approx 1.1537$ , an increase of 0.1537 over the original  $f^*$ .

On the other hand, Theorem 19.1 predicts that changing the right-hand side of the constraint by 0.3 unit would change the maximum value of the objective function by roughly

$$0.3 \cdot \mu = 0.3 \cdot 0.5 = 0.15 \text{ unit,}$$

an approximation correct to two decimal places.

## Several Equality Constraints

The statement and proof of the natural generalization of Theorem 19.1 to several variables and several equality constraints is straightforward.

**Theorem 19.2** Let  $f, h_1, \dots, h_m$  be  $C^1$  functions on  $\mathbf{R}^n$ . Let  $\mathbf{a} = (a_1, \dots, a_m)$  be an  $m$ -tuple of exogenous parameters, and consider the problem  $(P_{\mathbf{a}})$  of maximizing  $f(x_1, \dots, x_n)$  subject to the constraints

$$h_1(x_1, \dots, x_n) = a_1, \dots, h_m(x_1, \dots, x_n) = a_m.$$

Let  $x_1^*(\mathbf{a}), \dots, x_n^*(\mathbf{a})$  denote the solution of problem  $(P_{\mathbf{a}})$ , with corresponding Lagrange multipliers  $\mu_1^*(\mathbf{a}), \dots, \mu_m^*(\mathbf{a})$ . Suppose further that the  $x_i^*$ 's and  $\mu_j^*$ 's are differentiable functions of  $(a_1, \dots, a_m)$  and that NDCQ holds. Then, for each  $j = 1, \dots, m$ ,

$$\mu_j^*(a_1, \dots, a_m) = \frac{\partial}{\partial a_j} f(x_1^*(a_1, \dots, a_m), \dots, x_n^*(a_1, \dots, a_m)). \quad (6)$$

## Inequality Constraints

Theorems 19.1 and 19.2 hold equally well for inequality constraints, as the following theorem indicates. For ease of exposition, we will assume that all the constraints under consideration are inequality constraints. The statement of the corresponding result for mixed constraints is straightforward.

**Theorem 19.3** Let  $\mathbf{a}^* = (a_1^*, \dots, a_k^*)$  be a  $k$ -tuple. Consider the problem  $(Q_{\mathbf{a}^*})$  of maximizing  $f(x_1, \dots, x_n)$  subject to the  $k$  inequality constraints

$$g_1(x_1, \dots, x_n) \leq a_1^*, \dots, g_k(x_1, \dots, x_n) \leq a_k^*. \quad (7)$$

Let  $x_1^*(\mathbf{a}^*), \dots, x_n^*(\mathbf{a}^*)$  denote the solution of problem  $(Q_{\mathbf{a}^*})$ , and let  $\lambda_1^*(\mathbf{a}^*), \dots, \lambda_k^*(\mathbf{a}^*)$  be the corresponding Lagrange multipliers. Suppose that as  $\mathbf{a}$  varies near  $\mathbf{a}^*$ ,  $x_1^*, \dots, x_n^*$ , and  $\lambda_1^*, \dots, \lambda_k^*$  are differentiable functions of  $(a_1, \dots, a_k)$  and that the NDCQ holds at  $\mathbf{a}^*$ . Then, for each  $j = 1, \dots, k$ ,

$$\lambda_j^*(a_1^*, \dots, a_k^*) = \frac{\partial}{\partial a_j} f(x_1^*(a_1^*, \dots, a_k^*), \dots, x_n^*(a_1^*, \dots, a_k^*)). \quad (8)$$

*Proof (Sketch)* For ease of notation, we write  $\mathbf{a}^*$  simply as  $\mathbf{a}$ . As usual, we break the inequality constraints into two groups: the binding ones and the nonbinding ones. The binding constraints can be treated as equality constraints, and so we apply Theorem 19.2 to them. Let  $g_j$  be the constraint function for one of the nonbinding constraints:  $g_j(\mathbf{x}^*(\mathbf{a})) < a_j$ . Let  $C$  be the constraint set described by the inequalities (7). Let  $a_j'$  be any number such that

$$g_j(\mathbf{x}^*(\mathbf{a})) < a_j' < a_j;$$

and let  $C'$  be the constraint set described by inequalities (7) with  $g_j(\mathbf{x}) \leq a_j'$  replacing  $g_j(\mathbf{x}) \leq a_j$ .

Since  $\mathbf{x}^*(\mathbf{a})$  maximizes  $f$  on  $C$ , since  $C' \subset C$ , and since  $\mathbf{x}^*(\mathbf{a}) \in C'$ , it follows that  $\mathbf{x}^*(\mathbf{a})$  maximizes  $f$  on  $C'$ . In other words, if

$$\mathbf{a}' = (a_1, \dots, a_{j-1}, a_j', a_{j+1}, \dots, a_k),$$

then  $\mathbf{x}^*(\mathbf{a}') = \mathbf{x}^*(\mathbf{a})$  and therefore  $f(\mathbf{x}^*(\mathbf{a}')) = f(\mathbf{x}^*(\mathbf{a}))$ ; that is, varying  $a_j$  a bit does not affect the maximum value of  $f$ . This implies that

$$\frac{\partial}{\partial a_j} f(x_1^*(a_1, \dots, a_m), \dots, x_n^*(a_1, \dots, a_m)) = 0.$$

Since  $\lambda_j^*(\mathbf{a})$  is zero too by Theorem 18.4, equation (8) still holds for nonbinding inequality constraints. ■

**Example 19.2** In Example 18.9, we computed that the max of  $xyz$  on the set

$$x + y + z \leq 1, \quad x \geq 0, \quad y \geq 0, \quad z \geq 0$$

occurs at  $x = y = z = 1/3$ , where  $xyz = 1/27$ . The four multipliers are  $1/9$ ,  $0$ ,  $0$ , and  $0$ , respectively.

- (a) If we change the first constraint to  $x + y + z \leq 0.9$ , we compute that the solution occurs at  $x = y = z = 0.3$ , where  $xyz = 0.027$ . Theorem 19.3 predicts that the new optimal value would be

$$\frac{1}{27} + \frac{1}{9} \cdot \left(-\frac{1}{10}\right) \approx 0.0259,$$

an estimate that is off by only .0011 or four percent.

- (b) If, instead, we change the second constraint from  $x \geq 0$  to  $x \geq 0.1$ , we do not change the solution or the optimum value because the new region is a subset of the old region and it still contains the optimal point for the old region. This result is consistent with Theorem 19.3 since the multiplier for the (nonbinding) constraint  $x \geq 0$  was zero.

## Interpreting the Multiplier

Finally, we consider the central role that equation (8) plays in economic theory. Think of the objective function  $f(\mathbf{x})$  in problem  $(Q_a)$  of Theorem 19.3 as the profit function of a firm and think of the  $a_j$ 's on the right-hand sides of the constraints as representing the amounts available for inputs in the firm's production process. For example, in an activity analysis problem, we suppose that the firm's production process consists of  $n$  different productive activities and that  $x_i$  represents the level of intensity of activity  $i$ . Let  $g_j(x_1, \dots, x_n)$  denote the amount of input  $j$  that the firm requires to run activity one at level  $x_1$ , activity two at level  $x_2$ , and so forth. Let  $a_j$  denote the amount of input  $j$  available to the firm, leading to the inequality constraints  $g_j(x_1, \dots, x_n) \leq a_j$ . Let  $f(x_1, \dots, x_n)$  denote the profits that the firm realizes from its products when it runs the activities at levels  $x_1, \dots, x_n$ , respectively. In this situation,

$$\frac{\partial}{\partial a_j} f(x_1^*(\mathbf{a}), \dots, x_n^*(\mathbf{a}))$$

represents the change in the optimal profit resulting from the availability of one more unit of input  $j$ . By (8), the  $j$ th multiplier  $\lambda_j^*(\mathbf{a})$  represents this infinitesimal change. It tells how valuable another unit of input  $j$  would be to the firm's profits. Alternatively, it tells the maximum amount the firm would be willing to pay to acquire another unit of input  $j$ . For this reason,  $\lambda_j^*(\mathbf{a})$  is often called the **internal value** or **imputed value**, or more frequently, the **shadow price** of input  $j$ . It may be a more important index to the firm than the external market price of input  $j$ .

## EXERCISES

- 19.1 a) Find the maximum and minimum distance from the origin to the ellipse  $x^2 + xy + y^2 = 3.3$ .  
 b) Use Theorem 19.1 and your answer to Exercise 18.2 to estimate the answers to part a.
- 19.2 Find the maximum of  $x + y + z^2$  subject to the constraints  $x^2 + y^2 + z^2 = 0.8$ ,  $y = 0$ :  
 a) by using Theorem 19.1 and Exercise 18.6,  
 b) by doing the calculation from scratch.
- 19.3 If  $x$  thousand dollars is spent on labor and  $y$  thousand dollars is spent on equipment, a certain factory produces  $Q(x, y) = 50x^{1/2}y^2$  units of output.  
 a) How should \$80,000 be allocated between labor and equipment to yield the largest possible output?  
 b) Use Theorem 19.1 to estimate the change in maximum output if this allocation decreased by \$1000.  
 c) Compute the exact change in b).
- 19.4 Use Theorem 19.3 and Exercise 18.11 to estimate the maximum value of  $f(x, y) = 2y^2 - x$  on the constraint set  $x^2 + y^2 \leq 0.9$ ,  $x \geq 0$ , and  $y \geq 0$ .
- 19.5 Use Theorem 19.3 and Exercise 18.12 to estimate the maximum value of  $f(x, y, z) = xyz + z$  on the constraint set  $x^2 + y^2 + z \leq 6.2$ ,  $x \geq 0$ ,  $y \geq 0$ , and  $z \geq 0$ .
- 19.6 Use the Implicit Function Theorem to write out a specific inequality which would guarantee the validity of the assumption in Theorem 19.1 that the solution  $(x(a), y(a))$  of Problem (1) depends smoothly on  $a$ .
- 19.7 Prove Theorem 19.2.
- 19.8 Write out the statement of the theorem which corresponds to Theorems 19.2 and 19.3, but for both equality and inequality constraints.
- 19.9 Write out the statement of the theorem which corresponds to Theorems 19.2 and 19.3 for a constrained minimization problem.

## 19.2 ENVELOPE THEOREMS

Theorems 19.1, 19.2, and 19.3 are special cases of a class of theorems which describe how the optimal value of the objective function in a parameterized optimization problem changes as one of the parameters change. Such theorems are called **Envelope Theorems**. We begin with the Envelope Theorem for unconstrained problems.

## Unconstrained Problems

**Theorem 19.4** Let  $f(\mathbf{x}; a)$  be a  $C^1$  function of  $\mathbf{x} \in \mathbf{R}^n$  and the scalar  $a$ . For each choice of the parameter  $a$ , consider the unconstrained maximization problem

$$\text{maximize } f(\mathbf{x}; a) \quad \text{with respect to } \mathbf{x}. \quad (9)$$



Let  $\mathbf{x}^*(a)$  be a solution of this problem. Suppose that  $\mathbf{x}^*(a)$  is a  $C^1$  function of  $a$ . Then,

$$\frac{d}{da} f(\mathbf{x}^*(a); a) = \frac{\partial}{\partial a} f(\mathbf{x}^*(a); a). \quad (10)$$

*Proof* We compute via the Chain Rule that

$$\begin{aligned} \frac{d}{da} f(\mathbf{x}^*(a); a) &= \sum_i \frac{\partial f}{\partial x_i}(\mathbf{x}^*(a); a) \cdot \frac{dx_i^*}{da}(a) + \frac{\partial f}{\partial a}(\mathbf{x}^*(a); a) \cdot 1 \\ &= \frac{\partial f}{\partial a}(\mathbf{x}^*(a); a), \end{aligned}$$

since  $\frac{\partial f}{\partial x_i}(\mathbf{x}^*(a); a) = 0$  for  $i = 1, \dots, n$ , by the usual first order conditions from Chapter 17. ■

Notice the similarities between the proofs of Theorem 19.1 and Theorem 19.4. At first glance, conclusion (10) of Theorem 19.4 looks uninteresting because both sides of (10) seem so similar. But the *partial* derivative on the right-hand side of (10) is a lot easier to deal with than the *total* derivative on the left-hand side, as the following examples illustrate.

*Example 19.3* Consider the problem of maximizing

$$f(x; a) = -a^3 x^4 + 15x^3 - e^a x^2 + 17$$

around  $a = 1$ . Since  $f$  is a quartic polynomial in  $x$  with a negative leading coefficient when  $a = 1$ ,  $f(x) \rightarrow -\infty$  as  $x \rightarrow \pm\infty$ . Therefore,  $f$  does have a finite global maximizer  $x^*(a)$  for each value of  $a$  near 1. By (10),

$$\frac{d}{da} f(\mathbf{x}^*(a); a) = \frac{\partial}{\partial a} f(\mathbf{x}^*(a); a) = -3a^2 x^{*4} - e^a x^{*2},$$

which is negative at all  $a$  and all  $x \neq 0$ . So, even without solving for the optimal  $x^*(a)$ , we can tell that as  $a$  increases beyond 0,  $f(x^*(a); a)$  is a decreasing function of  $a$ . The peak of the graph of the function  $x \mapsto f(x; a)$  decreases as  $a$  increases.

*Example 19.4* What will be the effect of a unit increase in  $a$  on the maximum value of  $f(x; a) = -x^2 + 2ax + 4a^2$ , where we maximize  $f$  with respect to  $x$  for each  $a$ ? First, let's compute the answer directly. The equation for the maximizer of  $f$  is

$$f'(x) = -2x + 2a = 0;$$

so,  $x^*(a) = a$ . Plugging this into  $f(x; a)$  leads to

$$f(x^*(a); a) = f(a, a) = -a^2 + 2a \cdot a + 4a^2 = 5a^2, \quad (11)$$

which will increase at a rate of  $10a$  as  $a$  increases.

If, instead, we had applied the envelope theorem, we could have skipped step (11) and found directly that

$$\frac{df^*}{da} = \frac{\partial f}{\partial a}(x^*(a); a) = 2x + 8a = 10a,$$

since  $x^*(a) = a$ .

**Example 19.5** A Silicon Valley firm produces an output of microchips denoted by  $y$  and has a cost function  $c(y)$ , with  $c'(y) > 0$  and  $c''(y) > 0$ . Of the chips it produces, a fraction  $1 - \alpha$  are unavoidably defective and cannot be sold. Working chips can be sold at a price  $p$ , and the microchip market is highly competitive. How will an increase in production quality affect the firm's profit?

The firm's profit function is

$$\pi(p, \alpha) = \max_y [p\alpha y - c(y)],$$

where "max<sub>y</sub>" means the maximum value as  $y$  varies. The conditions on the cost function guarantee that there is a nonzero profit-maximizing output  $y^*(\alpha)$  which depends smoothly on  $\alpha$ . The derivative of optimal profit  $\pi$  with respect to  $\alpha$  is:

$$\frac{d\pi}{d\alpha} = \frac{\partial}{\partial \alpha}(p\alpha y - c(y)) = py > 0.$$

As one would suspect, increasing the fraction of nondefective chips will increase the firm's profit. Once again, we were able to determine this without actually solving for the optimal output.

Of course, Theorem 19.4 generalizes easily to the case where there is more than one parameter. One works with one parameter at a time and finds that

$$\frac{d}{da_i} f(x_1^*(\mathbf{a}), \dots, x_n^*(\mathbf{a}); a_1, \dots, a_k) = \frac{\partial}{\partial a_i} f(x_1^*(\mathbf{a}), \dots, x_n^*(\mathbf{a}); a_1, \dots, a_k).$$

## Constrained Problems

The most general envelope theorem deals with *constrained* problems in which there are parameters in both the objective function and in the constraints. For example, consider the problem of maximizing  $f(\mathbf{x}; a)$  subject to the constraints

$h_1(\mathbf{x}; a) = 0, \dots, h_k(\mathbf{x}; a) = 0$ . If  $f$  does not depend on  $a$  and if each  $h_i(\mathbf{x}; a)$  can be written as  $h_i(\mathbf{x}) - a$ , then we are back to the situation in Theorem 19.2. So the case under consideration is more general than the other two cases we have looked at. However, the answer is nearly as straightforward. As the next theorem indicates, the rate of change of  $f(\mathbf{x}^*(a); a)$  with respect to  $a$  equals the *partial* derivative with respect to  $a$ , not of  $f$ , but of the corresponding Lagrangian function.

**Theorem 19.5** Let  $f, h_1, \dots, h_k: \mathbf{R}^n \times \mathbf{R}^1 \rightarrow \mathbf{R}^1$  be  $C^1$  functions. Let  $\mathbf{x}^*(a) = (x_1^*(a), \dots, x_n^*(a))$  denote the solution of the problem of maximizing  $\mathbf{x} \mapsto f(\mathbf{x}; a)$  on the constraint set

$$h_1(\mathbf{x}; a) = 0, \dots, h_k(\mathbf{x}; a) = 0,$$

for any fixed choice of the parameter  $a$ . Suppose that  $\mathbf{x}^*(a)$  and the Lagrange multipliers  $\mu_1(a), \dots, \mu_k(a)$  are  $C^1$  functions of  $a$  and that the NDCQ holds. Then,

$$\frac{d}{da} f(\mathbf{x}^*(a); a) = \frac{\partial L}{\partial a}(\mathbf{x}^*(a), \mu(a); a), \quad (12)$$

where  $L$  is the natural Lagrangian for this problem.

Note, as in expression (10), that the left hand side of (12) is a total derivative while the right hand side is a partial derivative. The proof of Theorem 19.5 is similar to the proofs of Theorems 19.1 and Theorem 19.4 and will be left as an exercise.

**Example 19.6** Change the constraint in Example 18.7 from  $x^2 + y^2 \leq 1$  to  $x^2 + 1.1y^2 \leq 1$ , keeping the objective function  $f(x, y) = xy$ . If we write both constraints as  $x^2 + ay^2 \leq 1$ , the Lagrangian for the parameterized problem is

$$L(x, y, \lambda; a) = xy - \lambda(x^2 + ay^2 - 1).$$

The solution for the original ( $a = 1$ ) problem was  $x = y = 1/\sqrt{2}$ ,  $\lambda = 1/2$ . The Envelope Theorem tells us that as  $a$  changes from 1 to 1.1, the optimal value of  $f$  changes by approximately

$$\frac{\partial L}{\partial a} \left( \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{2}; 1 \right) \cdot (0.1).$$

Since

$$\frac{\partial L}{\partial a} = -\lambda y^2 = -\frac{1}{2} \cdot \left( \frac{1}{\sqrt{2}} \right)^2 = -\frac{1}{4},$$

the optimal value will decrease by approximately  $.1/4 = .025$  to  $.475$ . One can calculate directly that the solution to the new problem is  $x = 1/\sqrt{2}$ ,  $y = 1/\sqrt{2}$ , with maximum objective value of  $f$  approximately equal to  $0.4767$ .

All the theorems in the last two sections had two basic hypotheses: the smooth dependence of the maximizers and multipliers on the parameters and the nondegenerate constraint qualification (NDCQ). In Section 19.4, we will look at these hypotheses more carefully and restate them in terms of properties of the objective and constraint functions of the problem.

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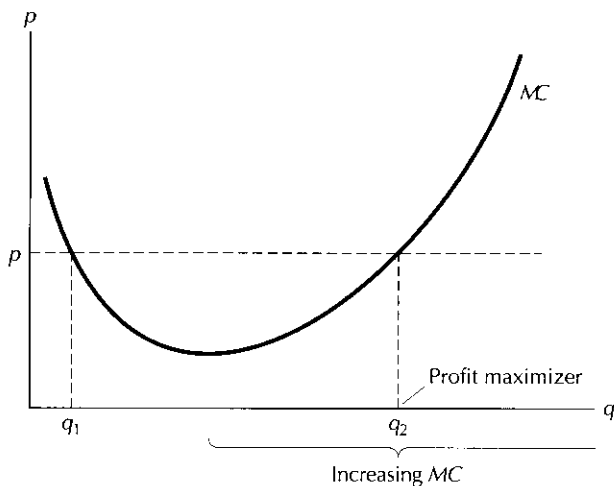
### EXERCISES

- 19.10** Write out a careful proof of Theorem 19.5.
- 19.11** Recover the statement of the Lagrange Multiplier Theorem (Theorem 19.1) from the statement of Theorem 19.5.
- 19.12** Use Exercise 18.2 and the Envelope Theorem to estimate the maximum and minimum distance from the origin to the ellipse  $x^2 + xy + 0.9y^2 = 3$ .
- 19.13** Use Example 18.13 and the Envelope Theorem to estimate the maximum value of  $x^2 + x + 4.1y^2$  on the constraint set:  $2x + 2y \leq 1$ ,  $x \geq 0$ ,  $y \geq 0$ .
- 

## 19.3 SECOND ORDER CONDITIONS

In the analysis of an economic model, the *first order* conditions for a maximization problem often yield an economic principle. The corresponding *second order* conditions provide some fine tuning on these principles. For example, as we noted in Section 3.6, for a firm in a competitive industry, the first order condition for profit maximization implies that marginal revenue equals marginal cost at the profit-maximizing output. The second order condition for profit maximization requires that at the profit-maximizing output, the firm must be experiencing *increasing marginal cost*, as illustrated in Figure 19.1. From a computational point of view, the second order condition can often help choose a maximizer from the set of candidates which satisfy the first order conditions. For example, second order conditions would rule out  $q = q_1$  as a profit-maximizing output in Figure 19.1.

Furthermore, the second order conditions in a maximization problem play a role in the comparative statics analysis of the solution of that problem. As just mentioned, the first order conditions describe the relationship that must occur between the exogenous variables and the endogenous variables at the optimizing solution. In comparative statics or sensitivity analysis, we ask how changes in the exogenous variables affect the optimal values of the endogenous variables. To answer this question, we call on the Implicit Function Theorem and compute total differentials of the first order conditions. We are then led naturally to working with



**Figure  
19.1**

*MC vs. price for a competitive industry.*

a linear system of equations whose coefficient matrix is the matrix of the *second order* conditions of the original maximization problem. The sign of the determinant of that matrix becomes important, for example, when one uses Cramer's rule to solve the resulting system for the differentials of the endogenous variables. The next section illustrates this use of the second order conditions and relates this approach to the theorems of the first two sections.

In Section 17.3, we saw that the second order condition for maximizing an *unconstrained* function  $f(x_1, \dots, x_n)$  of  $n$  variables is that the Hessian of  $f$  at the maximizer  $\mathbf{x}^*$

$$D^2 f(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

be negative definite. More precisely, at a maximum  $f(\mathbf{x}^*)$ ,  $Df(\mathbf{x}^*)$  must be zero and  $D^2 f(\mathbf{x}^*)$  must be *negative semidefinite* (necessary conditions):

$$\mathbf{v}^T (D^2 f(\mathbf{x}^*)) \mathbf{v} \leq 0 \quad \text{for all nonzero vectors } \mathbf{v}.$$

To *guarantee* that a point  $\mathbf{x}^*$  is a local maximizer, we need  $Df(\mathbf{x}^*) = \mathbf{0}$  and  $D^2 f(\mathbf{x}^*)$  *negative definite* (sufficient conditions):

$$\mathbf{v}^T (D^2 f(\mathbf{x}^*)) \mathbf{v} < 0 \quad \text{for all nonzero vectors } \mathbf{v}.$$

In this section we will focus on the sufficient second order conditions for *constrained* maximization and minimization problems. Just as the second order

conditions for unconstrained problems lead to consideration of the definiteness of certain quadratic forms, the second order conditions for constrained problems are closely related to the definiteness of a quadratic form *restricted to a linear subspace* — material covered in Section 16.3.

## Constrained Maximization Problems

Intuitively, the second order condition for a constrained maximization problem:

- (1) should involve the negative definiteness of some Hessian matrix, but
- (2) should only be concerned with directions along the constraint set.

For example, suppose that the objective function is a *quadratic* function  $Q(\mathbf{x}) = \mathbf{x}^T H \mathbf{x} = \sum h_{ij} x_i x_j$  for some symmetric matrix  $H = ((h_{ij}))$  and that the constraint set is defined by the system of *linear* equations  $A\mathbf{x} = \mathbf{0}$ . Since  $\mathbf{0}$  is in the constraint set and since it is a critical point of  $Q$ , it is natural to ask whether  $\mathbf{0}$  is the constrained max. Analytically, we want to know whether

$$0 \geq \mathbf{x}^T H \mathbf{x} \quad \text{for all } \mathbf{x} \text{ such that } A\mathbf{x} = \mathbf{0}.$$

As we saw in Chapter 16, we are asking whether  $Q$  is negative definite on the constraint set  $A\mathbf{x} = \mathbf{0}$ . The algebraic condition for determining whether or not  $\mathbf{x}^T H \mathbf{x}$  is negative on the linear constraint set  $A\mathbf{x} = \mathbf{0}$  involves the signs of certain leading principal minors of the *bordered matrix*:

$$\begin{pmatrix} \mathbf{0} & A \\ A^T & H \end{pmatrix}.$$

For the problem of maximizing a general  $f(\mathbf{x})$  subject to the possibly nonlinear equality constraints  $h_1(\mathbf{x}) = c_1, \dots, h_k(\mathbf{x}) = c_k$ , the first order conditions entail finding the critical points of the Lagrangian function

$$\begin{aligned} L(x_1, \dots, x_n, \mu_1, \dots, \mu_k) \\ = f(\mathbf{x}) - \mu_1(h_1(\mathbf{x}) - c_1) - \dots - \mu_k(h_k(\mathbf{x}) - c_k). \end{aligned} \tag{13}$$

Let  $(\mathbf{x}^*, \mu^*)$  be a critical point of  $L$ . We expect that the second order condition involves the negative definiteness of a quadratic form along a linear constraint set. A natural candidate for the quadratic form is the Hessian of the Lagrangian function with respect to  $x_1, \dots, x_n$ . The natural linear constraint set for this problem is the hyperplane which is tangent to the constraint set  $\{\mathbf{x} \in \mathbf{R}^n : \mathbf{h}(\mathbf{x}) = \mathbf{c}\}$  at the point  $\mathbf{x}^*$ .

The following theorem states that these natural candidates do indeed yield the proper second order sufficient conditions for a constrained max. First, recall (Theorem 15.6) that the tangent space to  $\{\mathbf{h}(\mathbf{x}) = \mathbf{c}\}$  at the point  $\mathbf{x}^*$  is the set of vectors  $\mathbf{v}$  such that  $D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = \mathbf{0}$  (considered as vectors with their tails at  $\mathbf{x}^*$ ). To see this, let  $\mathbf{x}(t)$  be a curve at  $\mathbf{x}^*$  on this constraint set with tangent vector  $\mathbf{v}$ , so

that  $\mathbf{x}(0) = \mathbf{x}^*$ ,  $\mathbf{x}'(0) = \mathbf{v}$ , and  $\mathbf{h}(\mathbf{x}(t)) = \mathbf{c}$ . Then, by the Chain Rule,

$$0 = \frac{d}{dt} \mathbf{h}(\mathbf{x}(t)) \Big|_{t=0} = D\mathbf{h}(\mathbf{x}(0)) \cdot \mathbf{x}'(0) = D\mathbf{h}(\mathbf{x}^*) \mathbf{v};$$

in words, the tangent vector  $\mathbf{v}$  to any curve in the constraint set  $\{\mathbf{h}(\mathbf{x}) = \mathbf{c}\}$  satisfies  $D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = \mathbf{0}$ .

**Theorem 19.6** Let  $f, h_1, \dots, h_k$  be  $C^2$  functions on  $\mathbf{R}^n$ . Consider the problem of maximizing  $f$  on the constraint set

$$C_h \equiv \{ \mathbf{x} : h_1(\mathbf{x}) = c_1, \dots, h_k(\mathbf{x}) = c_k \}.$$

Form the Lagrangian (13), and suppose that:

- (a)  $\mathbf{x}^*$  lies in the constraint set  $C_h$ ,
- (b) there exist  $\mu_1^*, \dots, \mu_k^*$  such that

$$\frac{\partial L}{\partial x_1} = 0, \dots, \frac{\partial L}{\partial x_n} = 0, \frac{\partial L}{\partial \mu_1} = 0, \dots, \frac{\partial L}{\partial \mu_k} = 0$$

at  $(x_1^*, \dots, x_n^*, \mu_1^*, \dots, \mu_k^*)$ .

- (c) the Hessian of  $L$  with respect to  $\mathbf{x}$  at  $(\mathbf{x}^*, \mu^*)$ ,  $D_{\mathbf{x}}^2 L(\mathbf{x}^*, \mu^*)$ , is negative definite on the linear constraint set  $\{\mathbf{v} : D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = \mathbf{0}\}$ ; that is,

$$\mathbf{v} \neq \mathbf{0} \quad \text{and} \quad D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = \mathbf{0} \implies \mathbf{v}^T (D_{\mathbf{x}}^2 L(\mathbf{x}^*, \mu^*)) \mathbf{v} < 0. \quad (14)$$

Then,  $\mathbf{x}^*$  is a strict local constrained max of  $f$  on  $C_h$ .

In Section 16.3 we learned the condition on bordered matrices for verifying second order condition (14). Border the  $n \times n$  Hessian  $D_{\mathbf{x}}^2 L(\mathbf{x}^*, \mu^*)$  with the  $k \times n$  constraint matrix  $D\mathbf{h}(\mathbf{x}^*)$ :

$$H \equiv \begin{pmatrix} \mathbf{0} & D\mathbf{h}(\mathbf{x}^*) \\ D\mathbf{h}(\mathbf{x}^*)^T & D_{\mathbf{x}}^2 L(\mathbf{x}^*, \mu^*) \end{pmatrix} = \left( \begin{array}{ccc|ccc} 0 & \cdots & 0 & \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \frac{\partial h_k}{\partial x_1} & \cdots & \frac{\partial h_k}{\partial x_n} \\ \hline \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_k}{\partial x_1} & \frac{\partial^2 L}{\partial x_1^2} & \cdots & \frac{\partial^2 L}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial h_1}{\partial x_n} & \cdots & \frac{\partial h_k}{\partial x_n} & \frac{\partial^2 L}{\partial x_1 \partial x_n} & \cdots & \frac{\partial^2 L}{\partial x_n^2} \end{array} \right). \quad (15)$$

If the last  $(n - k)$  leading principal minors of matrix (15) alternate in sign, with the sign of the determinant of the  $(k + n) \times (k + n)$  matrix  $H$  in (15) the same as the sign of  $(-1)^n$ , then condition  $c$  of Theorem 19.6 holds.

As one more indication of the naturalness of the statement of Theorem 19.6, note that the Hessian of the Lagrangian (13) with respect to all  $(n + k)$  variables  $\mu_1, \dots, \mu_k, x_1, \dots, x_n$  is

$$D^2_{(\mu, x)} L = \left( \begin{array}{ccc|ccc} 0 & \cdots & 0 & -\frac{\partial h_1}{\partial x_1} & \cdots & -\frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & -\frac{\partial h_k}{\partial x_1} & \cdots & -\frac{\partial h_k}{\partial x_n} \\ \hline -\frac{\partial h_1}{\partial x_1} & \cdots & -\frac{\partial h_k}{\partial x_1} & \frac{\partial^2 L}{\partial x_1^2} & \cdots & \frac{\partial^2 L}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ -\frac{\partial h_1}{\partial x_n} & \cdots & -\frac{\partial h_k}{\partial x_n} & \frac{\partial^2 L}{\partial x_1 \partial x_n} & \cdots & \frac{\partial^2 L}{\partial x_n^2} \end{array} \right), \quad (16)$$

since  $\frac{\partial^2 L}{\partial x_i \partial \mu_j} = -\frac{\partial h_j}{\partial x_i}$ . If we multiply each of the last  $n$  rows and each of the last  $n$  columns in (16) by  $-1$ , we will not change the sign of  $\det D^2 L$  or of any of its principal minors since this process involves an even number of multiplications by  $-1$  in every case. The result is the bordered Hessian (15). So, the bordered Hessian  $H$  in (15) has the same principal minors as the full Hessian (16) of the Lagrangian  $L$ . Recall, however, that the second order condition for the constrained maximization problem involves checking only the last  $n - k$  of the  $n + k$  leading principal minors of  $D^2_{(x, \mu)} L$ .

Let's work out the proof of Theorem 19.6 for the simplest constrained maximization problem: two variables and one equality constraint. The proof for the general case is presented in Chapter 30.

**Theorem 19.7** Let  $f$  and  $h$  be  $C^2$  functions on  $\mathbf{R}^2$ . Consider the problem of maximizing  $f$  on the constraint set  $C_h = \{(x, y) : h(x, y) = c\}$ . Form the Lagrangian

$$L(x, y, \mu) = f(x, y) - \mu(h(x, y) - c).$$

Suppose that  $(x^*, y^*, \mu^*)$  satisfies:

$$(a) \quad \frac{\partial L}{\partial x} = 0, \quad \frac{\partial L}{\partial y} = 0, \quad \frac{\partial L}{\partial \mu} = 0 \quad \text{at } (x^*, y^*, \mu^*), \quad \text{and}$$



$$(b) \det \begin{pmatrix} 0 & \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} \\ \frac{\partial h}{\partial x} & \frac{\partial^2 L}{\partial x^2} & \frac{\partial^2 L}{\partial x \partial y} \\ \frac{\partial h}{\partial y} & \frac{\partial^2 L}{\partial y \partial x} & \frac{\partial^2 L}{\partial y^2} \end{pmatrix} > 0 \quad \text{at } (x^*, y^*, \mu^*).$$

Then,  $(x^*, y^*)$  is a local *max* of  $f$  on  $C_h$ .

*Proof* Condition *b* implies that

$$\frac{\partial h}{\partial x}(x^*, y^*) \neq 0 \quad \text{or} \quad \frac{\partial h}{\partial y}(x^*, y^*) \neq 0.$$

We assume that  $(\partial h / \partial y)(x^*, y^*) \neq 0$ , without loss of generality. Then, by the Implicit Function Theorem (Theorem 15.1), the constraint set  $C_h$  can be considered as the graph of a  $C^1$  function  $y = \phi(x)$  around  $(x^*, y^*)$ ; in other words,

$$h(x, \phi(x)) = C \quad \text{for all } x \text{ near } x^*. \quad (17)$$

Differentiating (17) yields

$$\frac{\partial h}{\partial x}(x, \phi(x)) + \frac{\partial h}{\partial y}(x, \phi(x)) \phi'(x) = 0 \quad (18)$$

$$\text{or} \quad \phi'(x) = - \frac{\frac{\partial h}{\partial x}(x, \phi(x))}{\frac{\partial h}{\partial y}(x, \phi(x))}. \quad (19)$$

$$\text{Let} \quad F(x) \equiv f(x, \phi(x)) \quad (20)$$

be  $f$  evaluated on  $C_h$ , a function of *one* unconstrained variable. By the usual first and second order conditions for such functions, if  $F'(x^*) = 0$  and  $F''(x^*) < 0$ , then  $x^*$  will be a strict local *max* of  $F$  and  $(x^*, y^*) = (x^*, \phi(x^*))$  will be a local constrained *max* of  $f$ .

So, we compute  $F'(x^*)$  and  $F''(x^*)$ . Now,

$$F'(x) = \frac{\partial f}{\partial x}(x, \phi(x)) + \frac{\partial f}{\partial y}(x, \phi(x)) \phi'(x). \quad (21)$$

Multiply equation (18) by  $-\mu^*$  and add it to (21), evaluating both at  $x = x^*$ :

$$\begin{aligned} F'(x^*) &= \left( \frac{\partial f}{\partial x}(x^*, y^*) - \mu^* \frac{\partial h}{\partial x}(x^*, y^*) \right) \\ &\quad + \phi'(x^*) \left( \frac{\partial f}{\partial y}(x^*, y^*) - \mu^* \frac{\partial h}{\partial y}(x^*, y^*) \right) \\ &= \frac{\partial L}{\partial x}(x^*, y^*) + \phi'(x^*) \frac{\partial L}{\partial y}(x^*, y^*). \end{aligned} \quad (22)$$

By hypothesis *a* of the theorem,  $F'(x^*) = 0$ .

Now, take another derivative of  $F(x)$  at  $x^*$ , setting  $y^* = \phi(x^*)$  in (22):

$$\begin{aligned} F''(x^*) &= \frac{\partial^2 L}{\partial x^2} + 2 \frac{\partial^2 L}{\partial x \partial y} \phi'(x^*) + \frac{\partial^2 L}{\partial y^2} \phi'(x^*)^2 \\ &= \frac{\partial^2 L}{\partial x^2} + 2 \frac{\partial^2 L}{\partial x \partial y} \left( -\frac{\partial h / \partial x}{\partial h / \partial y} \right) + \frac{\partial^2 L}{\partial y^2} \left( -\frac{\partial h / \partial x}{\partial h / \partial y} \right)^2 \quad \text{by (19)} \\ &= \frac{1}{\left( \frac{\partial h}{\partial y} \right)^2} \left[ \frac{\partial^2 L}{\partial x^2} \left( \frac{\partial h}{\partial y} \right)^2 - 2 \frac{\partial^2 L}{\partial x \partial y} \frac{\partial h}{\partial x} \frac{\partial h}{\partial y} + \frac{\partial^2 L}{\partial y^2} \left( \frac{\partial h}{\partial x} \right)^2 \right] \end{aligned}$$

which is negative, by hypothesis *b* of the theorem.

Since  $F'(x^*) = 0$  and  $F''(x^*) < 0$ ,

$$x \mapsto F(x) = f(x, \phi(x))$$

has a local max at  $x^*$ , and therefore,  $f$  restricted to  $C_h$  has a local max at  $(x^*, y^*)$ .



## Minimization Problems

**Remark** We have been concentrating on constrained *maximization* problems, up to this point. The second order conditions for a constrained *minimization* problem involve the *positive* definiteness of  $D_x^2 L(x^*, \mu^*)$  on the nullspace of  $Dh(x^*)$ —replacing (14) by

$$v \neq 0 \quad \text{and} \quad Dh(x^*)v = 0 \implies v^T (D_x^2 L(x^*, \mu^*))v > 0 \quad (23)$$

in the statement of Theorem 19.6. By our discussion in Section 16.3, the bordered Hessian conditions for (23) are that the last  $(n - k)$  leading principal minors of (15) all have the *same* sign as  $(-1)^k$ , where  $k$  is the number of constraints. For the case  $n = 2$  and  $k = 1$  in Theorem 19.7, this positive definiteness condition requires that the determinant in Condition *b* of Theorem 19.7 be negative.

*Example 19.7* In Example 18.5, we considered the problem of maximizing  $f(x_1, x_2) = x_1^2 x_2$  on the constraint set  $h(x_1, x_2) = 2x_1^2 + x_2^2 = 3$ . There, we found six solutions to the first order conditions (18.11):

$$(x_1, x_2, \mu) = \begin{cases} (0, \pm\sqrt{3}, 0) \\ (\pm 1, +1, +.5) \\ (\pm 1, -1, -.5). \end{cases}$$

Let us use second order conditions to decide which of these points are local maxima and which are local minima.

Differentiate the first order conditions (18.11) once again to obtain the general bordered Hessian

$$H = \begin{pmatrix} 0 & h_{x_1} & h_{x_2} \\ h_{x_1} & L_{x_1 x_1} & L_{x_1 x_2} \\ h_{x_2} & L_{x_2 x_1} & L_{x_2 x_2} \end{pmatrix} = \begin{pmatrix} 0 & 4x_1 & 2x_2 \\ 4x_1 & 2x_2 - 4\mu & 2x_1 \\ 2x_2 & 2x_1 & -2\mu \end{pmatrix}.$$

This problem has  $n = 2$  variables and  $k = 1$  equality constraint. As Theorem 19.7 indicates, we need only check the sign of  $n - k = 1$  determinant — the determinant of  $H$  itself. If  $\det H$  has the same sign as  $(-1)^n = +1$ , that is, if  $\det H > 0$ , at a candidate point, that point is a local max. If  $\det H$  has the same sign as  $(-1)^k = -1$ , that is, if  $\det H < 0$ , at the candidate point, that point is a local min.

At the points  $(\pm 1, -1, -0.5)$ ,

$$H = \begin{pmatrix} 0 & \pm 4 & 2 \\ \pm 4 & 0 & \pm 2 \\ -2 & \pm 2 & 1 \end{pmatrix}.$$

In either case,  $\det H = -16$ ; so these two points are local minima.

At the points  $(\pm 1, 1, 0.5)$ ,

$$H = \begin{pmatrix} 0 & \pm 4 & 2 \\ \pm 4 & 0 & \pm 2 \\ 2 & \pm 2 & -1 \end{pmatrix}.$$

In either case,  $\det H = +48$ ; so these two points are local maxima.

These computations support the observations we made at the end of Example 18.5. However, we were not able to determine the character of  $(x_1, x_2) = (0, \pm\sqrt{3})$  by simply plugging these points into the objective function in Example 18.5. Since  $\mu = 0$  for these points, the corresponding bordered Hessian is

$$H = \begin{pmatrix} 0 & 0 & \pm 2\sqrt{3} \\ 0 & \pm 2\sqrt{3} & 0 \\ \pm 2\sqrt{3} & 0 & 0 \end{pmatrix}.$$

For  $(x_1, x_2) = (0, +\sqrt{3})$ ,  $\det H = -24\sqrt{3} < 0$ ; this point is a local min. For  $(x_1, x_2) = (0, -\sqrt{3})$ ,  $\det H = +24\sqrt{3} > 0$ ; this point is a local max. These calculations, which should agree with the conclusions of the geometric approach in Exercise 18.1, illustrate that extrema computed via first and second order conditions of Theorem 19.6 need not be *global* extrema.

**Example 19.8** Consider the problem of maximizing  $x^2y^2z^2$  subject to the constraint  $x^2 + y^2 + z^2 = 3$ , a special case of Exercise 18.9. The first order conditions are

$$\frac{\partial L}{\partial x} = 2xy^2z^2 - 2\mu x = 0$$

$$\frac{\partial L}{\partial y} = 2x^2yz^2 - 2\mu y = 0$$

$$\frac{\partial L}{\partial z} = 2x^2y^2z - 2\mu z = 0$$

$$-\frac{\partial L}{\partial \mu} = x^2 + y^2 + z^2 - 3 = 0$$

with solution  $x^2 = y^2 = z^2 = \mu = 1$ . The bordered Hessian for this problem is

$$\begin{pmatrix} 0 & 2x & 2y & 2z \\ 2x & 2y^2z^2 - 2\mu & 4xyz^2 & 4xy^2z \\ 2y & 4xyz^2 & 2x^2z^2 - 2\mu & 4x^2yz \\ 2z & 4xy^2z & 4x^2yz & 2x^2y^2 - 2\mu \end{pmatrix}.$$

At  $x = y = z = \mu = 1$ , the bordered Hessian becomes

$$\left( \begin{array}{ccc|c} 0 & 2 & 2 & 2 \\ 2 & 0 & 4 & 4 \\ 2 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 \end{array} \right) \quad (24)$$

Since  $n = 3$  and  $k = 1$ , we have to check the signs of two leading principal minors: the  $3 \times 3$  submatrix  $H_3$  above the dashed lines in (24) and the complete  $4 \times 4$  matrix  $H_4$  in (24). One computes that  $\det H_3 = 32$  and  $\det H_4 = -192$ . Since these determinants alternate in sign and since the sign of  $\det H_4$  is the sign of  $(-1)^3 = -1$ , the candidate  $x = y = z = 1$  is indeed local constrained max by Theorem 19.6.

## Inequality Constraints

To include inequality constraints in the statement of Theorem 19.6, we call on the natural techniques that we used at this stage of our discussion of first order conditions, in Section 18.3. Given a solution  $(\mathbf{x}^*, \lambda^*)$  of the first order conditions, divide the inequality constraints into binding constraints and nonbinding constraints at  $\mathbf{x}^*$ . On the one hand, we treat the binding inequality constraints like equality constraints; on the other hand, the multipliers for the nonbinding constraints must be zero and these constraints drop out of the Lagrangian. The following theorem summarizes these considerations for a constrained maximization problem.

**Theorem 19.8** Let  $f, g_1, \dots, g_m, h_1, \dots, h_k$  be  $C^2$  functions on  $\mathbf{R}^n$ . Consider the problem of maximizing  $f$  on the constraint set

$$C_{g,h} \equiv \{ \mathbf{x} : g_1(\mathbf{x}) \leq b_1, \dots, g_m(\mathbf{x}) \leq b_m, h_1(\mathbf{x}) = c_1, \dots, h_k(\mathbf{x}) = c_k \}.$$

Form the Lagrangian

$$\begin{aligned} L(x_1, \dots, x_n, \lambda_1, \dots, \lambda_m, \mu_1, \dots, \mu_k) \\ = f(\mathbf{x}) - \lambda_1(g_1(\mathbf{x}) - b_1) - \dots - \lambda_m(g_m(\mathbf{x}) - b_m) \\ - \mu_1(h_1(\mathbf{x}) - c_1) - \dots - \mu_k(h_k(\mathbf{x}) - c_k). \end{aligned}$$

- (a) Suppose that there exist  $\lambda_1^*, \dots, \lambda_m^*, \mu_1^*, \dots, \mu_k^*$  such that the first order conditions of Theorem 18.5 are satisfied; that is,

$$\begin{aligned} \frac{\partial L}{\partial x_1} = 0, \dots, \frac{\partial L}{\partial x_n} = 0, \quad \text{at } (\mathbf{x}^*, \lambda^*, \mu^*), \\ \lambda_1^* \geq 0, \dots, \lambda_m^* \geq 0, \\ \lambda_1^*(g_1(\mathbf{x}^*) - b_1) = 0, \dots, \lambda_m^*(g_m(\mathbf{x}^*) - b_m) = 0, \\ h_1(\mathbf{x}^*) = c_1, \dots, h_k(\mathbf{x}^*) = c_k. \end{aligned}$$

- (b) For notation's sake, suppose that  $g_1, \dots, g_e$  are binding at  $\mathbf{x}^*$  and  $g_{e+1}, \dots, g_m$  are not binding. Write  $(g_1, \dots, g_e)$  as  $\mathbf{g}_E$ . Suppose that the Hessian of  $L$  with respect to  $\mathbf{x}$  at  $(\mathbf{x}^*, \lambda^*, \mu^*)$  is negative definite on the linear constraint set

$$\{ \mathbf{v} : D\mathbf{g}_E(\mathbf{x}^*)\mathbf{v} = \mathbf{0} \quad \text{and} \quad D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = \mathbf{0} \}$$

that is,  $\mathbf{v} \neq \mathbf{0}, \quad D\mathbf{g}_E(\mathbf{x}^*)\mathbf{v} = \mathbf{0}, \quad D\mathbf{h}(\mathbf{x}^*)\mathbf{v} = \mathbf{0}$

$$\implies \mathbf{v}^T \cdot (D_{\mathbf{x}}^2 L(\mathbf{x}^*, \lambda^*, \mu^*)) \cdot \mathbf{v} < 0.$$

Then  $\mathbf{x}^*$  is a strict local constrained max of  $f$  on  $C_{g,h}$ .

To check condition (b), form the **bordered Hessian**

$$\left( \begin{array}{cccccc|ccc} 0 & \cdots & 0 & 0 & \cdots & 0 & \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_1}{\partial x_n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \frac{\partial g_e}{\partial x_1} & \cdots & \frac{\partial g_e}{\partial x_n} \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_1}{\partial x_n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \frac{\partial h_k}{\partial x_1} & \cdots & \frac{\partial h_k}{\partial x_n} \\ \hline \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_e}{\partial x_1} & \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_k}{\partial x_1} & \frac{\partial^2 L}{\partial x_1^2} & \cdots & \frac{\partial^2 L}{\partial x_1 \partial x_1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_1}{\partial x_n} & \cdots & \frac{\partial g_e}{\partial x_n} & \frac{\partial h_1}{\partial x_n} & \cdots & \frac{\partial h_k}{\partial x_n} & \frac{\partial^2 L}{\partial x_1 \partial x_n} & \cdots & \frac{\partial^2 L}{\partial x_n^2} \end{array} \right).$$

If the last  $n - (e + k)$  leading principal minors alternate in sign with the sign of the determinant of the largest matrix the same as the sign of  $(-1)^n$ , then condition c holds.

A little care has to be taken in writing the minimization version of Theorem 19.8. In particular, the minimization problem should be presented in standard form, as in Theorem 18.6. One makes the following changes in the wording of Theorem 19.8 for an inequality-constrained minimization problem:

- (1) change the word “maximizing” to “minimizing” on line two,
- (2) write the inequality constraints as  $g_i(\mathbf{x}) \geq b_i$  in the presentation of the constraint set  $C_{gh}$ ,
- (3) change “negative definite” and “ $< 0$ ” in condition (b) to “positive definite” and “ $> 0$ ,”
- (4) change “max” to “min” in the concluding sentence.

The bordered Hessian check requires that the last  $n - (e + k)$  leading principal minors all have the same sign as  $(-1)^{e+k}$ .

### Alternative Approaches to the Bordered Hessian Condition

The bordered Hessian condition for a constrained max or min can be presented in different ways. The two most common alternatives to our presentation involve the position of the border in the bordered Hessian and the rules for the signs of the leading principal minors to distinguish a max from a min. Many texts place

the Jacobian  $D\mathbf{h}(\mathbf{x}^*)$  of the constraint functions to the right and below the Hessian  $D_x^2 L(\mathbf{x}^*, \lambda^*)$  of the Lagrangian with respect to the  $x_i$ 's:

$$H = \left( \begin{array}{ccc|ccc} \frac{\partial^2 L}{\partial x_1^2} & \cdots & \frac{\partial^2 L}{\partial x_1 \partial x_n} & | & \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_k}{\partial x_1} \\ \vdots & \ddots & \vdots & | & \vdots & \ddots & \vdots \\ \frac{\partial^2 L}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 L}{\partial x_n^2} & | & \frac{\partial h_1}{\partial x_n} & \cdots & \frac{\partial h_k}{\partial x_n} \\ \hline \frac{\partial h_1}{\partial x_1} & \cdots & \frac{\partial h_1}{\partial x_n} & | & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & | & \vdots & \ddots & \vdots \\ \frac{\partial h_k}{\partial x_1} & \cdots & \frac{\partial h_k}{\partial x_n} & | & 0 & \cdots & 0 \end{array} \right). \quad (25)$$

Instead of examining the *last*  $n - k$  leading principal minors, this point of view looks at the  $n - k$  largest principal minors which “respect the borders”; after computing the determinant of all of  $H$ , one first throws away row  $n$  and column  $n$  of  $H$ , then row  $n - 1$  and column  $n - 1$  of  $H$ , and so on.

Furthermore, some texts state the sign conditions by emphasizing the sign of the *smallest* principal minor checked instead of the sign of the whole bordered Hessian  $H$ . For a constrained minimization problem, the  $n - k$  principal minors all have the same sign; for a constrained maximization problem, they alternate in sign. For a constrained minimization problem, the sign of the smallest principal minor checked is the same as the sign of  $(-1)^k$ ; for a constrained maximization problem, the sign of the smallest principal minor checked is the same as the sign of  $(-1)^{k+1}$ .

### Necessary Second Order Conditions

Theorems 19.6, 19.7, and 19.8 state *sufficient* second order conditions for a candidate point to be a solution of a constrained maximization problem, namely, that the Hessian  $D_x^2 L(\mathbf{x}^*, \lambda^*)$  of the Lagrangian with respect to the  $x_i$ 's be negative definite on the nullspace of the Jacobian  $D\mathbf{h}(\mathbf{x}^*)$  of the constraint functions — condition (14) in Theorem 19.6. The corresponding *necessary* condition that a constrained maximizer must satisfy is, of course, that  $D_x^2 L(\mathbf{x}^*, \lambda^*)$  be negative *semidefinite* on the nullspace of  $D\mathbf{h}(\mathbf{x}^*)$ . The complete characterization of constrained negative semidefiniteness in terms of principal submatrices of the bordered Hessian is a bit too complex to state here in all its generality. It certainly requires that each of the largest  $n - k$  leading principal minors must be zero or have the same sign as  $(-1)^k$ . If  $DL(\mathbf{x}^*) = \mathbf{0}$  but one of the last  $n - k$  leading principal minors of the bordered Hessian is nonzero and has the wrong sign for negative definiteness, then the candidate cannot be a local constrained max.

## EXERCISES

- 19.14** Check the second order conditions for the solutions of the first order conditions in Exercises 18.2, 18.3, 18.5, 18.6, and 18.7.
- 19.15** Redo the proof of Theorem 19.7 for the case  $(\partial h / \partial y)(x^*, y^*) = 0$ .
- 19.16** Rewrite the proof of Theorem 19.7 for the constrained minimization problem.
- 19.17** Write out a complete proof of Theorem 19.8 for the case of two variables and one inequality constraint.

## 19.4 SMOOTH DEPENDENCE ON THE PARAMETERS

All the results in the first two sections of this chapter had two basic hypotheses: the smooth dependence of the maximizers on the parameters of the problem and the nondegenerate constraint qualification (NDCQ). In this section, we will look at these hypotheses a little more carefully and show how to phrase them in terms of the problem's objective and constraint functions.

Consider first the parameterized unconstrained problem (9) of Theorem 19.5:

$$F(a) = \max_{\mathbf{x}} f(\mathbf{x}; a). \quad (26)$$

Since there are no nonnegativity constraints and we are assuming that a maximizer  $\mathbf{x}^*(a)$  exists, then  $\mathbf{x}^*(a)$  is a solution of the usual first order conditions

$$\begin{aligned} \frac{\partial f}{\partial x_1}(x_1, \dots, x_n; a) &= 0 \\ &\vdots \\ \frac{\partial f}{\partial x_n}(x_1, \dots, x_n; a) &= 0. \end{aligned} \quad (27)$$

By the Implicit Function Theorem (Theorem 15.7), we can solve these  $n$  equations (27) for the  $n$  unknowns  $x_1, \dots, x_n$  as  $C^1$  functions of the exogenous variable  $a$  provided that the Jacobian of the functions in (27), with respect to the endogenous variables  $x_1, \dots, x_n$ , is nonsingular at  $(\mathbf{x}^*(a); a)$ . But the Jacobian of the first order partial derivatives  $\partial f / \partial x_i$  is simply the Hessian of  $f$  at  $(\mathbf{x}^*(a); a)$ :

$$D^2 f_{(\mathbf{x}^*(a); a)} = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_1} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & \frac{\partial^2 f}{\partial x_2 \partial x_n} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}. \quad (28)$$



By our work with critical points of real-valued functions in Chapter 17, the Hessian matrix  $D^2f$  is generally nonsingular. In fact, since (26) is a maximization problem, the determinant of the Hessian should have the same sign as  $(-1)^n$  by the usual second order necessary conditions, as described at the end of Section 17.3.

To summarize, we can replace the hypothesis that  $\mathbf{x}^*(a)$  is a  $C^1$  function of  $a$  in Theorem 19.4 by the hypothesis that  $\mathbf{x}^*(a)$  is a *nondegenerate* critical point of  $f$  in that the Hessian matrix (28) of  $f$  is nonsingular at  $(\mathbf{x}^*(a); a)$ .

A similar analysis works for constrained problems. Consider the parameterized constrained maximization problem  $(S_a)$ :

$$\begin{aligned} & \text{maximize} && f(\mathbf{x}; a) \\ & \text{subject to} && h_1(\mathbf{x}; a) = 0, \dots, h_k(\mathbf{x}; a) = 0, \end{aligned} \tag{S_a}$$

as discussed in Theorem 19.5. Assuming that NDCQ

$$\text{rank} \begin{pmatrix} \frac{\partial h_1}{\partial x_1}(\mathbf{x}^*(a); a) & \cdots & \frac{\partial h_1}{\partial x_n}(\mathbf{x}^*(a); a) \\ \vdots & \ddots & \vdots \\ \frac{\partial h_k}{\partial x_1}(\mathbf{x}^*(a); a) & \cdots & \frac{\partial h_k}{\partial x_n}(\mathbf{x}^*(a); a) \end{pmatrix} = k \tag{29}$$

holds at  $\mathbf{x}^*(a)$ , we write the Lagrangian for Problem  $(S_a)$  as

$$\begin{aligned} L(x_1, \dots, x_n, \mu_1, \dots, \mu_k; a) = & f(\mathbf{x}; a) - \mu_1 h_1(\mathbf{x}; a) \\ & - \cdots - \mu_k h_k(\mathbf{x}; a). \end{aligned}$$

The constrained maximizer  $\mathbf{x}^*(a)$  must satisfy the first order conditions

$$\begin{aligned} \frac{\partial L}{\partial x_1}(\mathbf{x}, \mu; a) = 0, \dots, \frac{\partial L}{\partial x_n}(\mathbf{x}, \mu; a) = 0, \\ \frac{\partial L}{\partial \mu_1}(\mathbf{x}, \mu; a) = 0, \dots, \frac{\partial L}{\partial \mu_k}(\mathbf{x}, \mu; a) = 0, \end{aligned} \tag{30}$$

a system of  $n + k$  equations in  $n + k$  unknowns  $x_1, \dots, x_n, \mu_1, \dots, \mu_k$ . Once again, we call on the Implicit Function Theorem for conditions that will guarantee that  $\mathbf{x}^*(a)$  and  $\mu^*(a)$  depend smoothly on the exogenous variable  $a$ ; the Jacobian of the defining equations (30) with respect to the endogenous variables must be an  $(n + k) \times (n + k)$  nonsingular matrix at  $(\mathbf{x}^*(a), \mu^*(a); a)$ . This Jacobian is simply the *Hessian* of the Lagrangian

$$\begin{aligned}
 D^2L_{\mathbf{x},\mu} &= \begin{pmatrix} \frac{\partial^2 L}{\partial x_1^2} & \cdots & \frac{\partial^2 L}{\partial x_n \partial x_1} & \frac{\partial^2 L}{\partial \mu_1 \partial x_1} & \cdots & \frac{\partial^2 L}{\partial \mu_k \partial x_1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 L}{\partial x_1 \partial x_n} & \cdots & \frac{\partial^2 L}{\partial x_n^2} & \frac{\partial^2 L}{\partial \mu_1 \partial x_n} & \cdots & \frac{\partial^2 L}{\partial \mu_k \partial x_n} \\ \frac{\partial^2 L}{\partial x_1 \partial \mu_1} & \cdots & \frac{\partial^2 L}{\partial x_n \partial \mu_1} & \frac{\partial^2 L}{\partial \mu_1^2} & \cdots & \frac{\partial^2 L}{\partial \mu_k \partial \mu_1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 L}{\partial x_1 \partial \mu_k} & \cdots & \frac{\partial^2 L}{\partial x_n \partial \mu_k} & \frac{\partial^2 L}{\partial \mu_1 \partial \mu_k} & \cdots & \frac{\partial^2 L}{\partial \mu_k^2} \end{pmatrix} \\
 &= \begin{pmatrix} \frac{\partial^2 L}{\partial x_1^2} & \cdots & \frac{\partial^2 L}{\partial x_n \partial x_1} & -\frac{\partial h_1}{\partial x_1} & \cdots & -\frac{\partial h_k}{\partial x_1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 L}{\partial x_1 \partial x_n} & \cdots & \frac{\partial^2 L}{\partial x_n^2} & -\frac{\partial h_1}{\partial x_n} & \cdots & -\frac{\partial h_k}{\partial x_n} \\ -\frac{\partial h_1}{\partial x_1} & \cdots & -\frac{\partial h_1}{\partial x_n} & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ -\frac{\partial h_k}{\partial x_1} & \cdots & -\frac{\partial h_k}{\partial x_n} & 0 & \cdots & 0 \end{pmatrix}.
 \end{aligned} \tag{31}$$

evaluated at  $(\mathbf{x}^*(a), \mu^*(a); a)$ .

As we learned in Section 19.3, for example, in expression (25), this is one form of the matrix one uses to check the second order conditions for a constrained max. As in the unconstrained problem, this Hessian usually has nonzero determinant. In fact, for a *nondegenerate* constrained maximization problem  $(S_a)$ , its determinant has the same sign as  $(-1)^n$ .

Note that the Jacobian (29) of the constraint functions  $Dh$  plays a major role in this Hessian. In fact, as you are asked to verify in Exercise 19.9, the determinant of  $D^2L$  will be zero if  $Dh$  does not have maximal rank  $k$ . So, the natural condition that guarantees that  $\mathbf{x}^*(a)$  and the  $\mu_j^*(a)$ 's depend smoothly on the parameter  $a$  includes the nondegenerate constraint qualification NDCQ.

In summary, we can replace the condition in Theorem 19.5 that  $\mathbf{x}^*(a)$  and the  $\mu_i(a)$ 's be  $C^1$  functions of the parameter  $a$  and the requirement of the nondegenerate constraint qualification by the nondegenerate second order condition for constrained problems, namely that the Hessian of the Lagrangian has nonzero determinant at  $(\mathbf{x}^*(a), \mu^*(a); a)$ .

**Theorem 19.9** Let  $\mathbf{x}^*(a)$  be the solution of the parameterized constrained maximization problem  $(S_a)$ , and let  $\mu^*(a)$  be the corresponding Lagrange multiplier. Fix the value of the parameter at  $a = a_0$ . If the Hessian matrix (31) is nonsingular at the point  $(\mathbf{x}^*(a_0), \mu^*(a_0); a_0)$ , then:

- (a)  $\mathbf{x}^*(a)$  and  $\mu^*(a)$  are  $C^1$  functions of  $a$  at  $a = a_0$ ; and
- (b) the NDCQ holds at  $(\mathbf{x}^*(a_0), \mu^*(a_0); a_0)$ .

Finally, we remark that the envelope theorems and their variations are true even without such nondegeneracy conditions, in fact, even without smoothness assumptions on  $f$  and the  $h_j$ 's. In this case, we need to assume that all the functions involved are concave or convex functions.

### EXERCISES

- 19.18** Consider the problem of maximizing  $x_1^2 x_2$  on the constraint set  $2x_1^2 + x_2^2 = a$ , as in Example 18.5. Use the Implicit Function Theorem directly on the first order conditions of this problem to prove that the solutions  $x_1(a)$ ,  $x_2(a)$ ,  $\lambda(a)$  depend smoothly on the parameter  $a$  near  $a = 3$ .
- 19.19** Prove that if  $Dh$  in (29) does not have maximal rank, then  $D^2L$  in (31) has zero determinant.

## 19.5 CONSTRAINT QUALIFICATIONS

In applying the theorems of Chapter 18 to find the solutions of a constrained optimization problem, we have been checking the constraint qualification by calculating the critical points of the constraint functions. If none of these critical points lie in the constraint set, we write out the Lagrangian, use its partial derivatives to write out the system of first order conditions, and solve this system of equations. If the constraint set *does* contain critical points of the constraint functions, then we include these points among our candidates for a solution to the original constrained maximization problem, along with the solutions of the first order conditions.

When working with economic applications or studying general economic theory, one usually does not know the exact functional form of the constraint functions of the problem under study. Therefore, it is helpful to have versions of the theorems of Chapter 18 which do not impose hypotheses on the constraint functions and which incorporate both types of candidates for constrained maxima — critical points of the Lagrangian and critical points of the constraint function — into a

single criterion. Such a criterion does exist. However, it requires one more multiplier — a multiplier for the objective function itself.

To describe this variation, we begin, as usual, with the simplest case — the problem with two variables and one equality constraint. The following theorem summarizes this alternative approach.

**Theorem 19.10** Let  $f$  and  $h$  be  $C^1$  functions of two variables. Suppose that  $\mathbf{x}^* = (x_1^*, x_2^*)$  is a solution of the problem

$$\begin{aligned} &\text{maximize} && f(x_1, x_2) \\ &\text{on the constraint set} && \{(x_1, x_2) : h(x_1, x_2) = c\}. \end{aligned}$$

Construct the Lagrangian

$$L(x_1, x_2, \mu_0, \mu_1) \equiv \mu_0 f(x_1, x_2) - \mu_1 [h(x_1, x_2) - c],$$

including a multiplier  $\mu_0$  for the objective function. Then, there exist multipliers  $\mu_0^*$  and  $\mu_1^*$  such that:

- (a)  $\mu_0^*$  and  $\mu_1^*$  are not both zero,
- (b)  $\mu_0^*$  is either 0 or 1,
- (c) the quadruple  $(x_1^*, x_2^*, \mu_0^*, \mu_1^*)$  satisfies the equations

$$\begin{aligned} \frac{\partial L}{\partial x_1} &= \mu_0 \frac{\partial f}{\partial x_1}(x_1, x_2) - \mu_1 \frac{\partial h}{\partial x_1}(x_1, x_2) = 0 \\ \frac{\partial L}{\partial x_2} &= \mu_0 \frac{\partial f}{\partial x_2}(x_1, x_2) - \mu_1 \frac{\partial h}{\partial x_2}(x_1, x_2) = 0 \\ \frac{\partial L}{\partial \mu_1} &= c - h(x_1, x_2) = 0. \end{aligned} \tag{32}$$

*Proof* Suppose that  $(x_1^*, x_2^*)$  is a solution of the constrained maximization problem. If  $(x_1^*, x_2^*)$  is not a critical point of  $h$ , we can set  $\mu_0 = 1$  and use Theorem 18.1 to deduce that  $(x_1^*, x_2^*, \mu_1^*)$  satisfies system (32). On the other hand, if  $(x_1^*, x_2^*)$  is a critical point of  $h$  so that  $\partial h / \partial x_1$  and  $\partial h / \partial x_2$  in (32) are zero at  $(x_1^*, x_2^*)$ , we can take  $\mu_1^*$  to be any *nonzero* number and set  $\mu_0^*$  equal to zero. The resulting  $(x_1^*, x_2^*, \mu_0^*, \mu_1^*)$  will be a solution of system (32). ■

In concrete calculations with specified functional forms, it is easier to apply the method of Theorem 18.1 and of Example 18.5: first check for the critical points of  $h$  that lie in the constraint set, then calculate the critical points of the usual Lagrangian with  $\mu_0 = 1$ . The following example illustrates both methods.

**Example 19.9** Consider the problem of maximizing  $f(x, y) = x$  on the constraint set  $C_h$  defined by  $h(x, y) \equiv x^3 + y^2 = 0$ . The constraint set is the cusp drawn in Figure 19.2. (Check.) Since we are trying to maximize  $x$ , we want to find the point on  $C_h$  furthest to the right in Figure 19.2. The only solution is clearly the cusp point at the origin.

Now, let's try this analytically. If we use Theorem 18.1, we look first for the critical points of  $h$  by solving

$$\frac{\partial h}{\partial x} = 3x^2 = 0 \quad \text{and} \quad \frac{\partial h}{\partial y} = 2y = 0.$$

The only solution is  $(0, 0)$ , which does indeed lie on the constraint set. Now, find the critical points of the Lagrangian

$$L(x, y, \mu) = f(x, y) - \mu[h(x, y) - c] = x - \mu(x^3 + y^2),$$

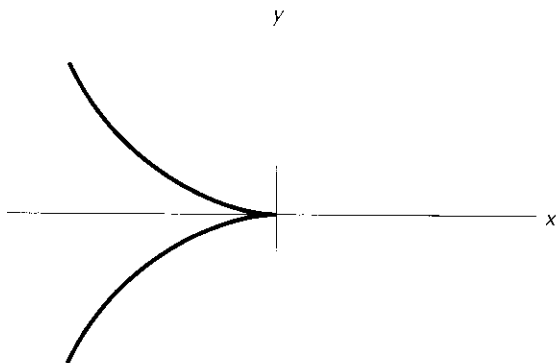
by solving the system

$$\frac{\partial L}{\partial x} = 1 - 3\mu x^2 = 0$$

$$\frac{\partial L}{\partial y} = -2\mu y = 0$$

$$\frac{\partial L}{\partial \mu} = -(x^3 + y^2) = 0.$$

This system has *no* solution. (Verify!) Consequently, the only candidate for a solution of the maximization problem is the critical point of  $h$ ,  $(0, 0)$ , which is indeed the solution of the problem.



**Figure**  
**19.2**

The constraint set  $x^3 + y^2 = 0$ .

Finally, we could have used Theorem 19.10 and worked with

$$L(x, y, \mu_0, \mu_1) = \mu_0 x - \mu_1 (x^3 + y^2).$$

In this case, we are looking for solutions of

$$\frac{\partial L}{\partial x} = \mu_0 - 3\mu_1 x^2 = 0$$

$$\frac{\partial L}{\partial y} = -2\mu_1 y = 0$$

$$\frac{\partial L}{\partial \mu_1} = -(x^3 + y^2) = 0,$$

where  $(\mu_0, \mu_1) \neq (0, 0)$ . The only solution with  $(\mu_0, \mu_1) \neq (0, 0)$  is  $x = y = \mu_0 = 0$  and  $\mu_1 = \text{anything nonzero}$ . (Check.) This does indeed give the constrained maximizer.

This example shows that we really do require a constraint qualification in the statement of Theorem 18.1. Of course, Theorem 19.10 holds for the general constrained optimization problem with equality and/or inequality constraints on  $\mathbf{R}^n$ . We state the version for inequality constraints here. This theorem is sometimes called the **Fritz John Theorem**.

**Theorem 19.11** Suppose that  $f, g_1, \dots, g_k$  are  $C^1$  functions of  $n$  variables. Suppose that  $\mathbf{x}^*$  is a local maximizer of  $f$  on the constraint set defined by the  $k$  inequalities

$$g_1(x_1, \dots, x_n) \leq b_1, \dots, g_k(x_1, \dots, x_n) \leq b_k.$$

Form the Lagrangian

$$L(x_1, \dots, x_n, \lambda_0, \lambda_1, \dots, \lambda_k) \equiv \lambda_0 f(\mathbf{x}) - \lambda_1 [g_1(\mathbf{x}) - b_1] - \dots - \lambda_k [g_k(\mathbf{x}) - b_k],$$

with a multiplier  $\lambda_0$  for the objective function. Then, there exist multipliers  $\lambda^* = (\lambda_0^*, \lambda_1^*, \dots, \lambda_k^*)$  such that:

$$(a) \quad \frac{\partial L}{\partial x_1}(\mathbf{x}^*, \lambda^*) = 0, \quad \dots, \quad \frac{\partial L}{\partial x_n}(\mathbf{x}^*, \lambda^*) = 0,$$

$$(b) \quad \lambda_1^* [g_1(\mathbf{x}^*) - b_1] = 0, \quad \dots, \quad \lambda_k^* [g_k(\mathbf{x}^*) - b_k] = 0,$$

$$(c) \quad \lambda_i^* \geq 0, \quad \dots, \quad \lambda_k^* \geq 0,$$

$$(d) \quad g_1(\mathbf{x}^*) \leq b_1, \quad \dots, \quad g_k(\mathbf{x}^*) \leq b_k,$$

$$(e) \quad \lambda_0^* = 0 \text{ or } 1, \text{ and}$$

$$(f) \quad (\lambda_0^*, \lambda_1^*, \dots, \lambda_k^*) \neq (0, 0, \dots, 0).$$

**Example 19.10** Let us solve the problem in Example 18.9 using Theorem 19.11.

Without checking any constraint qualification, we would write the Lagrangian

$$L(x, y, \lambda_0, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = \lambda_0 xyz - \lambda_1(x + y + z - 1) + \lambda_2 x + \lambda_3 y + \lambda_4 z.$$

The fifteen first order conditions in Example 18.9 would remain the same, except that the first three would each begin with a  $\lambda_0$ . We would also add conditions  $e$  and  $f$  from Theorem 19.11. If  $\lambda_0^* = 0$ , then equation 20 in Chapter 18 becomes

$$\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4,$$

none of which can be zero by Condition  $f$  in Theorem 19.11. But if  $\lambda_1, \lambda_2, \lambda_3$ , and  $\lambda_4$  are positive, all the constraints would have to be binding:  $x = 0, y = 0, z = 0, x + y + z = 1$  — an impossibility. We conclude that  $\lambda_0^* = 1$  and proceed as in Example 18.9.

We would clearly prefer that  $\lambda_0^* \neq 0$ ; otherwise, the objective function — the very function we are maximizing — drops completely out of the first order conditions. For this reason, we prefer to have some qualification on the constraint functions which guarantees that we can take  $\lambda_0^* = 1$ , hence the name “constraint qualification.”

The first order conditions of the last chapter emphasized the nondegenerate constraint qualification (NDCQ). However, there are other constraint qualifications which guarantee that  $\lambda_0^*$  can be set equal to 1. The following theorem lists some of the more commonly used constraint qualifications for the inequality constrained maximization problem.

**Theorem 19.12** Let  $f, g_1, \dots, g_k$  be as in Theorem 19.11, and suppose that  $\mathbf{x}^* \in \mathbf{R}^n$  is a local maximizer of  $f$  on the constraint set defined by

$$g_1(\mathbf{x}) \leq b_1, \dots, g_k(\mathbf{x}) \leq b_k.$$

For ease of notation, suppose that  $g_1, \dots, g_h$  yield binding constraints at  $\mathbf{x}^*$  and that  $g_{h+1}, \dots, g_k$  are not binding at  $\mathbf{x}^*$ . Suppose that the *binding* constraint functions satisfy one of the following properties:

- (a) (**NDCQ**) the  $h \times n$  Jacobian matrix  $(\partial g_i / \partial x_j(\mathbf{x}^*)), i = 1, \dots, h, j = 1, \dots, n$ , has maximal rank  $h$ ,
- (b) (**Karush–Kuhn–Tucker CQ**) For any vector  $\mathbf{v}$  in  $\mathbf{R}^n$  with the property that  $Dg_i(\mathbf{x}^*)(\mathbf{v}) \leq 0$  for  $i = 1, \dots, h$ , there exists  $\epsilon > 0$  and a  $C^1$  curve  $\alpha: [0, \epsilon) \rightarrow \mathbf{R}^n$  such that:
  - (i)  $\alpha(0) = \mathbf{x}^*$ ,
  - (ii)  $\alpha'(0) = \mathbf{v}$ , and
  - (iii)  $g_i(\alpha(t)) \leq b_i$  for all  $i = 1, 2, \dots, k$  and all  $t \in [0, \epsilon)$ .

- (c) (**Slater CQ**) There is a ball  $U$  about  $\mathbf{x}^*$  in  $\mathbf{R}^n$  such that  $g_1, \dots, g_h$  are convex functions on  $U$  and there exists  $\mathbf{z} \in U$  so that each  $g_i(\mathbf{z}) < b_i$ .
- (d)  $g_1, \dots, g_h$  are concave functions.
- (e)  $g_1, \dots, g_h$  are linear functions.

Then, we can take  $\lambda_0^* = 1$  in the conclusion of Theorem 19.11.

The first constraint qualification in the statement of Theorem 19.12 is the NDCQ which we used throughout Chapter 18. The second constraint qualification is the condition which Kuhn and Tucker used in their pioneering work on inequality constrained problems. It is designed to rule out cusps in the constraint set, like the one in Example 19.9. (See Exercise 19.23.) The third constraint qualification requires that the constraint set have a nonempty interior and that the binding constraint functions be convex functions.

The last three constraint qualifications in Theorem 19.12 are particularly helpful in economics applications because they are assumptions on the general character of the constraint functions — assumptions which are common in economic theory. Condition *e* in Theorem 19.12 is especially important because linear constraints are common in economics models. For example, we can automatically take  $\lambda_0^* = 1$  in the utility maximization problem of Example 18.1 because all of its constraint functions are linear. The same statement can be made for the problems in Examples 18.9 and 18.13. Entire books are devoted to linear programming problems — constrained optimization problems in which the objective function and all the constraint functions are linear. By Theorem 19.12, we can take  $\lambda_0 = 1$  in all these problems. We can take  $\lambda_0^* = 1$  in Example 18.7 because its constraint function is a convex function. The proof of Theorem 19.12 is an application of Farkas' lemma, which we omit at this point.

For minimization problems with inequality constraints of the form  $g_i(x) \geq b_i$ , Theorem 19.12 still holds provided that we reverse the two  $\leq$ 's in Condition *b*, and interchange the words "concave" and "convex" in Conditions *c* and *d*.

## EXERCISES

- 19.20** Following the guidelines of Example 19.10, carry out the analogous application of Theorem 19.11 for Example 18.13.
- 19.21** Which of the last three constraint qualifications in Theorem 19.12 hold for the constraint functions in Exercises 18.10, 18.11, 18.12, 18.17 and 18.18?
- 19.22** Consider the problem of maximizing  $x$  subject to  $y - x^4 \leq 0$ ,  $x^3 - y \leq 0$ , and  $x \leq 1/2$ . Try solving this problem with and without using a multiplier  $\lambda_0$  for the objective function.
- 19.23** Verify that constraint qualification *b* in Theorem 19.12 is not satisfied in Example 19.9.



## 19.6 PROOFS OF FIRST ORDER CONDITIONS

In this concluding section, we present complete proofs of the major theorems of Chapter 18. For some of these theorems, we have already presented geometric arguments for lower dimensional versions. However, the results of Chapter 18 are important enough to warrant complete, careful proofs.

Furthermore, the proofs we present here are good examples of the mathematical proofs one encounters in advanced economic theory. They use much of the mathematical theory developed in the earlier chapters of this book. In particular, they rely heavily on:

- (1) the Implicit Function Theorem (Section 15.3),
- (2) the Chain Rule for functions of several variables (Section 14.5), and
- (3) the fact that the rows of an  $m \times n$  matrix ( $m \leq n$ ) are linearly independent  $n$ -vectors if and only if the matrix has rank  $m$ .

Let's say a little more about this last statement. In Section 7.4, we defined the *rank* of an  $m \times n$  matrix  $A$  as the number of nonzero rows in its row echelon form  $A_R$ . In particular, if  $A$  has rank  $m$ , the  $m$  rows of  $A_R$  are all nonzero. Since each row in  $A_R$  starts with more zeros than does the row above it, it is easy to see that the rows of  $A_R$  are linearly independent (Lemma 27.1). However, one can go back and forth between the rows of  $A$  and the rows of  $A_R$  just by adding multiples of one row to another. It follows that all the rows of  $A_R$  are linearly independent if and only if all the rows of  $A$  are linearly independent. Statement 3 above now follows. See Section 27.3 for a complete discussion.

### Proof of Theorems 18.1 and 18.2: Equality Constraints

We are assuming that:

- (1)  $\mathbf{x}^*$  maximizes  $f$  on the constraint set

$$h_1(\mathbf{x}^*) = c_1, \dots, h_m(\mathbf{x}^*) = c_m; \quad (33)$$

- (2) the  $m \times n$  Jacobian  $D\mathbf{h}(\mathbf{x}^*)$  of the constraint functions  $\mathbf{h}$  has maximal rank  $m$  at  $\mathbf{x}^*$ :

$$m = \text{rank } D\mathbf{h}(\mathbf{x}^*) = \text{rank} \begin{pmatrix} \frac{\partial h_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial h_1}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial h_m}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial h_m}{\partial x_n}(\mathbf{x}^*) \end{pmatrix}. \quad (34)$$

We first claim that the  $(m + 1) \times n$  Jacobian matrix

$$\begin{pmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial f}{\partial x_n}(\mathbf{x}^*) \\ \frac{\partial h_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial h_1}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial h_m}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial h_m}{\partial x_n}(\mathbf{x}^*) \end{pmatrix}. \quad (35)$$

does *not* have maximal rank. Let  $c_0 = f(\mathbf{x}^*)$ . Consider the system of equations

$$\begin{aligned} f(x_1, \dots, x_n) &= c_0 \\ h_1(x_1, \dots, x_n) &= c_1 \\ &\vdots \\ h_m(x_1, \dots, x_n) &= c_m. \end{aligned} \quad (36)$$

We know that  $(x_1^*, \dots, x_n^*)$  is a solution of system (36). Think of the right-hand sides  $c_0, c_1, \dots, c_m$  as exogenous variables. Then, the matrix (35) is simply the Jacobian of the system (36) with respect to the endogenous variables  $x_1, \dots, x_n$ .

Suppose that matrix (35) *does* have maximal rank  $m + 1$ . Then, by the Implicit Function Theorem, we can vary the  $c_i$ 's to  $c_i'$ 's a little bit and still find a solution  $x_1', \dots, x_n'$  to the revised system (36) with the  $c_i'$ 's on the right-hand side. In particular, we could find a solution  $x_1^{**}, \dots, x_n^{**}$  to the perturbed system

$$\begin{aligned} f(x_1, \dots, x_n) &= c_0 + \varepsilon \\ h_1(x_1, \dots, x_n) &= c_1 \\ &\vdots \\ h_m(x_1, \dots, x_n) &= c_m, \end{aligned} \quad (37)$$

where  $\varepsilon$  is a small *positive* number. Looking at the last  $m$  equations in (37), we see that  $(x_1^{**}, \dots, x_n^{**})$  still satisfies the constraints (33). However, comparing the first equation of (36) to the first equation of (37), we see that

$$f(\mathbf{x}^{**}) > f(\mathbf{x}^*)$$

since  $c_0 + \varepsilon > c_0$ . This contradicts our assumption that  $\mathbf{x}^*$  maximizes  $f$  on constraint set (33). We conclude that matrix (35) does not have rank  $m + 1$ . This means that its  $m + 1$  rows are linearly dependent; that is, there exist scalars

$\alpha_0, \alpha_1, \dots, \alpha_n$ , not all zero, so that

$$\alpha_0 \begin{pmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}^*) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\mathbf{x}^*) \end{pmatrix} + \alpha_1 \begin{pmatrix} \frac{\partial h_1}{\partial x_1}(\mathbf{x}^*) \\ \vdots \\ \frac{\partial h_1}{\partial x_n}(\mathbf{x}^*) \end{pmatrix} + \cdots + \alpha_m \begin{pmatrix} \frac{\partial h_m}{\partial x_1}(\mathbf{x}^*) \\ \vdots \\ \frac{\partial h_m}{\partial x_n}(\mathbf{x}^*) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \quad (38)$$

We next show that NDCQ (34) implies that  $\alpha_0 \neq 0$ . For, if  $\alpha_0 = 0$  in (38), then  $\nabla h_1(\mathbf{x}^*), \dots, \nabla h_m(\mathbf{x}^*)$  are linearly dependent by (38). This means that matrix (34) which has the  $\nabla h_i(\mathbf{x}^*)$ 's as its  $m$  rows does *not* have maximal rank — a contradiction to hypothesis *b*. We conclude that  $\alpha_0 \neq 0$ .

Finally, divide (38) through by the nonzero number  $\alpha_0$  and let  $\mu_i = -\alpha_i/\alpha_0$  for  $i = 0, \dots, m$ . Then, (38) becomes

$$\nabla f(\mathbf{x}^*) - \mu_1 \nabla h_1(\mathbf{x}^*) - \cdots - \mu_m \nabla h_m(\mathbf{x}^*) = \mathbf{0};$$

this is precisely the conclusion of Theorem 18.2.

Note that if we do not assume the NDCQ (34), then we obtain the conclusion of the Fritz John Theorem, Theorem 19.11, for equality constraints.

### Proof of Theorems 18.3 and 18.4: Inequality Constraints

We turn to the proofs of the corresponding theorems for *inequality* constraints. Now, we are assuming that:

- (1)  $\mathbf{x}^* \in \mathbf{R}^n$  maximizes  $f$  on the constraint set

$$g_1(\mathbf{x}) \leq b_1, \dots, g_k(\mathbf{x}) \leq b_k; \quad (39)$$

- (2) only  $g_1, \dots, g_e$  are binding at  $\mathbf{x}^*$ :

$$g_1(\mathbf{x}^*) = b_1, \dots, g_e(\mathbf{x}^*) = b_e, \quad (40)$$

$$g_{e+1}(\mathbf{x}^*) < b_{e+1}, \dots, g_k(\mathbf{x}^*) < b_k; \quad (41)$$

- (3) the  $e \times n$  Jacobian matrix  $Dg_E(\mathbf{x}^*)$  has maximal rank  $e$ ; that is,

$$\text{rank } Dg_E(\mathbf{x}^*) = \text{rank} \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial g_1}{\partial x_n}(\mathbf{x}^*) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_e}{\partial x_1}(\mathbf{x}^*) & \cdots & \frac{\partial g_e}{\partial x_n}(\mathbf{x}^*) \end{pmatrix}. \quad (42)$$

Since the  $g_j$ 's are continuous functions, there is a open ball  $B = B_r(\mathbf{x}^*)$  of radius  $r > 0$  about  $\mathbf{x}^*$  such that  $g_j(\mathbf{x}) < b_j$  for all  $\mathbf{x} \in B$  and for  $j = e + 1, \dots, n$ . We will work in the open set  $B$  for the rest of this proof.

Note that  $\mathbf{x}^*$  maximizes  $f$  in  $B$  over the constraint set

$$g_1(\mathbf{x}) = b_1, \dots, g_e(\mathbf{x}) = b_e, \quad (43)$$

because if there were another point  $\mathbf{x}^{**}$  in  $B$  that satisfied (43) and gave a higher value of  $f$ , then this point would yield a higher value of  $f$  on the original constraint set (39) and contradict the definition of  $\mathbf{x}^*$ . Furthermore, by (42),  $\mathbf{x}^*$  satisfies the NDCQ for the problem of maximizing  $f$  on the constraint set (43). Therefore, by Theorem 18.2, there exist  $\mu_1^*, \dots, \mu_e^*$  such that

$$\begin{aligned} \frac{\partial \hat{L}}{\partial x_1}(\mathbf{x}^*, \mu^*) &= 0, \dots, \frac{\partial \hat{L}}{\partial x_n}(\mathbf{x}^*, \mu^*) = 0, \\ g_1(\mathbf{x}^*) - b_1 &= 0, \dots, g_e(\mathbf{x}^*) - b_e = 0, \end{aligned} \quad (44)$$

where  $\hat{L}(\mathbf{x}, \mu) \equiv f(\mathbf{x}) - \mu_1[g_1(\mathbf{x}) - b_1] - \dots - \mu_e[g_e(\mathbf{x}) - b_e]$ .

Now, consider the usual Lagrangian

$$L(\mathbf{x}, \lambda_1, \dots, \lambda_k) \equiv f(\mathbf{x}) - \lambda_1[g_1(\mathbf{x}) - b_1] - \dots - \lambda_k[g_k(\mathbf{x}) - b_k]. \quad (45)$$

Let  $\lambda_i^* = \mu_i^*$  for  $i = 1, \dots, e$  and set  $\lambda_j^* = 0$  for  $j = e + 1, \dots, m$ . Using this choice of the  $\lambda_j^*$ 's and noting equation (44), we see that  $(\mathbf{x}^*, \lambda^*)$  is a solution of the  $n + k$  equations in  $n + k$  unknowns:

$$\begin{aligned} \frac{\partial L}{\partial x_1}(\mathbf{x}^*, \lambda^*) &= 0, \dots, \frac{\partial L}{\partial x_n}(\mathbf{x}^*, \lambda^*) = 0, \\ \lambda_1^* \cdot [g_1(\mathbf{x}^*) - b_1] &= 0, \dots, \lambda_e^* \cdot [g_e(\mathbf{x}^*) - b_e] = 0, \\ \lambda_{e+1}^* \cdot [g_{e+1}(\mathbf{x}^*) - b_{e+1}] &= 0, \dots, \lambda_k^* \cdot [g_k(\mathbf{x}^*) - b_k] = 0. \end{aligned} \quad (46)$$

Except for the condition that all the  $\lambda_i$ 's must be  $\geq 0$ , we have completed the proof of the first order conditions of Theorem 18.4. We turn now to the proof that the  $\lambda_i$ 's must be  $\geq 0$ .

Consider the system of  $e$  equations in  $n + e$  variables:

$$\begin{aligned} g_1(x_1, \dots, x_n) &= b_1 \\ &\vdots \\ g_e(x_1, \dots, x_n) &= b_e. \end{aligned} \quad (47)$$

By the rank condition (42) and the Implicit Function Theorem (Theorem 15.7), there exist  $e$  coordinates  $x_{j_1}, \dots, x_{j_e}$  such that we can consider system (47) as implicitly defining  $x_{j_1}, \dots, x_{j_e}$  in terms of the rest of the  $x_i$ 's and all the  $b_j$ 's. In this latter set of exogenous variables, hold  $b_2, \dots, b_e$  constant, hold the exogenous  $x_j$ 's constant, and let  $b_1$  decrease linearly:  $t \mapsto b_1 - t$  for  $t \geq 0$ . By the Implicit Function Theorem, as the exogenous variable  $b_1$  varies, we can still solve system (47) for  $x_{j_1}, \dots, x_{j_e}$ . This means, in particular, that there is a  $C^1$  curve  $\mathbf{x}(t)$  defined for  $t \in [0, \varepsilon)$ , such that  $\mathbf{x}(0) = \mathbf{x}^*$  and, for all  $t \in [0, \varepsilon)$ ,

$$g_1(\mathbf{x}(t)) = b_1 - t \quad \text{and} \quad g_j(\mathbf{x}(t)) = b_j \quad \text{for } j = 2, \dots, e. \quad (48)$$

Let  $\mathbf{v} = \mathbf{x}'(0)$ . Applying the Chain Rule to (48), we conclude that

$$Dg_1(\mathbf{x}^*)\mathbf{v} = -1, \quad Dg_j(\mathbf{x}^*)\mathbf{v} = 0 \quad \text{for } j = 2, \dots, e. \quad (49)$$

Since  $\mathbf{x}(t)$  lies in the constraint set for all  $t$  and  $\mathbf{x}^*$  maximizes  $f$  in the constraint set,  $f$  must be nonincreasing along  $\mathbf{x}(t)$ . Therefore,

$$\left. \frac{d}{dt} f(\mathbf{x}(t)) \right|_{t=0} = Df(\mathbf{x}^*)\mathbf{v} \leq 0.$$

Let  $D_{\mathbf{x}}L(\mathbf{x}^*)$  denote the derivative of the Lagrangian (45) with respect to  $\mathbf{x}$ . By our first order conditions (46) and by (49),

$$\begin{aligned} 0 &= D_{\mathbf{x}}L(\mathbf{x}^*)\mathbf{v} \\ &= Df(\mathbf{x}^*)\mathbf{v} - \sum_i \lambda_i Dg_i(\mathbf{x}^*)\mathbf{v} \\ &= Df(\mathbf{x}^*)\mathbf{v} - \lambda_1 Dg_1(\mathbf{x}^*)\mathbf{v} \\ &= Df(\mathbf{x}^*)\mathbf{v} + \lambda_1. \end{aligned}$$

Since  $Df(\mathbf{x}^*)\mathbf{v} \leq 0$ , we conclude that  $\lambda_1 \geq 0$ . A similar argument shows that  $\lambda_j \geq 0$  for  $j = 1, \dots, e$ . This finishes the proof of Theorem 18.4.

The argument just presented works equally well when the problem contains both inequality constraints and equality constraints, provided that NDCQ is valid at  $\mathbf{x}^*$  for the combined equality and *binding* inequality constraints.

## EXERCISES

- 19.24** Write out the proof that  $\lambda_2 \geq 0$  in the proof of Theorem 18.4.  
**19.25** Write out a careful proof of Theorem 18.5 for mixed constraints.